

Macrolides and Clindamycin  
Whitney J. Nesbitt  
Haley A. Pritchard  
David M. Aronoff

#### Short View Summary

##### Macrolides

##### Erythromycin

- Usual adult doses
- Oral: base—250 to 500 mg every 6 to 12 hours; maximum: 4 g daily
- Ethylsuccinate: 400 to 800 mg every 6 to 12 hours; maximum: 4 g daily
- IV: lactobionate—15 to 20 mg/kg/day divided every 6 hours, or 500 mg to 1 g every 6 hours, or given as a continuous infusion over 24 hours; maximum: 4 g daily
- Usual pediatric doses
- Ethylsuccinate: 10 to 12.5 mg/kg/dose every 6 hours; maximum: 2 g daily
- Renal or hepatic failure
- In renal failure, slightly dialyzable (5%–20%); supplemental dose is not necessary in hemodialysis or peritoneal dialysis or in continuous arteriovenous or venovenous hemofiltration
- No dose adjustment in hepatic failure
- Cerebrospinal fluid (CSF) penetration
- CSF-to-blood level ratio: normal meninges—2% to 13%; inflamed meninges—7% to 25%
- Adverse effects
- Cardiovascular corrected QT (QTc) interval prolongation, torsades de pointes, gastrointestinal (GI) upset, pruritus, erythema multiforme, drug-induced pemphigus, cholestatic jaundice, postnatal pyloric stenosis
- Contraindications
- Caution for patients with underlying heart disease or for those taking other drugs that may prolong the QT interval regarding potential for inducing fatal ventricular arrhythmias
- Hypersensitivity to erythromycin, any macrolide antibiotics, or any component of the formulation
- Concomitant use with cisapride, ergotamine, terfenadine, lovastatin, or simvastatin
- Drug-drug interactions
- Major inhibitor of cytochrome P-450 isoenzyme 3A (CYP3A4)
- Indications
- Bartonella spp. infections, chancroid, granuloma inguinale, legionnaires' disease, lymphogranuloma venereum, nongonococcal urethritis, pertussis, gastric and intestinal motility disorders
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- Azithromycin
- • Usual adult dose
- • 500 mg on day 1, followed by 250 mg once daily on days 2 to 5
- • Usual pediatric dose
- • 10 to 12 mg/kg/dose on day 1 (maximum: 500 mg/dose), followed by 5 to 6 mg/kg/dose (maximum: 250 mg/dose) on days 2 through 5
- • Renal or hepatic failure
- • No dosage adjustment required; use with caution in patients with severe renal impairment (glomerular filtration rate <10 mL/min)
- • CSF penetration
- • Poor central nervous system (CNS) penetration
- • Adverse effects
- • Cardiovascular (QTc) prolongation, torsades de pointes, GI upset, pruritus, erythema multiforme, Stevens-Johnson syndrome, fixed drug eruptions, vaginitis, cholestatic jaundice, drug-induced liver injury
- • Contraindications
- • Caution for patients with underlying heart disease or for those taking other drugs that may prolong the QT interval regarding potential for inducing fatal ventricular arrhythmias
- • Hypersensitivity to azithromycin, any macrolide antibiotics, or any component of the formulation
- • History of cholestatic jaundice/hepatic dysfunction associated with prior azithromycin use
- • Drug-drug interactions
- • Amiodarone and other QTc-prolonging agents, quinine
- • Indications
- • Treatment of acute otitis media; pharyngitis/tonsillitis; community-acquired pneumonia; pelvic inflammatory disease; antimicrobial prophylaxis for premature rupture of membranes or unplanned Caesarean section; genital ulcer disease (in men) due to Haemophilus ducreyi (chancroid); acute bacterial exacerbations of chronic obstructive pulmonary disease; acute bacterial sinusitis; prevention of Mycobacterium avium complex (alone or in combination with rifabutin) in patients with advanced human immunodeficiency virus (HIV) infection; treatment of disseminated M. avium complex (in combination with ethambutol) in patients with advanced HIV infection; treatment of nontuberculous mycobacterial infections; skin and skin structure infections (uncomplicated) due to Staphylococcus aureus, Streptococcus pyogenes, or Streptococcus agalactiae; infections caused by Babesia microti (in combination with atovaquone); urethritis and cervicitis due to Chlamydia trachomatis or Neisseria gonorrhoeae; trachoma (C. trachomatis); and yaws (Treponema pallidum subsp. pertenue)
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- • Creatinine clearance <30 mL/min: decrease clarithromycin dose by 50%

- • Hemodialysis: administer after hemodialysis session is completed; in hepatic failure, no dosing adjustment is necessary as long as renal function is normal
- • CSF penetration
- • Poor CNS penetration
- • Adverse effects
- • Cardiovascular (QTc) prolongation, torsades de pointes, hypersensitivity reactions
- • Contraindications
- • Caution for patients with underlying heart disease or for those taking other drugs that may prolong the QT interval regarding potential for inducing fatal ventricular arrhythmias
- • Hypersensitivity to azithromycin, any macrolide antibiotics, or any component of the formulation
- • History of cholestatic jaundice/hepatic dysfunction associated with prior azithromycin use
- • Drug-drug interactions
- • Colchicine toxicity, major inhibitor of CYP3A4
- • Indications
- • Pharyngitis/tonsillitis; acute maxillary sinusitis; acute exacerbation of chronic bronchitis; community-acquired pneumonia; uncomplicated skin/skin structure infections due to susceptible *S. aureus*, *S. pyogenes*; disseminated mycobacterial infections due to *M. avium* or *Mycobacterium intracellulare*; prevention of disseminated mycobacterial infections due to *M. avium* complex disease (e.g., patients with advanced HIV infection); duodenal ulcer disease due to *Helicobacter pylori* in regimens with other drugs, including amoxicillin and lansoprazole or omeprazole
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- Ketolides
- • Solithromycin (not FDA approved)
- Clindamycin
- • Usual adult doses
- • Oral: 150 to 450 mg/dose every 6 to 8 hours; maximum dose: 1800 mg daily
- • IM, IV: 1.2 to 2.7 g/day in 2 to 4 divided doses; maximum dose: 4800 mg daily
- • Usual pediatric doses
- • Oral: 20 to 30 mg/kg every day divided every 6 to 8 hours; maximum dose: 450 mg/dose
- • IV: 20 to 40 mg/kg every day divided every 6 to 8 hours; maximum dose: 900 mg/dose
- • Renal or hepatic failure
- • Renal failure: no adjustment required
- • CSF penetration
- • Poor
- • Adverse effects
- • Diarrhea, abdominal pain, vaginitis, drug exanthem
- • Contraindications
- • Hypersensitivity to clindamycin, lincomycin, or any component of the formulation
- • Drug-drug interactions
- • Lincosamide antibiotics may diminish the therapeutic effects of erythromycin
- • Indications

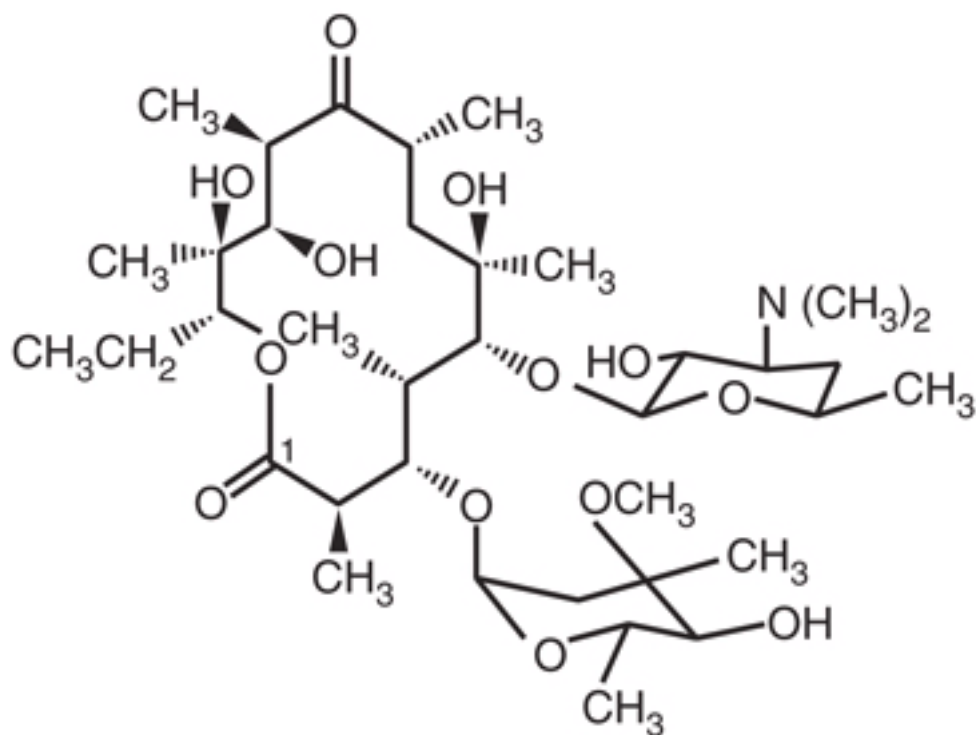
Macrolides were among the first major classes of microbial products to be discovered that possess antibiotic properties, after  $\beta$ -lactams, tetracyclines, and aminoglycosides.<sup>1</sup> The macrolide antibiotics (erythromycin, azithromycin, and clarithromycin) and the lincosamide antibiotics (lincomycin and clindamycin) are chemically unrelated but possess many similar biologic properties in terms of mechanisms of action and resistance, antimicrobial activity, and clinical pharmacology. Erythromycin, the macrolide in longest use, is sometimes useful as an alternative to penicillin G and other antibiotics. Azithromycin and clarithromycin have advantages over erythromycin related to their antimicrobial activity, pharmacokinetics, fewer gastrointestinal side effects, and effectiveness in certain infections. Although more expensive, they have largely replaced erythromycin in clinical use. Azithromycin is noteworthy among the macrolides for fewer drug interactions and antiinflammatory activities. Dirithromycin, a semisynthetic derivative of erythromycin, is not available in the United States, is not widely used in Europe, but is prescribed in China; it is not included in this chapter but reviews of its use are available.<sup>2,3</sup> All of the macrolide antibiotics in clinical use have the potential for inducing cardiac arrhythmias.<sup>4</sup> Telithromycin, the first member of the ketolide class, exhibited increased activity against many bacterial strains that were resistant to macrolides. It was associated with rare, yet severe hepatic toxicity that substantially limited its clinical usefulness, and manufacturing has therefore been

discontinued. Solithromycin is a ketolide antibiotic that was developed for the treatment of community-acquired pneumonia and other infections (including urogenital gonorrhea), particularly those caused by macrolide-resistant pathogens.<sup>5-7</sup> However, it is not yet approved for use in the United States or Europe. Nafithromycin (WCK 4873), a novel lactone ketolide class of antibiotic, which overcomes primary modes of macrolide resistance in *Streptococcus pneumoniae* and appears to be active against multidrug-resistant typical and atypical respiratory pathogens, is in phase three clinical studies for community-acquired pneumonia.<sup>8</sup> Clindamycin has been restricted in use by its association with *Clostridioides difficile* (formerly *Clostridium difficile*) colitis; it remains mostly as an alternative-choice antibiotic in the treatment of several infections, especially certain anaerobic infections. Lincomycin is now mainly of historic interest.

## Erythromycin

### Derivation, Chemistry, and Preparations

Erythromycin was derived in 1952 from a strain of *Saccharopolyspora erythraea* (originally named *Streptomyces erythraeus*) obtained from soil from the Philippines. It consists of a mixture of antibiotics (such as erythromycins A, B, C, and D), in which erythromycin A is the most active



component.<sup>9</sup> The structure (Fig. 28.1) of erythromycin A consists of a 14-member macrocyclic lactone ring—therefore the class name macrolide—attached to two sugar moieties, desosamine and L-cladinose. Erythromycin base is poorly soluble in water, has a pKa of 8.8, is rapidly inactivated by gastric acid, and is often inconsistently absorbed after oral administration. Pharmaceutical preparations for oral use have been made with an aim to diminish destruction by gastric acid and to promote better absorption. Oral preparations available include (1) enteric-coated tablets, enteric-coated pellets in capsules for delayed release, and “film”-coated tablets of the base; (2) the stearate salt (formed in association with the amino group on desosamine), available as film-coated tablets; and (3) the ethylsuccinate ester (formed with the hydroxyl group

on desosamine), available in tablet, chewable, and liquid forms. The water-soluble salt of erythromycin prepared for intravenous use is erythromycin lactobionate. Erythromycin lactobionate is not given intramuscularly because of injection site pain. Erythromycin base is also available in 0.5%, 2%, and 3% topical solutions, gels, and creams for treatment of acne vulgaris and in an ophthalmic ointment for treatment of bacterial conjunctivitis and prevention of neonatal gonococcal and chlamydial conjunctivitis.

#### Mechanisms of Action

Erythromycin inhibits RNA-dependent protein synthesis at the step of chain elongation in susceptible organisms. X-ray crystallography studies indicate that several functional groups on erythromycin A bind to sequences on domain V of the 23S ribosomal RNA (rRNA) that is a component of the 50S subunit of the bacterial ribosome.<sup>10-12</sup> That binding site is near the peptidyl transferase center, and peptide chain elongation is thereby prevented by blocking of the polypeptide exit tunnel.<sup>10,11,13</sup> As a result, peptidyl transfer RNA (peptidyl-tRNA) is dissociated from the ribosome.<sup>10,11</sup> Data provided by macrolide-resistant mutants indicate that there are also important interactions between macrolides and specific ribosomal proteins or bases of the 50S subunit near the peptidyl transferase center.<sup>13,14</sup>

Studies in *Escherichia coli* and *Staphylococcus aureus* have demonstrated that erythromycin also inhibits the formation of the 50S ribosomal subunit.<sup>10,13,15</sup> In some bacteria, erythromycin interferes with the ribosomal binding of other macrolides, lincomycin, and chloramphenicol, suggesting common or overlapping binding sites for these antibiotics.

#### Mechanisms of Resistance

Macrolide resistance is commonly achieved through three major mechanisms: (1) efflux of the drug out of the bacterial cell, (2) target modification (either ribosomal RNA or protein) resulting in impaired ability to inhibit protein synthesis, and (3) enzymatic drug modification.<sup>16-18</sup> Drug resistance to macrolides is a major clinical problem among most pathogens for which these drugs are used, including *S. pneumoniae*, *Campylobacter jejuni*, *Helicobacter pylori*, *Mycoplasma pneumoniae*, *Neisseria gonorrhoeae*, *S. aureus*, and others.<sup>18</sup>

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- Contraindications

- Caution for patients with underlying heart disease or for those taking other drugs that may prolong the QT interval regarding potential for inducing fatal ventricular arrhythmias
- Hypersensitivity to erythromycin, any macrolide antibiotics, or any component of the formulation
- Concomitant use with cisapride, ergotamine, terfenadine, lovastatin, or simvastatin
- Drug-drug interactions
- Major inhibitor of cytochrome P-450 isoenzyme 3A (CYP3A4)

#### Indications

- *Bartonella* spp. infections, chancroid, granuloma inguinale, legionnaires' disease, lymphogranuloma venereum, nongonococcal urethritis, pertussis, gastric and intestinal motility disorders

#### Azithromycin

- Usual adult dose
- 500 mg on day 1, followed by 250 mg once daily on days 2 to 5
- Usual pediatric dose
- 10 to 12 mg/kg/dose on day 1 (maximum: 500 mg/dose), followed by 5 to 6 mg/kg/dose (maximum: 250 mg/dose) on days 2 through 5
- Renal or hepatic failure
- No dosage adjustment required; use with caution in patients with severe renal impairment (glomerular filtration rate <10 mL/min)
- CSF penetration
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- Cardiovascular (QTc) prolongation, torsades de pointes, GI upset, pruritus, erythema multiforme, Stevens-Johnson syndrome, fixed drug eruptions, vaginitis, cholestatic jaundice, drug-induced liver injury

#### Contraindications

- Caution for patients with underlying heart disease or for those taking other drugs that may prolong the QT interval regarding potential for inducing fatal ventricular arrhythmias
- Hypersensitivity to azithromycin, any macrolide antibiotics, or any component of the formulation
- History of cholestatic jaundice/hepatic dysfunction associated with prior azithromycin use
- Drug-drug interactions

- Amiodarone and other QTc-prolonging agents, quinine

#### Indications

- Treatment of acute otitis media; pharyngitis/tonsillitis; community-acquired pneumonia; pelvic inflammatory disease; antimicrobial prophylaxis for premature rupture of membranes or unplanned Caesarean section; genital ulcer disease (in men) due to *Haemophilus ducreyi* (chancroid); acute bacterial exacerbations of chronic obstructive pulmonary disease; acute bacterial sinusitis; prevention of *Mycobacterium avium* complex (alone or in combination with rifabutin) in patients with advanced human immunodeficiency virus (HIV) infection; treatment of disseminated *M. avium* complex (in combination with ethambutol) in patients with advanced HIV infection; treatment of nontuberculous mycobacterial infections; skin and skin structure infections (uncomplicated) due to *Staphylococcus aureus*, *Streptococcus pyogenes*, or *Streptococcus agalactiae*; infections caused by *Babesia microti* (in combination with atovaquone); urethritis and cervicitis due to *Chlamydia trachomatis* or *Neisseria gonorrhoeae*; trachoma (*C. trachomatis*); and yaws (*Treponema pallidum* subsp. *pertenue*)

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- Usual adult dose
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- Creatinine clearance <30 mL/min: decrease clarithromycin dose by 50%
- Hemodialysis: administer after hemodialysis session is completed; in hepatic failure, no dosing adjustment is necessary as long as renal function is normal
- CSF penetration
- Poor CNS penetration
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- Caution for patients with underlying heart disease or for those taking other drugs that may prolong the QT interval regarding potential for inducing fatal ventricular arrhythmias
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#### Ketolides

- Solithromycin (not FDA approved)

#### Clindamycin

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- Contraindications
- Hypersensitivity to clindamycin, lincomycin, or any component of the formulation
- Drug-drug interactions
- Lincosamide antibiotics may diminish the therapeutic effects of erythromycin
- Indications
- Treatment of susceptible bacterial infections, mainly those caused by anaerobes, streptococci, pneumococci, and staphylococci; pelvic inflammatory disease (IV); bacterial vaginosis; hidradenitis suppurativa; toxin production mediation in necrotizing or invasive skin and soft tissue infection; component of regimens for malaria, toxoplasmosis, babesiosis (in combination with quinine)

Macrolides were the among the first major classes of microbial products to be discovered that possess antibiotic properties, after  $\beta$ -lactams, tetracyclines, and aminoglycosides.<sup>1</sup> The macrolide antibiotics (erythromycin, azithromycin, and clarithromycin) and the lincosamide antibiotics (lincomycin and clindamycin) are chemically unrelated but possess many similar biologic properties in terms of mechanisms of action and resistance, antimicrobial activity, and clinical pharmacology. Erythromycin, the macrolide in longest use, is sometimes useful as an alternative to penicillin G and other antibiotics. Azithromycin and clarithromycin have advantages over erythromycin related to their antimicrobial activity, pharmacokinetics, fewer gastrointestinal side effects, and effectiveness in certain infections. Although more expensive, they have largely replaced erythromycin in clinical use. Azithromycin is noteworthy among the macrolides for fewer drug interactions and antiinflammatory activities. Dirithromycin, a semisynthetic derivative of erythromycin, is not available in the United States, is not widely used in Europe, but is prescribed in China; it is not included in this chapter but reviews of its use are available.<sup>2,3</sup> All of the macrolide antibiotics in clinical use have the potential for inducing cardiac arrhythmias.<sup>4</sup> Telithromycin, the first member of the ketolide class, exhibited increased activity against many bacterial strains that were resistant to macrolides. It was associated with rare, yet severe hepatic toxicity that substantially limited its clinical usefulness, and manufacturing has therefore been discontinued. Solithromycin is a ketolide antibiotic that was developed for the treatment of community-acquired pneumonia and other infections (including urogenital gonorrhea), particularly those caused by macrolide-resistant pathogens.<sup>5-7</sup> However, it is not yet approved for use in the

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### FIG. 28.1 Erythromycin base.

Follow for extended description

#### Mechanisms of Action

Erythromycin inhibits RNA-dependent protein synthesis at the step of chain elongation in susceptible organisms. X-ray crystallography studies indicate that several functional groups on erythromycin A bind to sequences on domain V of the 23S ribosomal RNA (rRNA) that is a component of the 50S subunit of the bacterial ribosome.<sup>10-12</sup> That binding site is near the peptidyl transferase center, and peptide chain elongation is thereby prevented by blocking of the polypeptide exit tunnel.<sup>10,11,13</sup> As a result, peptidyl transfer RNA (peptidyl-tRNA) is dissociated from the ribosome.<sup>10,11</sup> Data provided by macrolide-resistant mutants indicate that there are also important interactions between macrolides and specific ribosomal proteins or bases of the 50S subunit near the peptidyl transferase center.<sup>13,14</sup>

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#### Mechanisms of Resistance

Macrolide resistance is commonly achieved through three major mechanisms: (1) efflux of the drug out of the bacterial cell, (2) target modification (either ribosomal RNA or protein) resulting in impaired ability to inhibit protein synthesis, and (3) enzymatic drug modification.<sup>16-18</sup> Drug resistance to macrolides is a major clinical problem among most pathogens for which these drugs are used, including *S. pneumoniae*, *Campylobacter jejuni*, *Helicobacter pylori*, *Mycoplasma pneumoniae*, *Neisseria gonorrhoeae*, *S. aureus*, and others.<sup>18</sup>

#### Decreased Microbial Entry or Export of Drug

Enterobacteriales exhibit decreased permeability of the outer cell envelope to macrolides.<sup>19-21</sup> This intrinsic resistance is also exhibited by *Pseudomonas* spp. and *Acinetobacter* spp. In



addition, chromosomally encoded efflux pumps of several families can provide macrolide and, in some cases, lincosamide and streptogramin resistance.<sup>14,18</sup> A different efflux system for erythromycin and other 14- and 15-member macrolides (but not 16-member macrolides, lincosamides, or analogues of streptogramin B), called the M phenotype, has been elucidated in erythromycin-resistant strains of *Streptococcus pyogenes*, *S. pneumoniae*, group C streptococci, and enterococcal species. This efflux system is encoded by the *mef(A)* gene, is carried on a transposable element, and consists of transmembrane domains across the cytoplasmic membrane.<sup>10</sup> The M phenotype resistance is expressed by *S. pneumoniae* at moderate levels with erythromycin minimal inhibitory concentrations (MICs) of 1 to 64 µg/mL.<sup>22</sup>

#### Target Site Alterations

Mutations in genes for 50S ribosomal proteins or bases of critical domains of the 23S rRNA receptor site confer resistance to erythromycin and sometimes to other macrolides (M), lincosamides (L), and streptogramin type B (SB); in some but not all strains, this is associated with a decreased binding affinity for erythromycin in *S. pneumoniae*, *H. pylori*, *Mycobacterium avium*, *Bacillus subtilis*, *S. pyogenes*, *Campylobacter* spp., *M. pneumoniae*, *E. coli*, and *S. aureus*.<sup>10,13,14,23-27</sup> This pattern of resistance, referred to as the MLSB phenotype, is mediated by the *erm* (erythromycin ribosome methylation) genes on plasmids or transposons on chromosomes, which are self-transferable. There are many identified classes of *erm* genes, the most important of which are the *ermA*, *ermB*, *ermC*, and *ermF* classes. The resistance results from decreased binding of the antibiotics to their overlapping targets on the ribosome. It is the most widespread mechanism of resistance to the macrolides and the lincosamides and can be exhibited by strains of *S. aureus* (particularly having *ermA* or *ermC* genes), *S. pyogenes* (with *ermA* or *ermB* genes), *S. pneumoniae*, *Enterococcus* spp., *Corynebacterium diphtheriae*, *Campylobacter* spp., *Bacteroides fragilis* (*ermF* gene), *Clostridium perfringens*, *Listeria* spp., *M. pneumoniae*, and *Legionella* spp. The MLSB resistance phenotype may be constitutive or inducible by subinhibitory concentrations of erythromycin or other macrolides that bring about induction of the methylating enzyme. Lincosamides are not inducers of this system. When of the inducible type, macrolide resistance is manifest, but clindamycin susceptibility is often found in vitro; however, in that latter case, resistance will often emerge in vivo by selection of mutants that are constitutively producing a methylating enzyme, especially in infections with high bacterial density.<sup>25</sup> Strains of *S. pyogenes* with the *ermB* gene are constitutive producers of methylase and therefore are usually resistant to both the macrolides and clindamycin.<sup>28</sup> Strains of *S. pyogenes* with the *ermA* gene are usually inducible producers of methylase that consistently exhibit macrolide resistance and both macrolide and clindamycin resistance when the strain is exposed to those antibiotics.<sup>25,28</sup>

#### Drug Inactivation

Enzymatic inactivation of erythromycin and some other 14-, 15-, and 16-member ring macrolides can render them unable to bind to the 50S ribosome.<sup>18</sup> There are two major classes of bacterial enzymes that accomplish this drug inactivation: macrolide phosphotransferases and macrolide esterases.<sup>18</sup> Phosphotransferases have been described in strains of *S. aureus*, *E. coli*, *Pseudomonas*, *Pasteurella*, *Klebsiella*, *Serratia*, *Shigella*, and *Nocardia* spp. and are encoded by genes designated *mph(A)*, *mph(B)*, and *mph(C)*.<sup>14,18</sup> Esterase genes (*ere[A]* and *ere[B]*) on plasmids encode for the hydrolysis of the macrocyclic lactone of erythromycin and have been found in strains of *E. coli*, *Klebsiella* spp., *Citrobacter* spp., *Proteus* spp., *Pseudomonas* spp., *Enterobacter* spp., *Salmonella* spp., *Vibrio cholerae*, and rare strains of *S. aureus*.<sup>14,18</sup>

Nucleic acid amplification methods that allow for relatively rapid detection of the different mechanisms of macrolide resistance among clinical isolates (genotypic testing) have been developed.<sup>29,30</sup> A given bacterial strain may possess more than one type of macrolide resistance mechanism, resulting in complex resistant phenotypes.

#### Antimicrobial Activity

The antimicrobial activity of erythromycin is broad in spectrum, being exhibited against gram-positive and gram-negative bacteria, including actinomycetes and mycobacteria, as well as against treponemes, mycoplasmas, Chlamydia, and rickettsiae. Depending on drug concentration, bacterial species, phase of growth, and density of the inoculum, erythromycin is essentially bacteriostatic. Bacterial killing is favored by higher antibiotic concentrations, higher pH, lower bacterial density, and rapid growth.<sup>31</sup> The activity of erythromycin, which is a weak

base, increases markedly with increasing pH over the range of 5.5 to 8.5 for both gram-positive and gram-negative bacteria,<sup>32,33</sup> possibly reflecting increased entry into the bacterial cell of the un-ionized drug that is more plentiful at the higher pH.

The in vitro susceptibilities of potential pathogens to erythromycin are listed in Table 28.1.14,27,34–45 Macrolide resistance is increasing in pneumococci<sup>46</sup> and group A streptococci isolated in the United States, and resistant clinical isolates have been increasingly encountered worldwide.<sup>34,47–50</sup> In the United States (2011–20), the prevalence of erythromycin resistance among *S. pneumoniae* strains was approximately 38%.<sup>46</sup> Rates of macrolide resistance were higher among isolates from noninvasive pneumococcal disease (44%) compared with invasive pneumococcal disease (27.4%).<sup>46</sup> *S. pneumoniae* strains demonstrate complete cross-resistance among the macrolides,<sup>51,52</sup> but cross-resistance extending from the macrolides to clindamycin is variable, depending on whether the resistance mechanism is of the MLSB or M phenotype.<sup>52,53</sup> A recent US surveillance study identified clindamycin resistance in 14.7% of 3487 *S. pneumoniae* isolates but erythromycin resistance in 35.9%.<sup>54</sup>

Worldwide, erythromycin resistance in *S. pyogenes* varies from as low as 2% in Europe to as high as 90% in China,<sup>55,56</sup> with rates of macrolide nonsusceptibility of 14.5% to 23% reported in the United States.<sup>57,58</sup> As with *S. pneumoniae*, there is cross-resistance among the macrolides in *S. pyogenes*; however, resistance does not extend to clindamycin if resistance is due to the M phenotype.<sup>59,60</sup> Clindamycin resistance among *S. pyogenes* is increasing in the United States, from an estimated 0.5% in 2003<sup>28</sup> to 15% in children.<sup>61</sup> Isolates from invasive infections are more commonly clindamycin resistant, approaching 1 in 4 by 2017.<sup>57</sup>

Resistance to erythromycin by *S. aureus* may be selected by its use in hospitals.<sup>62</sup> Most methicillin-resistant strains and many methicillin-susceptible clinical isolates are now resistant to this agent.<sup>63,64</sup> Analyses of staphylococcal isolates from pediatric and adult patients revealed that across all ages, strains that were methicillin susceptible were on average 35% resistant to erythromycin, whereas strains that were methicillin resistant were 88% resistant.<sup>65</sup> In addition, there is a potential for the emergence, during treatment in an individual patient, of erythromycin resistance by *S. aureus*.<sup>66,67</sup> These strains may demonstrate the emergence of high-level resistance to erythromycin alone, or they may show cross-resistance to other macrolides and to lincomycin and clindamycin<sup>25</sup> due to the inducible type of MLSB resistance, described earlier.

The viridans group of streptococci has traditionally been considered generally susceptible to erythromycin. However, macrolide resistance may be increasing in some areas, with 57.8% susceptible in North America, 62.7% in Asia and the Pacific, and 73.7% in Europe.<sup>68,69</sup>

Other susceptible gram-positive organisms include the majority of strains of *Listeria monocytogenes* and *C. diphtheriae*.<sup>70,71</sup> Appreciable in vitro activity has been demonstrated against *Nocardia asteroides* when combined with ampicillin.<sup>72</sup>

Erythromycin displays activity against a wide range of gram-positive anaerobes, including *Actinomyces israelii*,<sup>73</sup> *Peptostreptococcus* spp., *Propionibacterium*, *Lactobacillus*, *Eubacterium*, *Bifidobacterium*, and many strains of *Peptococcus*.<sup>74</sup> *Clostridium tetani* and *C. perfringens* are generally susceptible<sup>75</sup>; however, many strains of *C. perfringens* may be only moderately susceptible<sup>76</sup> due to erythromycin target-site alterations.<sup>77</sup> A recent global surveillance study identified MLSB-mediated resistance genotypes in 19.2% of *C. difficile* strains tested.<sup>78</sup> Page 378

Other surveys have

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Macrolides and Clindamycin  
Whitney J. Nesbitt  
Haley A. Pritchard  
David M. Aronoff

Short View Summary

Macrolides

Erythromycin

- Usual adult doses
- Oral: base—250 to 500 mg every 6 to 12 hours; maximum: 4 g daily

- Ethylsuccinate: 400 to 800 mg every 6 to 12 hours; maximum: 4 g daily
  - IV: lactobionate— 15 to 20 mg/kg/day divided every 6 hours, or 500 mg to 1 g every 6 hours, or given as a continuous infusion over 24 hours; maximum: 4 g daily
  - Usual pediatric doses
  - Ethylsuccinate: 10 to 12.5 mg/kg/dose every 6 hours; maximum: 2 g daily
  - Renal or hepatic failure
  - In renal failure, slightly dialyzable (5%–20%); supplemental dose is not necessary in hemodialysis or peritoneal dialysis or in continuous arteriovenous or venovenous hemofiltration
  - No dose adjustment in hepatic failure
  - Cerebrospinal fluid (CSF) penetration
  - CSF-to-blood level ratio: normal meninges—2% to 13%; inflamed meninges—7% to 25%
  - Adverse effects
  - Cardiovascular corrected QT (QTc) interval prolongation, torsades de pointes, gastrointestinal (GI) upset, pruritus, erythema multiforme, drug-induced pemphigus, cholestatic jaundice, postnatal pyloric stenosis
  - Contraindications
  - Caution for patients with underlying heart disease or for those taking other drugs that may prolong the QT interval regarding potential for inducing fatal ventricular arrhythmias
  - Hypersensitivity to erythromycin, any macrolide antibiotics, or any component of the formulation
  - Concomitant use with cisapride, ergotamine, terfenadine, lovastatin, or simvastatin
  - Drug-drug interactions
  - Major inhibitor of cytochrome P-450 isoenzyme 3A (CYP3A4)
  - Indications
  - Bartonella spp. infections, chancroid, granuloma inguinale, legionnaires' disease, lymphogranuloma venereum, nongonococcal urethritis, pertussis, gastric and intestinal motility disorders
- Azithromycin**
- Usual adult dose
  - 500 mg on day 1, followed by 250 mg once daily on days 2 to 5
  - Usual pediatric dose
  - 10 to 12 mg/kg/dose on day 1 (maximum: 500 mg/dose), followed by 5 to 6 mg/kg/dose (maximum: 250 mg/dose) on days 2 through 5
  - Renal or hepatic failure
  - No dosage adjustment required; use with caution in patients with severe renal impairment (glomerular filtration rate <10 mL/min)
  - CSF penetration
  - Poor central nervous system (CNS) penetration
  - Adverse effects
  - Cardiovascular (QTc) prolongation, torsades de pointes, GI upset, pruritus, erythema multiforme, Stevens-Johnson syndrome, fixed drug eruptions, vaginitis, cholestatic jaundice, drug-induced liver injury
  - Contraindications
  - Caution for patients with underlying heart disease or for those taking other drugs that may prolong the QT interval regarding potential for inducing fatal ventricular arrhythmias
  - Hypersensitivity to azithromycin, any macrolide antibiotics, or any component of the formulation
  - History of cholestatic jaundice/hepatic dysfunction associated with prior azithromycin use
  - Drug-drug interactions
  - Amiodarone and other QTc-prolonging agents, quinine
  - Indications
  - Treatment of acute otitis media; pharyngitis/tonsillitis; community-acquired pneumonia; pelvic inflammatory disease; antimicrobial prophylaxis for premature rupture of membranes or unplanned Caesarean section; genital ulcer disease (in men) due to Haemophilus ducreyi (chancroid); acute bacterial exacerbations of chronic obstructive pulmonary disease; acute bacterial sinusitis; prevention of Mycobacterium avium complex (alone or in combination with rifabutin) in patients with advanced human immunodeficiency virus (HIV) infection; treatment of disseminated M. avium complex (in combination with ethambutol) in patients with advanced HIV infection; treatment of nontuberculous mycobacterial infections; skin and skin structure infections (uncomplicated) due to Staphylococcus aureus, Streptococcus pyogenes, or Streptococcus agalactiae; infections caused by Babesia microti (in combination with atovaquone); urethritis and

cervicitis due to *Chlamydia trachomatis* or *Neisseria gonorrhoeae*; trachoma (*C. trachomatis*); and yaws (*Treponema pallidum* subsp. *pertenue*)

#### Clarithromycin

- Usual adult dose
- Oral: 250 to 500 mg every 12 hours or 1000 mg (two 500-mg extended-release tablets) once daily for 7 to 14 days
- Usual pediatric dose
- Oral: 7.5 mg/kg/dose every 12 hours; maximum: 500 mg/dose
- Renal or hepatic failure
- Creatinine clearance <30 mL/min: decrease clarithromycin dose by 50%
- Hemodialysis: administer after hemodialysis session is completed; in hepatic failure, no dosing adjustment is necessary as long as renal function is normal
- CSF penetration
- Poor CNS penetration
- Adverse effects
- Cardiovascular (QTc) prolongation, torsades de pointes, hypersensitivity reactions
- Contraindications
- Caution for patients with underlying heart disease or for those taking other drugs that may prolong the QT interval regarding potential for inducing fatal ventricular arrhythmias
- Hypersensitivity to azithromycin, any macrolide antibiotics, or any component of the formulation
- History of cholestatic jaundice/hepatic dysfunction associated with prior azithromycin use
- Drug-drug interactions
- Colchicine toxicity, major inhibitor of CYP3A4
- Indications
- Pharyngitis/tonsillitis; acute maxillary sinusitis; acute exacerbation of chronic bronchitis; community-acquired pneumonia; uncomplicated skin/skin structure infections due to susceptible *S. aureus*, *S. pyogenes*; disseminated mycobacterial infections due to *M. avium* or *Mycobacterium intracellulare*; prevention of disseminated mycobacterial infections due to *M. avium* complex disease (e.g., patients with advanced HIV infection); duodenal ulcer disease due to *Helicobacter pylori* in regimens with other drugs, including amoxicillin and lansoprazole or omeprazole

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#### Ketolides

- Solithromycin (not FDA approved)

#### Clindamycin

- Usual adult doses
- Oral: 150 to 450 mg/dose every 6 to 8 hours; maximum dose: 1800 mg daily
- IM, IV: 1.2 to 2.7 g/day in 2 to 4 divided doses; maximum dose: 4800 mg daily
- Usual pediatric doses
- Oral: 20 to 30 mg/kg every day divided every 6 to 8 hours; maximum dose: 450 mg/dose
- IV: 20 to 40 mg/kg every day divided every 6 to 8 hours; maximum dose: 900 mg/dose
- Renal or hepatic failure
- Renal failure: no adjustment required
- CSF penetration
- Poor
- Adverse effects
- Diarrhea, abdominal pain, vaginitis, drug exanthem
- Contraindications
- Hypersensitivity to clindamycin, lincomycin, or any component of the formulation
- Drug-drug interactions
- Lincosamide antibiotics may diminish the therapeutic effects of erythromycin
- Indications
- Treatment of susceptible bacterial infections, mainly those caused by anaerobes, streptococci, pneumococci, and staphylococci; pelvic inflammatory disease (IV); bacterial vaginosis; hidradenitis suppurativa; toxin production mediation in necrotizing or invasive skin and soft tissue infection; component of regimens for malaria, toxoplasmosis, babesiosis (in combination with quinine)

Macrolides were the among the first major classes of microbial products to be discovered that possess antibiotic properties, after  $\beta$ -lactams, tetracyclines, and aminoglycosides.<sup>1</sup> The macrolide antibiotics (erythromycin, azithromycin, and clarithromycin) and the lincosamide

antibiotics (lincomycin and clindamycin) are chemically unrelated but possess many similar biologic properties in terms of mechanisms of action and resistance, antimicrobial activity, and clinical pharmacology. Erythromycin, the macrolide in longest use, is sometimes useful as an alternative to penicillin G and other antibiotics. Azithromycin and clarithromycin have advantages over erythromycin related to their antimicrobial activity, pharmacokinetics, fewer gastrointestinal side effects, and effectiveness in certain infections. Although more expensive, they have largely replaced erythromycin in clinical use. Azithromycin is noteworthy among the macrolides for fewer drug interactions and antiinflammatory activities. Dirithromycin, a semisynthetic derivative of erythromycin, is not available in the United States, is not widely used in Europe, but is prescribed in China; it is not included in this chapter but reviews of its use are available.<sup>2,3</sup> All of the macrolide antibiotics in clinical use have the potential for inducing cardiac arrhythmias.<sup>4</sup> Telithromycin, the first member of the ketolide class, exhibited increased activity against many bacterial strains that were resistant to macrolides. It was associated with rare, yet severe hepatic toxicity that substantially limited its clinical usefulness, and manufacturing has therefore been discontinued. Solithromycin is a ketolide antibiotic that was developed for the treatment of community-acquired pneumonia and other infections (including urogenital gonorrhea), particularly those caused by macrolide-resistant pathogens.<sup>5-7</sup> However, it is not yet approved for use in the United States or Europe. Nafithromycin (WCK 4873), a novel lactone ketolide class of antibiotic, which overcomes primary modes of macrolide resistance in *Streptococcus pneumoniae* and appears to be active against multidrug-resistant typical and atypical respiratory pathogens, is in phase three clinical studies for community-acquired pneumonia.<sup>8</sup> Clindamycin has been restricted in use by its association with *Clostridioides difficile* (formerly *Clostridium difficile*) colitis; it remains mostly as an alternative-choice antibiotic in the treatment of several infections, especially certain anaerobic infections. Lincomycin is now mainly of historic interest.

## Erythromycin

### Derivation, Chemistry, and Preparations

Erythromycin was derived in 1952 from a strain of *Saccharopolyspora erythraea* (originally named *Streptomyces erythraeus*) obtained from soil from the Philippines. It consists of a mixture of antibiotics (such as erythromycins A, B, C, and D), in which erythromycin A is the most active component.<sup>9</sup> The structure (Fig. 28.1) of erythromycin A consists of a 14-member macrocyclic lactone ring—therefore the class name macrolide—attached to two sugar moieties, desosamine and L-cladinose. Erythromycin base is poorly soluble in water, has a pKa of 8.8, is rapidly inactivated by gastric acid, and is often inconsistently absorbed after oral administration. Pharmaceutical preparations for oral use have been made with an aim to diminish destruction by gastric acid and to promote better absorption. Oral preparations available include (1) enteric-coated tablets, enteric-coated pellets in capsules for delayed release, and “film”-coated tablets of the base; (2) the stearate salt (formed in association with the amino group on desosamine), available as film-coated tablets; and (3) the ethylsuccinate ester (formed with the hydroxyl group on desosamine), available in tablet, chewable, and liquid forms. The water-soluble salt of erythromycin prepared for intravenous use is erythromycin lactobionate. Erythromycin lactobionate is not given intramuscularly because of injection site pain. Erythromycin base is also available in 0.5%, 2%, and 3% topical solutions, gels, and creams for treatment of acne vulgaris and in an ophthalmic ointment for treatment of bacterial conjunctivitis and prevention of neonatal gonococcal and chlamydial conjunctivitis.

### FIG. 28.1 Erythromycin base.

Follow for extended description

#### Mechanisms of Action

Erythromycin inhibits RNA-dependent protein synthesis at the step of chain elongation in susceptible organisms. X-ray crystallography studies indicate that several functional groups on erythromycin A bind to sequences on domain V of the 23S ribosomal RNA (rRNA) that is a component of the 50S subunit of the bacterial ribosome.<sup>10-12</sup> That binding site is near the peptidyl transferase center, and peptide chain elongation is thereby prevented by blocking of the polypeptide exit tunnel.<sup>10,11,13</sup> As a result, peptidyl transfer RNA (peptidyl-tRNA) is dissociated from the ribosome.<sup>10,11</sup> Data provided by macrolide-resistant mutants indicate that there are also important interactions between macrolides and specific ribosomal proteins or bases of the 50S subunit near the peptidyl transferase center.<sup>13,14</sup>

Studies in *Escherichia coli* and *Staphylococcus aureus* have demonstrated that erythromycin also inhibits the formation of the 50S ribosomal subunit.<sup>10,13,15</sup> In some bacteria, erythromycin interferes with the ribosomal binding of other macrolides, lincomycin, and chloramphenicol, suggesting common or overlapping binding sites for these antibiotics.

#### Mechanisms of Resistance

Macrolide resistance is commonly achieved through three major mechanisms: (1) efflux of the drug out of the bacterial cell, (2) target modification (either ribosomal RNA or protein) resulting in impaired ability to inhibit protein synthesis, and (3) enzymatic drug modification.<sup>16-18</sup> Drug resistance to macrolides is a major clinical problem among most pathogens for which these drugs are used, including *S. pneumoniae*, *Campylobacter jejuni*, *Helicobacter pylori*, *Mycoplasma pneumoniae*, *Neisseria gonorrhoeae*, *S. aureus*, and others.<sup>18</sup>

#### Decreased Microbial Entry or Export of Drug

Enterobacteriales exhibit decreased permeability of the outer cell envelope to macrolides.<sup>19-21</sup> This intrinsic resistance is also exhibited by *Pseudomonas* spp. and *Acinetobacter* spp. In addition, chromosomally encoded efflux pumps of several families can provide macrolide and, in some cases, lincosamide and streptogramin resistance.<sup>14,18</sup> A different efflux system for erythromycin and other 14- and 15-member macrolides (but not 16-member macrolides, lincosamides, or analogues of streptogramin B), called the M phenotype, has been elucidated in erythromycin-resistant strains of *Streptococcus pyogenes*, *S. pneumoniae*, group C streptococci, and enterococcal species. This efflux system is encoded by the *mef(A)* gene, is carried on a transposable element, and consists of transmembrane domains across the cytoplasmic membrane.<sup>10</sup> The M phenotype resistance is expressed by *S. pneumoniae* at moderate levels with erythromycin minimal inhibitory concentrations (MICs) of 1 to 64 µg/mL.<sup>22</sup>

#### Target Site Alterations

Mutations in genes for 50S ribosomal proteins or bases of critical domains of the 23S rRNA receptor site confer resistance to erythromycin and sometimes to other macrolides (M), lincosamides (L), and streptogramin type B (SB); in some but not all strains, this is associated with a decreased binding affinity for erythromycin in *S. pneumoniae*, *H. pylori*, *Mycobacterium avium*, *Bacillus subtilis*, *S. pyogenes*, *Campylobacter* spp., *M. pneumoniae*, *E. coli*, and *S. aureus*.<sup>10,13,14,23-27</sup> This pattern of resistance, referred to as the MLSB phenotype, is mediated by the *erm* (erythromycin ribosome methylation) genes on plasmids or transposons on chromosomes, which are self-transferable. There are many identified classes of *erm* genes, the most important of which are the *ermA*, *ermB*, *ermC*, and *ermF* classes. The resistance results from decreased binding of the antibiotics to their overlapping targets on the ribosome. It is the most widespread mechanism of resistance to the macrolides and the lincosamides and can be exhibited by strains of *S. aureus* (particularly having *ermA* or *ermC* genes), *S. pyogenes* (with *ermA* or *ermB* genes), *S. pneumoniae*, *Enterococcus* spp., *Corynebacterium diphtheriae*, *Campylobacter* spp., *Bacteroides fragilis* (*ermF* gene), *Clostridium perfringens*, *Listeria* spp., *M. pneumoniae*, and *Legionella* spp. The MLSB resistance phenotype may be constitutive or inducible by subinhibitory concentrations of erythromycin or other macrolides that bring about induction of the methylating enzyme. Lincosamides are not inducers of this system. When of the inducible type, macrolide resistance is manifest, but clindamycin susceptibility is often found in vitro; however, in that latter case, resistance will often emerge in vivo by selection of mutants that are constitutively producing a methylating enzyme, especially in infections with high bacterial density.<sup>25</sup> Strains of *S. pyogenes* with the *ermB* gene are constitutive producers of methylase and therefore are usually resistant to both the macrolides and clindamycin.<sup>28</sup> Strains of *S. pyogenes* with the *ermA* gene are usually inducible producers of methylase that consistently exhibit macrolide resistance and both macrolide and clindamycin resistance when the strain is exposed to those antibiotics.<sup>25,28</sup>

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The in vitro susceptibilities of potential pathogens to erythromycin are listed in Table 28.1.<sup>14,27,34–45</sup> Macrolide resistance is increasing in pneumococci<sup>46</sup> and group A streptococci isolated in the United States, and resistant clinical isolates have been increasingly encountered worldwide.<sup>34,47–50</sup> In the United States (2011–20), the prevalence of erythromycin resistance among *S. pneumoniae* strains was approximately 38%.<sup>46</sup> Rates of macrolide resistance were higher among isolates from noninvasive pneumococcal disease (44%) compared with invasive pneumococcal disease (27.4%).<sup>46</sup> *S. pneumoniae* strains demonstrate complete cross-resistance among the macrolides,<sup>51,52</sup> but cross-resistance extending from the macrolides to clindamycin is variable, depending on whether the resistance mechanism is of the MLSB or M phenotype.<sup>52,53</sup> A recent US surveillance study identified clindamycin resistance in 14.7% of 3487 *S. pneumoniae* isolates but erythromycin resistance in 35.9%.<sup>54</sup>

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In vitro Susceptibilities <sup>a</sup> to Erythromycin, Azithromycin, and Clarithromycin

| Organism                                       | Erythromycin      |                    | Azithromycin      |                   | Clarithromycin    |                     |
|------------------------------------------------|-------------------|--------------------|-------------------|-------------------|-------------------|---------------------|
|                                                | MIC <sub>50</sub> | MIC <sub>90</sub>  | MIC <sub>50</sub> | MIC <sub>90</sub> | MIC <sub>50</sub> | MIC <sub>90</sub>   |
| <i>Streptococcus pneumoniae</i>                |                   |                    |                   |                   |                   |                     |
| Penicillin-susceptible (MIC ≤ 0.06 µg/mL)      | 0.03              | 1.0                | 0.125             | 1.0               | 0.016             | 0.25                |
| Penicillin-intermediate (MIC = 0.12–1.0 µg/mL) | 0.03              | >64.0              | 0.125             | >64.0             | 0.03              | >64.0               |
| Penicillin-resistant (MIC ≥ 2.0 µg/mL)         | >8                | >64.0              | 1.0               | >64.0             | 0.5               | >64.0               |
| <i>Streptococcus pyogenes</i>                  | ≤0.06             | 0.06               | 0.12              | 0.25              | 0.03              | 0.06                |
| <i>Streptococcus agalactiae</i>                | 0.06              | 8                  | 0.06              | 0.12              | 0.03              | 0.06                |
| <i>Viridans streptococci</i>                   | 0.12              | >128               | 2.0               | 8.0               | 0.5               | 8                   |
| <i>Enterococcus</i> spp.                       |                   |                    |                   |                   |                   |                     |
| Vancomycin-susceptible                         | 1                 | 2                  | >8                | >8                | 0.5               | 1                   |
| Vancomycin-resistant                           | >128              | >128               | >8                | >8                | >128              | >128                |
| <i>Staphylococcus aureus</i>                   |                   |                    |                   |                   |                   |                     |
| Methicillin-susceptible                        | 0.25              | >128               | —                 | —                 | 0.06              | >128                |
| Methicillin-resistant                          | >128              | >128               | >128              | >128              | >128              | >128                |
| <i>Staphylococcus epidermidis</i>              | 32                | >128               | 16                | 128               | 16                | >128                |
| <i>Corynebacterium diphtheriae</i>             | 0.015             | 0.026              | 0.044             | 0.058             | 0.006             | 0.008               |
| <i>Listeria monocytogenes</i>                  | 0.125             | 0.25               | 1                 | 1                 | 0.06              | 0.125               |
| <i>Moraxella catarrhalis</i>                   | ≤0.25             | ≤0.25              | ≤0.06             | 0.06              | ≤0.25             | ≤0.25               |
| <i>Haemophilus influenzae</i>                  | 4                 | 8                  | 1                 | 2                 | 8                 | 16                  |
| <i>Bordetella pertussis</i>                    | 0.03              | 0.06               | 0.03              | 0.06              | 0.06              | 0.06                |
| <i>Neisseria gonorrhoeae</i>                   | 0.5               | 2                  | 0.12              | 0.25              | 0.25              | 1                   |
| <i>Neisseria meningitidis</i>                  | 1                 | 1                  | 0.5               | 1                 | 0.12              | 0.5                 |
| <i>Campylobacter jejuni</i>                    | 0.5–2             | 1–4                | 0.25              | 0.12–0.5          | 0.5–2             | 1–8                 |
| <i>Helicobacter pylori</i>                     | 0.12              | 0.25               | 0.25              | 0.5               | 0.008             | 0.015               |
| <i>Mycoplasma pneumoniae</i>                   | ≤0.015            | ≤0.015             | ≤0.015            | 0.015             | ≤0.015            | ≤0.015              |
| <i>Chlamydia trachomatis</i>                   | —                 | ≤0.25 <sup>b</sup> | —                 | 0.25 <sup>b</sup> | —                 | ≤0.015 <sup>b</sup> |
| <i>Chlamydia pneumoniae</i>                    | 0.125             | 0.25               | 0.125             | 0.25              | NA                | 0.03                |
| <i>Legionella pneumophila</i>                  | 0.125             | 0.5                | 0.12              | 0.5               | 0.032             | 0.046               |
| <i>Bacteroides fragilis</i>                    | 32                | >32                | >32               | >32               | 4                 | 8                   |
| <i>Peptococcus</i> , <i>Peptostreptococcus</i> | 2                 | 16                 | 1                 | >64               | 1                 | 4                   |
| <i>Clostridium perfringens</i>                 | 2                 | 2                  | 4                 | 4                 | 0.125             | 0.125               |
| <i>Propionibacterium</i> spp.                  | ≤0.06             | 0.5                | 0.125             | 2                 | ≤0.06             | ≤0.06               |
| <i>Mycobacterium avium</i> complex             | —                 | ≥64                | 8 <sup>c</sup>    | —                 | 2 <sup>c</sup>    | —                   |

<sup>a</sup> MIC<sub>50</sub> (MIC<sub>90</sub>), Minimal inhibitory concentration for 50% (90%) of isolates (µg/mL); values are ranges reported in referenced publications. <sup>14</sup> 27; <sup>34</sup> 43; <sup>54</sup> 70; <sup>76</sup> 306; <sup>489</sup> 491

<sup>b</sup> Reported as MIC<sub>100</sub>.

<sup>c</sup> Reported as median MIC.

Erythromycin displays activity against a wide range of gram-positive anaerobes, including *Actinomyces israelii*,<sup>73</sup> *Peptostreptococcus* spp., *Propionibacterium*, *Lactobacillus*, *Eubacterium*, *Bifidobacterium*, and many strains of *Peptococcus*.<sup>74</sup> *Clostridium tetani* and *C. perfringens* are generally susceptible<sup>75</sup>; however, many strains of *C. perfringens* may be only moderately susceptible<sup>76</sup> due to erythromycin target-site alterations.<sup>77</sup> A recent global surveillance study identified MLSB-mediated resistance genotypes in 19.2% of *C. difficile* strains tested.<sup>78</sup> Page 378 Other surveys have noted increasing resistance among *C. difficile* strains over time, with resistance rates ranging from 27% to 63%<sup>79–81</sup> primarily because of the increase in prevalence of certain ribotypes, such as polymerase chain reaction ribotype 001.<sup>80</sup> With gram-negative bacteria, erythromycin displays activity against *Bordetella pertussis*, for which reports of resistance are rare,<sup>82</sup> and *Moraxella catarrhalis*<sup>83</sup>; moderate activity against *Neisseria meningitidis*<sup>84</sup> and *N. gonorrhoeae*<sup>85,86</sup>; and relatively poor activity against *Haemophilus influenzae*.<sup>85</sup> Enterobacteriales are resistant to erythromycin.<sup>87</sup> However, erythromycin activity against gram-negative organisms is affected by pH: as the pH rises to 8.5, so does erythromycin's activity against organisms such as *E. coli* and *Klebsiella pneumoniae*.<sup>33</sup> In 2004, only 0.3% of *C. jejuni* isolates tested in the United States were resistant (MIC ≥32 µg/mL) to erythromycin.<sup>88</sup>



Breakpoint interpretive criteria for macrolide activity against anaerobic bacteria has not been established, but erythromycin generally has moderate activity against some species of gram-negative anaerobes, such as *Prevotella* and *Porphyromonas*. However, *B. fragilis* strains are usually resistant.<sup>89</sup> Erythromycin also demonstrates clinically useful activity against such diverse organisms as *Legionella pneumophila*,<sup>42</sup> *M. pneumoniae*,<sup>42</sup> *Ureaplasma urealyticum*,<sup>90</sup> *Chlamydia trachomatis*,<sup>40</sup> and *C. pneumoniae*.<sup>91</sup> Extracellular and intracellular *L. pneumophila* strains show substantial susceptibility to erythromycin.<sup>42,92</sup> Erythromycin is about 30 times more potent against *M. pneumoniae* than is levofloxacin<sup>42</sup> and 50 times more potent than tetracycline.<sup>93</sup> Macrolide-resistant variants of *M. pneumoniae* range in incidence between 2% and 26% in European countries,<sup>94,95</sup> 8.3% and 13.2% in the United States,<sup>96-98</sup> and up to 90% in some areas of Japan and China.<sup>99,100</sup> Macrolide resistance is a growing concern in *Mycoplasma genitalium* infections, with macrolide-resistance mutations in the 23S rRNA gene being identified in a majority of isolates worldwide.<sup>101</sup>

Nontuberculous mycobacteria are erythromycin susceptible; however, both clarithromycin and azithromycin are more active than erythromycin against mycobacteria.<sup>102</sup> *Mycobacterium tuberculosis* is resistant to erythromycin.<sup>103</sup>

### Clinical Pharmacology

The peak serum levels obtained after single doses of various erythromycin preparations are given in Table 28.2.<sup>104-106</sup> Erythromycin base is subject to destruction by gastric acid, and preparations of the base have been made with an acid-resistant coating to delay dissolution of the drug until it reaches the small bowel. The esters and ester salts of erythromycin used in the liquid suspension are more acid stable, form a stable suspension in water, and are tasteless. Erythromycin base (absorbed intact), stearate (absorbed as the base), and ethylsuccinate (absorbed both as the intact ester and as the free base after hydrolysis in the intestine) are usually

Serum Levels of Erythromycin in Adults

| Preparation    | DOSE (mg)        | ROUTE       | Peak Serum Levels |                                      |
|----------------|------------------|-------------|-------------------|--------------------------------------|
|                |                  |             | Hours After Dose  | Concentration (µg/mL)                |
| Base           | 250              | Oral        | 4                 | 0.3–1.0 <sup>a</sup>                 |
|                | 500              |             | 4                 | 0.3–1.9                              |
| Stearate       | 250 (fasting)    | Oral        | 3                 | 0.2–1.3                              |
|                | 500 (fasting)    |             | 3                 | 0.4–1.8                              |
|                | 500 (after food) |             | 3                 | 0.1–0.4 <sup>b</sup>                 |
| Ethylsuccinate | 500              | Oral        | 0.5–2.5           | 1.5 <sup>c</sup> (0.6 <sup>d</sup> ) |
| Lactobionate   | 500              | Intravenous | 1                 | 9.9                                  |

<sup>a</sup> Somewhat higher levels reported with some enteric-coated preparations after repeated doses. <sup>62</sup>

<sup>b</sup> One study demonstrated higher levels (to 2.8 µg/mL) with dose taken during a meal.

<sup>c</sup> Total drug (inactive ester and free base).

<sup>d</sup> Free base.

absorbed more completely in the fasting state, although one study demonstrated increased absorption of a stearate preparation when it was taken with a meal.<sup>107</sup> After absorption, about 45% of the ethylsuccinate preparation is present in the serum as the inactive ester and about 55% as the active base.<sup>Page 379</sup> Average serum levels achieved under fasting conditions with these preparations are similar; however, results with the base may be erratic. Erythromycin base has become available in a capsule containing enteric-coated granules; this preparation is promoted as giving more uniform absorption,<sup>107,108</sup> but some enteric-coated tablets may provide similar blood levels.<sup>109</sup> The clinical significance of the much less active esterified form of the drug that is present in serum in appreciable concentration is controversial. It would seem that in treatment of infections of only moderate severity by organisms highly susceptible to erythromycin, differences in therapeutic results using the various oral preparations would be insignificant. Limited clinical comparisons confirm that suspicion.<sup>110</sup> Intravenous preparations of erythromycin achieve appreciably higher serum levels and should be used to treat serious infections requiring erythromycin.

Erythromycin is distributed through total body water.<sup>111</sup> Values given for protein binding vary from 40% to 90%. The drug persists in tissues longer than in the blood. The ratios of tissue or body fluid concentrations to simultaneous serum concentrations (usually at peak) are as follows: aqueous humor, 0.3; ascites, 0.4; bile, 28; middle ear exudate in otitis media, 0.3 to 0.7; pleural fluid, 0.7; prostatic

### Short View Summary

#### Macrolides

##### Erythromycin

- Usual adult doses
- Oral: base—250 to 500 mg every 6 to 12 hours; maximum: 4 g daily
- Ethylsuccinate: 400 to 800 mg every 6 to 12 hours; maximum: 4 g daily
- IV: lactobionate—15 to 20 mg/kg/day divided every 6 hours, or 500 mg to 1 g every 6 hours, or given as a continuous infusion over 24 hours; maximum: 4 g daily
- Usual pediatric doses
- Ethylsuccinate: 10 to 12.5 mg/kg/dose every 6 hours; maximum: 2 g daily
- Renal or hepatic failure
- In renal failure, slightly dialyzable (5%–20%); supplemental dose is not necessary in hemodialysis or peritoneal dialysis or in continuous arteriovenous or venovenous hemofiltration
- No dose adjustment in hepatic failure
- Cerebrospinal fluid (CSF) penetration
- CSF-to-blood level ratio: normal meninges—2% to 13%; inflamed meninges—7% to 25%
- Adverse effects
- Cardiovascular corrected QT (QTc) interval prolongation, torsades de pointes, gastrointestinal (GI) upset, pruritus, erythema multiforme, drug-induced pemphigus, cholestatic jaundice, postnatal pyloric stenosis
- Contraindications
- Caution for patients with underlying heart disease or for those taking other drugs that may prolong the QT interval regarding potential for inducing fatal ventricular arrhythmias
- Hypersensitivity to erythromycin, any macrolide antibiotics, or any component of the formulation
- Concomitant use with cisapride, ergotamine, terfenadine, lovastatin, or simvastatin
- Drug-drug interactions
- Major inhibitor of cytochrome P-450 isoenzyme 3A (CYP3A4)
- Indications
- *Bartonella* spp. infections, chancroid, granuloma inguinale, legionnaires' disease, lymphogranuloma venereum, nongonococcal urethritis, pertussis, gastric and intestinal motility disorders

##### Azithromycin

- Usual adult dose
- 500 mg on day 1, followed by 250 mg once daily on days 2 to 5
- Usual pediatric dose
- 10 to 12 mg/kg/dose on day 1 (maximum: 500 mg/dose), followed by 5 to 6 mg/kg/dose (maximum: 250 mg/dose) on days 2 through 5
- Renal or hepatic failure
- No dosage adjustment required; use with caution in patients with severe renal impairment (glomerular filtration rate <10 mL/min)
- CSF penetration
- Poor central nervous system (CNS) penetration
- Adverse effects
- Cardiovascular (QTc) prolongation, torsades de pointes, GI upset, pruritus, erythema multiforme, Stevens-Johnson syndrome, fixed drug eruptions, vaginitis, cholestatic jaundice, drug-induced liver injury
- Contraindications
- Caution for patients with underlying heart disease or for those taking other drugs that may prolong the QT interval regarding potential for inducing fatal ventricular arrhythmias
- Hypersensitivity to azithromycin, any macrolide antibiotics, or any component of the formulation
- History of cholestatic jaundice/hepatic dysfunction associated with prior azithromycin use
- Drug-drug interactions
- Amiodarone and other QTc-prolonging agents, quinine

- Indications
- Treatment of acute otitis media; pharyngitis/tonsillitis; community-acquired pneumonia; pelvic inflammatory disease; antimicrobial prophylaxis for premature rupture of membranes or unplanned Caesarean section; genital ulcer disease (in men) due to *Haemophilus ducreyi* (chancroid); acute bacterial exacerbations of chronic obstructive pulmonary disease; acute bacterial sinusitis; prevention of *Mycobacterium avium* complex (alone or in combination with rifabutin) in patients with advanced human immunodeficiency virus (HIV) infection; treatment of disseminated *M. avium* complex (in combination with ethambutol) in patients with advanced HIV infection; treatment of nontuberculous mycobacterial infections; skin and skin structure infections (uncomplicated) due to *Staphylococcus aureus*, *Streptococcus pyogenes*, or *Streptococcus agalactiae*; infections caused by *Babesia microti* (in combination with atovaquone); urethritis and cervicitis due to *Chlamydia trachomatis* or *Neisseria gonorrhoeae*; trachoma (*C. trachomatis*); and yaws (*Treponema pallidum* subsp. *pertenue*)

#### Clarithromycin

- Usual adult dose
- Oral: 250 to 500 mg every 12 hours or 1000 mg (two 500-mg extended-release tablets) once daily for 7 to 14 days
- Usual pediatric dose
- Oral: 7.5 mg/kg/dose every 12 hours; maximum: 500 mg/dose
- Renal or hepatic failure
- Creatinine clearance <30 mL/min: decrease clarithromycin dose by 50%
- Hemodialysis: administer after hemodialysis session is completed; in hepatic failure, no dosing adjustment is necessary as long as renal function is normal
- CSF penetration
- Poor CNS penetration
- Adverse effects
- Cardiovascular (QTc) prolongation, torsades de pointes, hypersensitivity reactions
- Contraindications
- Caution for patients with underlying heart disease or for those taking other drugs that may prolong the QT interval regarding potential for inducing fatal ventricular arrhythmias
- Hypersensitivity to azithromycin, any macrolide antibiotics, or any component of the formulation
- History of cholestatic jaundice/hepatic dysfunction associated with prior azithromycin use
- Drug-drug interactions
- Colchicine toxicity, major inhibitor of CYP3A4
- Indications
- Pharyngitis/tonsillitis; acute maxillary sinusitis; acute exacerbation of chronic bronchitis; community-acquired pneumonia; uncomplicated skin/skin structure infections due to susceptible *S. aureus*, *S. pyogenes*; disseminated mycobacterial infections due to *M. avium* or *Mycobacterium intracellulare*; prevention of disseminated mycobacterial infections due to *M. avium* complex disease (e.g., patients with advanced HIV infection); duodenal ulcer disease due to *Helicobacter pylori* in regimens with other drugs, including amoxicillin and lansoprazole or omeprazole

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#### Ketolides

- Solithromycin (not FDA approved)

#### Clindamycin

- Usual adult doses
- Oral: 150 to 450 mg/dose every 6 to 8 hours; maximum dose: 1800 mg daily
- IM, IV: 1.2 to 2.7 g/day in 2 to 4 divided doses; maximum dose: 4800 mg daily
- Usual pediatric doses
- Oral: 20 to 30 mg/kg every day divided every 6 to 8 hours; maximum dose: 450 mg/dose
- IV: 20 to 40 mg/kg every day divided every 6 to 8 hours; maximum dose: 900 mg/dose
- Renal or hepatic failure
- Renal failure: no adjustment required
- CSF penetration
- Poor
- Adverse effects
- Diarrhea, abdominal pain, vaginitis, drug exanthem
- Contraindications
- Hypersensitivity to clindamycin, lincomycin, or any component of the formulation

- Drug-drug interactions
- Lincosamide antibiotics may diminish the therapeutic effects of erythromycin
- Indications
- Treatment of susceptible bacterial infections, mainly those caused by anaerobes, streptococci, pneumococci, and staphylococci; pelvic inflammatory disease (IV); bacterial vaginosis; hidradenitis suppurativa; toxin production mediation in necrotizing or invasive skin and soft tissue infection; component of regimens for malaria, toxoplasmosis, babesiosis (in combination with quinine)

Macrolides were the among the first major classes of microbial products to be discovered that possess antibiotic properties, after  $\beta$ -lactams, tetracyclines, and aminoglycosides.<sup>1</sup> The macrolide antibiotics (erythromycin, azithromycin, and clarithromycin) and the lincosamide antibiotics (lincomycin and clindamycin) are chemically unrelated but possess many similar biologic properties in terms of mechanisms of action and resistance, antimicrobial activity, and clinical pharmacology. Erythromycin, the macrolide in longest use, is sometimes useful as an alternative to penicillin G and other antibiotics. Azithromycin and clarithromycin have advantages over erythromycin related to their antimicrobial activity, pharmacokinetics, fewer gastrointestinal side effects, and effectiveness in certain infections. Although more expensive, they have largely replaced erythromycin in clinical use. Azithromycin is noteworthy among the macrolides for fewer drug interactions and antiinflammatory activities. Dirithromycin, a semisynthetic derivative of erythromycin, is not available in the United States, is not widely used in Europe, but is prescribed in China; it is not included in this chapter but reviews of its use are available.<sup>2,3</sup> All of the macrolide antibiotics in clinical use have the potential for inducing cardiac arrhythmias.<sup>4</sup> Telithromycin, the first member of the ketolide class, exhibited increased activity against many bacterial strains that were resistant to macrolides. It was associated with rare, yet severe hepatic toxicity that substantially limited its clinical usefulness, and manufacturing has therefore been discontinued. Solithromycin is a ketolide antibiotic that was developed for the treatment of community-acquired pneumonia and other infections (including urogenital gonorrhea), particularly those caused by macrolide-resistant pathogens.<sup>5-7</sup> However, it is not yet approved for use in the United States or Europe. Nafithromycin (WCK 4873), a novel lactone ketolide class of antibiotic, which overcomes primary modes of macrolide resistance in *Streptococcus pneumoniae* and appears to be active against multidrug-resistant typical and atypical respiratory pathogens, is in phase three clinical studies for community-acquired pneumonia.<sup>8</sup> Clindamycin has been restricted in use by its association with *Clostridioides difficile* (formerly *Clostridium difficile*) colitis; it remains mostly as an alternative-choice antibiotic in the treatment of several infections, especially certain anaerobic infections. Lincomycin is now mainly of historic interest.

## Erythromycin

### Derivation, Chemistry, and Preparations

Erythromycin was derived in 1952 from a strain of *Saccharopolyspora erythraea* (originally named *Streptomyces erythraeus*) obtained from soil from the Philippines. It consists of a mixture of antibiotics (such as erythromycins A, B, C, and D), in which erythromycin A is the most active component.<sup>9</sup> The structure (Fig. 28.1) of erythromycin A consists of a 14-member macrocyclic lactone ring—therefore the class name macrolide—attached to two sugar moieties, desosamine and L-cladinose. Erythromycin base is poorly soluble in water, has a pKa of 8.8, is rapidly inactivated by gastric acid, and is often inconsistently absorbed after oral administration. Pharmaceutical preparations for oral use have been made with an aim to diminish destruction by gastric acid and to promote better absorption. Oral preparations available include (1) enteric-coated tablets, enteric-coated pellets in capsules for delayed release, and “film”-coated tablets of the base; (2) the stearate salt (formed in association with the amino group on desosamine), available as film-coated tablets; and (3) the ethylsuccinate ester (formed with the hydroxyl group on desosamine), available in tablet, chewable, and liquid forms. The water-soluble salt of erythromycin prepared for intravenous use is erythromycin lactobionate. Erythromycin lactobionate is not given intramuscularly because of injection site pain. Erythromycin base is also available in 0.5%, 2%, and 3% topical solutions, gels, and creams for treatment of acne vulgaris and in an ophthalmic ointment for treatment of bacterial conjunctivitis and prevention of neonatal gonococcal and chlamydial conjunctivitis.

FIG. 28.1 Erythromycin base.

Follow for extended description

#### Mechanisms of Action

Erythromycin inhibits RNA-dependent protein synthesis at the step of chain elongation in susceptible organisms. X-ray crystallography studies indicate that several functional groups on erythromycin A bind to sequences on domain V of the 23S ribosomal RNA (rRNA) that is a component of the 50S subunit of the bacterial ribosome.<sup>10-12</sup> That binding site is near the peptidyl transferase center, and peptide chain elongation is thereby prevented by blocking of the polypeptide exit tunnel.<sup>10,11,13</sup> As a result, peptidyl transfer RNA (peptidyl-tRNA) is dissociated from the ribosome.<sup>10,11</sup> Data provided by macrolide-resistant mutants indicate that there are also important interactions between macrolides and specific ribosomal proteins or bases of the 50S subunit near the peptidyl transferase center.<sup>13,14</sup>

Studies in *Escherichia coli* and *Staphylococcus aureus* have demonstrated that erythromycin also inhibits the formation of the 50S ribosomal subunit.<sup>10,13,15</sup> In some bacteria, erythromycin interferes with the ribosomal binding of other macrolides, lincomycin, and chloramphenicol, suggesting common or overlapping binding sites for these antibiotics.

#### Mechanisms of Resistance

Macrolide resistance is commonly achieved through three major mechanisms: (1) efflux of the drug out of the bacterial cell, (2) target modification (either ribosomal RNA or protein) resulting in impaired ability to inhibit protein synthesis, and (3) enzymatic drug modification.<sup>16-18</sup> Drug resistance to macrolides is a major clinical problem among most pathogens for which these drugs are used, including *S. pneumoniae*, *Campylobacter jejuni*, *Helicobacter pylori*, *Mycoplasma pneumoniae*, *Neisseria gonorrhoeae*, *S. aureus*, and others.<sup>18</sup>

#### Decreased Microbial Entry or Export of Drug

Enterobacteriales exhibit decreased permeability of the outer cell envelope to macrolides.<sup>19-21</sup> This intrinsic resistance is also exhibited by *Pseudomonas* spp. and *Acinetobacter* spp. In addition, chromosomally encoded efflux pumps of several families can provide macrolide and, in some cases, lincosamide and streptogramin resistance.<sup>14,18</sup> A different efflux system for erythromycin and other 14- and 15-member macrolides (but not 16-member macrolides, lincosamides, or analogues of streptogramin B), called the M phenotype, has been elucidated in erythromycin-resistant strains of *Streptococcus pyogenes*, *S. pneumoniae*, group C streptococci, and enterococcal species. This efflux system is encoded by the *mef(A)* gene, is carried on a transposable element, and consists of transmembrane domains across the cytoplasmic membrane.<sup>10</sup> The M phenotype resistance is expressed by *S. pneumoniae* at moderate levels with erythromycin minimal inhibitory concentrations (MICs) of 1 to 64 µg/mL.<sup>22</sup>

#### Target Site Alterations

Mutations in genes for 50S ribosomal proteins or bases of critical domains of the 23S rRNA receptor site confer resistance to erythromycin and sometimes to other macrolides (M), lincosamides (L), and streptogramin type B (SB); in some but not all strains, this is associated with a decreased binding affinity for erythromycin in *S. pneumoniae*, *H. pylori*, *Mycobacterium avium*, *Bacillus subtilis*, *S. pyogenes*, *Campylobacter* spp., *M. pneumoniae*, *E. coli*, and *S. aureus*.<sup>10,13,14,23-27</sup> This pattern of resistance, referred to as the MLSB phenotype, is mediated by the *erm* (erythromycin ribosome methylation) genes on plasmids or transposons on chromosomes, which are self-transferable. There are many identified classes of *erm* genes, the most important of which are the *ermA*, *ermB*, *ermC*, and *ermF* classes. The resistance results from decreased binding of the antibiotics to their overlapping targets on the ribosome. It is the most widespread mechanism of resistance to the macrolides and the lincosamides and can be exhibited by strains of *S. aureus* (particularly having *ermA* or *ermC* genes), *S. pyogenes* (with *ermA* or *ermB* genes), *S. pneumoniae*, *Enterococcus* spp., *Corynebacterium diphtheriae*, *Campylobacter* spp., *Bacteroides fragilis* (*ermF* gene), *Clostridium perfringens*, *Listeria* spp., *M. pneumoniae*, and *Legionella* spp. The MLSB resistance phenotype may be constitutive or inducible by subinhibitory concentrations of erythromycin or other macrolides that bring about induction of the methylating enzyme. Lincosamides are not inducers of this system. When of the inducible type, macrolide resistance is manifest, but clindamycin susceptibility is often found in vitro; however, in that latter case, resistance will often emerge in vivo by selection of mutants that are constitutively producing a methylating enzyme, especially in infections with high bacterial density.<sup>25</sup> Strains of *S. pyogenes* with the *ermB* gene are constitutive producers of methylase and

therefore are usually resistant to both the macrolides and clindamycin.<sup>28</sup> Strains of *S. pyogenes* with the *ermA* gene are usually inducible producers of methylase that consistently exhibit macrolide resistance and both macrolide and clindamycin resistance when the strain is exposed to those antibiotics.<sup>25,28</sup>

#### Drug Inactivation

Enzymatic inactivation of erythromycin and some other 14-, 15-, and 16-member ring macrolides can render them unable to bind to the 50S ribosome.<sup>18</sup> There are two major classes of bacterial enzymes that accomplish this drug inactivation: macrolide phosphotransferases and macrolide esterases.<sup>18</sup> Phosphotransferases have been described in strains of *S. aureus*, *E. coli*, *Pseudomonas*, *Pasteurella*, *Klebsiella*, *Serratia*, *Shigella*, and *Nocardia* spp. and are encoded by genes designated *mph(A)*, *mph(B)*, and *mph(C)*.<sup>14,18</sup> Esterase genes (*ere[A]* and *ere[B]*) on plasmids encode for the hydrolysis of the macrocyclic lactone of erythromycin and have been found in strains of *E. coli*, *Klebsiella* spp., *Citrobacter* spp., *Proteus* spp., *Pseudomonas* spp., *Enterobacter* spp., *Salmonella* spp., *Vibrio cholerae*, and rare strains of *S. aureus*.<sup>14,18</sup>

Nucleic acid amplification methods that allow for relatively rapid detection of the different mechanisms of macrolide resistance among clinical isolates (genotypic testing) have been developed.<sup>29,30</sup> A given bacterial strain may possess more than one type of macrolide resistance mechanism, resulting in complex resistant phenotypes.

#### Antimicrobial Activity

The antimicrobial activity of erythromycin is broad in spectrum, being exhibited against gram-positive and gram-negative bacteria, including actinomycetes and mycobacteria, as well as against treponemes, mycoplasmas, Chlamydia, and rickettsiae. Depending on drug concentration, bacterial species, phase of growth, and density of the inoculum, erythromycin is essentially bacteriostatic. Bacterial killing is favored by higher antibiotic concentrations, higher pH, lower bacterial density, and rapid growth.<sup>31</sup> The activity of erythromycin, which is a weak base, increases markedly with increasing pH over the range of 5.5 to 8.5 for both gram-positive and gram-negative bacteria,<sup>32,33</sup> possibly reflecting increased entry into the bacterial cell of the un-ionized drug that is more plentiful at the higher pH.

The in vitro susceptibilities of potential pathogens to erythromycin are listed in Table 28.1.<sup>14,27,34–45</sup> Macrolide resistance is increasing in pneumococci<sup>46</sup> and group A streptococci isolated in the United States, and resistant clinical isolates have been increasingly encountered worldwide.<sup>34,47–50</sup> In the United States (2011–20), the prevalence of erythromycin resistance among *S. pneumoniae* strains was approximately 38%.<sup>46</sup> Rates of macrolide resistance were higher among isolates from noninvasive pneumococcal disease (44%) compared with invasive pneumococcal disease (27.4%).<sup>46</sup> *S. pneumoniae* strains demonstrate complete cross-resistance among the macrolides,<sup>51,52</sup> but cross-resistance extending from the macrolides to clindamycin is variable, depending on whether the resistance mechanism is of the MLSB or M phenotype.<sup>52,53</sup> A recent US surveillance study identified clindamycin resistance in 14.7% of 3487 *S. pneumoniae* isolates but erythromycin resistance in 35.9%.<sup>54</sup>

Worldwide, erythromycin resistance in *S. pyogenes* varies from as low as 2% in Europe to as high as 90% in China,<sup>55,56</sup> with rates of macrolide nonsusceptibility of 14.5% to 23% reported in the United States.<sup>57,58</sup> As with *S. pneumoniae*, there is cross-resistance among the macrolides in *S. pyogenes*; however, resistance does not extend to clindamycin if resistance is due to the M phenotype.<sup>59,60</sup> Clindamycin resistance among *S. pyogenes* is increasing in the United States, from an estimated 0.5% in 2003<sup>28</sup> to 15% in children.<sup>61</sup> Isolates from invasive infections are more commonly clindamycin resistant, approaching 1 in 4 by 2017.<sup>57</sup>

Resistance to erythromycin by *S. aureus* may be selected by its use in hospitals.<sup>62</sup> Most methicillin-resistant strains and many methicillin-susceptible clinical isolates are now resistant to this agent.<sup>63,64</sup> Analyses of staphylococcal isolates from pediatric and adult patients revealed that across all ages, strains that were methicillin susceptible were on average 35% resistant to erythromycin, whereas strains that were methicillin resistant were 88% resistant.<sup>65</sup> In addition, there is a potential for the emergence, during treatment in an individual patient, of erythromycin resistance by *S. aureus*.<sup>66,67</sup> These strains may demonstrate the emergence of high-level

resistance to erythromycin alone, or they may show cross-resistance to other macrolides and to lincomycin and clindamycin<sup>25</sup> due to the inducible type of MLSB resistance, described earlier.

The viridans group of streptococci has traditionally been considered generally susceptible to erythromycin. However, macrolide resistance may be increasing in some areas, with 57.8% susceptible in North America, 62.7% in Asia and the Pacific, and 73.7% in Europe.<sup>68,69</sup>

Other susceptible gram-positive organisms include the majority of strains of *Listeria monocytogenes* and *C. diphtheriae*.<sup>70,71</sup> Appreciable in vitro activity has been demonstrated against *Nocardia asteroides* when combined with ampicillin.<sup>72</sup>

Erythromycin displays activity against a wide range of gram-positive anaerobes, including *Actinomyces israelii*,<sup>73</sup> *Peptostreptococcus* spp., *Propionibacterium*, *Lactobacillus*, *Eubacterium*, *Bifidobacterium*, and many strains of *Peptococcus*.<sup>74</sup> *Clostridium tetani* and *C. perfringens* are generally susceptible<sup>75</sup>; however, many strains of *C. perfringens* may be only moderately susceptible<sup>76</sup> due to erythromycin target-site alterations.<sup>77</sup> A recent global surveillance study identified MLSB-mediated resistance genotypes in 19.2% of *C. difficile* strains tested.<sup>78</sup> Page 378 Other surveys have noted increasing resistance among *C. difficile* strains over time, with resistance rates ranging from 27% to 63%<sup>79-81</sup> primarily because of the increase in prevalence of certain ribotypes, such as polymerase chain reaction ribotype 001.<sup>80</sup> With gram-negative bacteria, erythromycin displays activity against *Bordetella pertussis*, for which reports of resistance are rare,<sup>82</sup> and *Moraxella catarrhalis*<sup>83</sup>; moderate activity against *Neisseria meningitidis*<sup>84</sup> and *N. gonorrhoeae*<sup>85,86</sup>; and relatively poor activity against *Haemophilus influenzae*.<sup>85</sup> Enterobacteriales are resistant to erythromycin.<sup>87</sup> However, erythromycin activity against gram-negative organisms is affected by pH: as the pH rises to 8.5, so does erythromycin's activity against organisms such as *E. coli* and *Klebsiella pneumoniae*.<sup>33</sup> In 2004, only 0.3% of *C. jejuni* isolates tested in the United States were resistant (MIC  $\geq 32$   $\mu\text{g/mL}$ ) to erythromycin.<sup>88</sup>

TABLE 28.1

In vitro Susceptibilities<sup>a</sup> to Erythromycin, Azithromycin, and Clarithromycin

| Organism                                                   | Erythromycin      |                   | Azithromycin      |                   | Clarithromycin    |             |
|------------------------------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------|
| MIC <sub>50</sub>                                          | MIC <sub>90</sub> | MIC <sub>50</sub> | MIC <sub>90</sub> | MIC <sub>50</sub> | MIC <sub>90</sub> |             |
| <i>Streptococcus pneumoniae</i>                            |                   |                   |                   |                   |                   |             |
| Penicillin-susceptible (MIC $\leq 0.06\mu\text{g/mL}$ )    |                   |                   |                   |                   |                   |             |
| 0.03                                                       | 1.0               | 0.125             | 1.0               | 0.016             | 0.25              |             |
| Penicillin-intermediate (MIC = 0.12–1.0 $\mu\text{g/mL}$ ) |                   |                   |                   |                   |                   |             |
| 0.03                                                       | >64.0             | 0.125             | >64.0             | 0.03              | >64.0             |             |
| Penicillin-resistant (MIC $\geq 2.0\mu\text{g/mL}$ )       |                   |                   |                   |                   |                   |             |
| >8                                                         | >64.0             | 1.0               | >64.0             | 0.5               | >64.0             |             |
| <i>Streptococcus pyogenes</i>                              |                   | $\leq 0.06$       | 0.06              | 0.12              | 0.25              | 0.03 0.06   |
| <i>Streptococcus agalactiae</i>                            |                   | 0.06              | 8                 | 0.06              | 0.12              | 0.03 0.06   |
| Viridans streptococci                                      | 0.12              | >128              | 2.0               | 8.0               | 0.5               | 8           |
| <i>Enterococcus</i> spp.                                   |                   |                   |                   |                   |                   |             |
| Vancomycin-susceptible                                     |                   |                   |                   |                   |                   |             |
| 1                                                          | 2                 | >8                | >8                | 0.5               | 1                 |             |
| Vancomycin-resistant                                       |                   |                   |                   |                   |                   |             |
| >128                                                       | >128              | >8                | >8                | >128              | >128              |             |
| <i>Staphylococcus aureus</i>                               |                   |                   |                   |                   |                   |             |
| Methicillin-susceptible                                    |                   |                   |                   |                   |                   |             |
| 0.25                                                       | >128              | —                 | —                 | 0.06              | >128              |             |
| Methicillin-resistant                                      |                   |                   |                   |                   |                   |             |
| >128                                                       | >128              | >128              | >128              | >128              | >128              |             |
| <i>Staphylococcus epidermidis</i>                          | 32                | >128              | 16                | 128               | 16                | >128        |
| <i>Corynebacterium diphtheriae</i>                         | 0.015             | 0.026             | 0.044             | 0.058             | 0.006             | 0.008       |
| <i>Listeria monocytogenes</i>                              | 0.125             | 0.25              | 1                 | 1                 | 0.06              | 0.125       |
| <i>Moraxella catarrhalis</i>                               | $\leq 0.25$       | $\leq 0.25$       | $\leq 0.06$       | 0.06              | $\leq 0.25$       | $\leq 0.25$ |
| <i>Haemophilus influenzae</i>                              | 4                 | 8                 | 1                 | 2                 | 8                 | 16          |
| <i>Bordetella pertussis</i>                                | 0.03              | 0.06              | 0.03              | 0.06              | 0.06              | 0.06        |
| <i>Neisseria gonorrhoeae</i>                               | 0.5               | 2                 | 0.12              | 0.25              | 0.25              | 1           |

|                                                |        |                    |                |                   |                |                     |  |
|------------------------------------------------|--------|--------------------|----------------|-------------------|----------------|---------------------|--|
| <i>Neisseria meningitidis</i>                  | 1      | 0.5                | 1              | 0.12              | 0.5            |                     |  |
| <i>Campylobacter jejuni</i>                    | 0.5–2  | 1–4                | 0.25           | 0.12–0.5          | 0.5–2          | 1–8                 |  |
| <i>Helicobacter pylori</i>                     | 0.12   | 0.25               | 0.25           | 0.5               | 0.008          | 0.015               |  |
| <i>Mycoplasma pneumoniae</i>                   | ≤0.015 | ≤0.015             | ≤0.015         | 0.015             | ≤0.015         | ≤0.015              |  |
| <i>Chlamydia trachomatis</i>                   | —      | ≤0.25 <sup>b</sup> | —              | 0.25 <sup>b</sup> | —              | ≤0.015 <sup>b</sup> |  |
| <i>Chlamydia pneumoniae</i>                    | 0.125  | 0.25               | 0.125          | 0.25              | NA             | 0.03                |  |
| <i>Legionella pneumophila</i>                  | 0.125  | 0.5                | 0.12           | 0.5               | 0.032          | 0.046               |  |
| <i>Bacteroides fragilis</i>                    | 32     | >32                | >32            | >32               | 4              | 8                   |  |
| <i>Peptococcus</i> , <i>Peptostreptococcus</i> | 2      | 16                 | 1              | >64               | 1              | 4                   |  |
| <i>Clostridium perfringens</i>                 | 2      | 2                  | 4              | 4                 | 0.125          | 0.125               |  |
| <i>Propionibacterium</i> spp.                  | ≤0.06  | 0.5                | 0.125          | 2                 | ≤0.06          | ≤0.06               |  |
| <i>Mycobacterium avium</i> complex             | —      | ≥64                | 8 <sup>c</sup> | —                 | 2 <sup>c</sup> | —                   |  |

a MIC<sub>50</sub> (MIC<sub>90</sub>), Minimal inhibitory concentration for 50% (90%) of isolates (μg/mL); values are ranges reported in referenced publications.14,27,34-43,54,70,76,306,489-491

b Reported as MIC<sub>100</sub>.

c Reported as median MIC.

Breakpoint interpretive criteria for macrolide activity against anaerobic bacteria has not been established, but erythromycin generally has moderate activity against some species of gram-negative anaerobes, such as *Prevotella* and *Porphyromonas*. However, *B. fragilis* strains are usually resistant.<sup>89</sup> Erythromycin also demonstrates clinically useful activity against such diverse organisms as *Legionella pneumophila*,<sup>42</sup> *M. pneumoniae*,<sup>42</sup> *Ureaplasma urealyticum*,<sup>90</sup> *Chlamydia trachomatis*,<sup>40</sup> and *C. pneumoniae*.<sup>91</sup> Extracellular and intracellular *L. pneumophila* strains show substantial susceptibility to erythromycin.<sup>42,92</sup> Erythromycin is about 30 times more potent against *M. pneumoniae* than is levofloxacin<sup>42</sup> and 50 times more potent than tetracycline.<sup>93</sup> Macrolide-resistant variants of *M. pneumoniae* range in incidence between 2% and 26% in European countries,<sup>94,95</sup> 8.3% and 13.2% in the United States,<sup>96-98</sup> and up to 90% in some areas of Japan and China.<sup>99,100</sup> Macrolide resistance is a growing concern in *Mycoplasma genitalium* infections, with macrolide-resistance mutations in the 23S rRNA gene being identified in a majority of isolates worldwide.<sup>101</sup>

Nontuberculous mycobacteria are erythromycin susceptible; however, both clarithromycin and azithromycin are more active than erythromycin against mycobacteria.<sup>102</sup> *Mycobacterium tuberculosis* is resistant to erythromycin.<sup>103</sup>

#### Clinical Pharmacology

The peak serum levels obtained after single doses of various erythromycin preparations are given in Table 28.2.<sup>104-106</sup> Erythromycin base is subject to destruction by gastric acid, and preparations of the base have been made with an acid-resistant coating to delay dissolution of the drug until it reaches the small bowel. The esters and ester salts of erythromycin used in the liquid suspension are more acid stable, form a stable suspension in water, and are tasteless. Erythromycin base (absorbed intact), stearate (absorbed as the base), and ethylsuccinate (absorbed both as the intact ester and as the free base after hydrolysis in the intestine) are usually absorbed more completely in the fasting state, although one study demonstrated increased absorption of a stearate preparation when it was taken with a meal.<sup>107</sup> After absorption, about 45% of the ethylsuccinate preparation is present in the serum as the inactive ester and about 55% as the active base. Page 379

TABLE 28.2

| Serum Levels of Erythromycin in Adults |               |       |                  |                       | Peak Serum Levels |  |
|----------------------------------------|---------------|-------|------------------|-----------------------|-------------------|--|
| Preparation                            | DOSE (mg)     | ROUTE | Hours After Dose | Concentration (μg/mL) |                   |  |
| Base                                   | 250           | Oral  | 4                | 0.3–1.0 <sup>a</sup>  |                   |  |
| 500                                    | 4             |       |                  | 0.3–1.9               |                   |  |
| Stearate                               | 250 (fasting) | Oral  | 3                | 0.2–1.3               |                   |  |
| 500 (fasting)                          | 3             |       |                  | 0.4–1.8               |                   |  |
| 500 (after food)                       | 3             |       |                  | 0.1–0.4 <sup>b</sup>  |                   |  |



|                |     |             |         |                                      |
|----------------|-----|-------------|---------|--------------------------------------|
| Ethylsuccinate | 500 | Oral        | 0.5–2.5 | 1.5 <sup>c</sup> (0.6 <sup>d</sup> ) |
| Lactobionate   | 500 | Intravenous | 1       | 9.9                                  |

a Somewhat higher levels reported with some enteric-coated preparations after repeated doses.<sup>62</sup>

b One study demonstrated higher levels (to 2.8 µg/mL) with dose taken during a meal.

c Total drug (inactive ester and free base).

d Free base.

Average serum levels achieved under fasting conditions with these preparations are similar; however, results with the base may be erratic. Erythromycin base has become available in a capsule containing enteric-coated granules; this preparation is promoted as giving more uniform absorption,<sup>107,108</sup> but some enteric-coated tablets may provide similar blood levels.<sup>109</sup> The clinical significance of the much less active esterified form of the drug that is present in serum in appreciable concentration is controversial. It would seem that in treatment of infections of only moderate severity by organisms highly susceptible to erythromycin, differences in therapeutic results using the various oral preparations would be insignificant. Limited clinical comparisons confirm that suspicion.<sup>110</sup> Intravenous preparations of erythromycin achieve appreciably higher serum levels and should be used to treat serious infections requiring erythromycin.

Erythromycin is distributed through total body water.<sup>111</sup> Values given for protein binding vary from 40% to 90%. The drug persists in tissues longer than in the blood. The ratios of tissue or body fluid concentrations to simultaneous serum concentrations (usually at peak) are as follows: aqueous humor, 0.3; ascites, 0.4; bile, 28; middle ear exudate in otitis media, 0.3 to 0.7; pleural fluid, 0.7; prostatic fluid, 0.4; cerebrospinal fluid (CSF) without meningitis, 0 to 0.02, and with meningitis, 0.05 to 0.1; infected maxillary paranasal sinus, 0.4 to 0.8; and tonsil, 0.3.

Concentrations achieved in the middle ear in otitis media are adequate to treat pneumococcal and group A streptococcal infections involving susceptible strains of these species but are not adequate to consistently eradicate *H. influenzae*.<sup>112,113</sup> High concentrations of erythromycin are achieved in alveolar macrophages<sup>114</sup> and polymorphonuclear leukocytes<sup>115</sup> compared with those in extracellular fluid.

Erythromycin does not enter the CSF in the setting of normal (uninflamed) meninges.<sup>116</sup> However, concentrations of erythromycin achieved in the CSF of patients with meningitis suggest that large parenteral doses may be effective against highly susceptible organisms such as *S. pneumoniae*.<sup>117</sup> Limited data from patients with septic arthritis suggest poor penetration of synovial fluid. Erythromycin is transferred across the placenta; fetal serum concentrations are about 2% of those in maternal serum, but higher concentrations accumulate in fetal tissue and amniotic fluid.<sup>118</sup> The drug is excreted in breast milk with a ratio of 0.41 between maternal milk and plasma levels, but it is considered safe if the infant is of an age to receive erythromycin directly.<sup>119</sup>

Up to 4.5% of an oral dose and 15% of a parenteral dose of erythromycin are recoverable in the urine. Urine concentrations after oral doses are often high but quite variable. Erythromycin is concentrated by the liver and excreted into the bile in high concentrations; however, only about 1.5% of the dose of the base and 0.2% of the ester can be recovered from bile in the first 8 hours, and some of this is reabsorbed from the intestine.<sup>120</sup> After an oral dose, large concentrations of the antibiotic are found in feces, probably representing ingested drug that was never absorbed, as well as some that was excreted in bile. A large proportion of absorbed drug cannot be accounted for by urinary or biliary excretion or by tissue binding and may be inactivated in the liver by demethylation.<sup>121</sup>

The normal serum half-life of erythromycin is 1.4 hours, and appreciable serum levels are maintained for 6 hours. In anuric patients, the half-life is prolonged to about 5 hours, but dosage reduction in patients with renal failure is generally not necessary.<sup>122</sup> Erythromycin is not removed by peritoneal dialysis or hemodialysis.

## Adverse Reactions

Although frequently used specifically for its motility effects, gastrointestinal adverse effects (30%) are the most common adverse effects caused by erythromycin, likely reflecting local effects of the drugs on gut motility (via motilin receptors) and/or vagus nerve actions.<sup>12,123</sup> Symptoms including abdominal pain (16%), nausea and vomiting (14%), and diarrhea occur more commonly in children and young adults than in older persons and may be associated with either intravenous or oral administration.<sup>124</sup> Pseudomembranous colitis caused by overgrowth of toxin-producing *C. difficile* is uncommon but can occur with the use of erythromycin.<sup>125,126</sup> Superinfection, especially of the gastrointestinal tract or vagina, with *Candida* spp. or gram-negative bacilli may occur, as with other antibiotics. Thrombophlebitis with intravenous use can be decreased by appropriate dilution of the dose in at least 200 mL of solution and by avoidance of rapid infusions. Infusion should take place over 45 to 60 minutes.

Allergic reactions including skin rash, fever, and eosinophilia are rare. Severe reactions such as Stevens-Johnson syndrome and pemphigus foliaceus have been reported.<sup>127,128</sup> Cholestatic hepatitis occurs rarely,<sup>129</sup> and chiefly in adults.<sup>130</sup> The syndrome typically begins after 20 days of therapy, but more rapidly in those previously treated, and consists of nausea, vomiting, and abdominal pain followed by jaundice, fever, and abnormal liver function tests consistent with cholestatic hepatitis. These findings are sometimes accompanied by rash, leukocytosis, and eosinophilia. The abnormalities usually clear within days to a few weeks after the drug is stopped but may return rapidly on rechallenge. However, hepatocyte toxicity induced by the drug or its metabolites, as well as allergy to altered hepatocyte components, may be contributory.<sup>131</sup> Reversible hepatotoxicity, including jaundice, has occurred with the stearate salt and with the ethylsuccinate ester of erythromycin.<sup>132</sup>

Ototoxicity has been reported rarely in association with the use of large intravenous doses of erythromycin lactobionate or large doses of oral erythromycin.<sup>133,134</sup> This may occur more commonly in older adults, in patients with hepatic or renal insufficiency, with receipt of higher doses, and with concurrent use of ototoxic medications.<sup>135,136</sup> Symptoms are usually reversible; however, irreversible tinnitus and hearing loss have been reported.<sup>127,137</sup>Page 380

Mitochondrial toxicities have also been reported with the use of macrolides. When simulating concentrations of locally applied antibiotics in a cell culture model of primary human osteoblasts, erythromycin-inhibited mitochondrial energetics, namely, proliferation and metabolic activity, of the osteoblasts, but to a lesser degree than azithromycin.<sup>138</sup> Mitochondria isolated from rat cardiomyocytes have also been affected by exposure to macrolide antibiotics. The resulting induced cascade of reactive oxygen species formation, mitochondrial membrane permeability, mitochondrial swell, and cytochrome c release within the cardiomyocytes with exposure to clarithromycin, azithromycin, and erythromycin has been proposed as a starting point for cardiotoxicities, including arrhythmias, QT prolongation, and torsades de pointes.<sup>139</sup>

Polymorphic ventricular tachycardia (torsades de pointes) with QT prolongation has been reported in association with treatment with intravenous and oral erythromycin.<sup>140-142</sup> The drug has been shown to affect repolarization in the isolated heart and to block electrical current in guinea pig ventricular myocytes in a way consistent with the observed arrhythmia.<sup>143</sup> The possibility for interaction with potential-lengthening drugs (classes IA and III antiarrhythmics) and for increased risk in the presence of electrolyte abnormality or prolonged QT interval should be kept in mind. Other major contributing factors to macrolide-associated torsades de pointes include the coadministration of cytochrome P-450 isoenzyme 3A4 (CYP3A4) inhibitors (resulting in increased drug exposure). One study reported that coadministration of CYP3A4 inhibitors and erythromycin led to a fivefold greater risk of cardiac sudden death.<sup>142</sup> (For further discussion of the potential for macrolide-induced arrhythmias and sudden death, see “Adverse Effects” under “Azithromycin and Clarithromycin.”)

Infantile hypertrophic pyloric stenosis has been epidemiologically linked to early exposure to erythromycin in the neonatal period. One meta-analysis showed that macrolide exposure in the first 120 days of life was associated with an odds ratio of 2.005 (95% CI, 1.125 to 3.575; *P* = .018) for development of hypertrophic pyloric stenosis compared with exposure to nonmacrolide antibiotics.<sup>144</sup> The same study also compared exposure to erythromycin versus nonerythromycin macrolides in the first 120 days of life and found an odds ratio of 2.384 (95% CI, 1.055 to 5.386; *P* = .037) for the development of hypertrophic pyloric stenosis with erythromycin.<sup>144</sup> It has been

hypothesized that erythromycin interacts with motilin receptors, inducing strong gastric and pyloric contractions leading to pyloric hypertrophy.<sup>145,146</sup> There is no substantive evidence of a risk associated with prenatal exposure or exposure from breast milk when erythromycin is taken by the lactating parent.<sup>144,147</sup> Erythromycin is classified as pregnancy category B.

#### Drug Interactions

Incompatibility during administration between intravenous preparations of erythromycin and other drugs has been reported; the latter include vitamin B complex and vitamin C, cephalothin, tetracycline, chloramphenicol, colistin, and heparin.

Erythromycin may produce interactions with other drugs by interfering with their hepatic metabolism through the CYP3A subclass of the cytochrome P-450 enzyme system.<sup>148,149</sup> The resulting increased drug levels may result in serious toxicity (Table 28.3). For example, elevations of midazolam serum concentrations have led to unconsciousness. The reverse of that process, in which drugs that elevate levels of erythromycin may promote its proarrhythmic effects, has been discussed earlier (see “Adverse Reactions” under “Erythromycin”).

Erythromycin can increase the bioavailability of digoxin, possibly by interfering with its inactivation by gut flora.<sup>148</sup> Sequential use of erythromycin and clindamycin should be avoided when possible because of the potential for the development of cross-resistance or dissociated resistance.

#### Uses of Erythromycin

Erythromycin has few indications for use as the drug of choice because of the availability of other macrolides with improved pharmacokinetic profiles, in addition to fewer side effects and drug interactions (Table 28.4).<sup>14,86,150-161</sup> Modified from Kim RB and the Editors of the Medical Letter. The Medical Letter Handbook of Adverse Drug Interactions. The Medical Letter on Drugs and Therapeutics; 2003.

Although erythromycin continues to be useful in the treatment of community-acquired respiratory infections, its utility has become limited to an alternative to penicillin G for group A  $\beta$ -hemolytic streptococci in the setting of penicillin hypersensitivity. Use is further limited in areas in which the incidence of penicillin-resistant pneumococci is high, as resistance to erythromycin is also common.<sup>162,163</sup> Erythromycin is also effective for the treatment of infections due to *M. catarrhalis*. Erythromycin is not consistently effective in treatment of infections caused by *H. influenzae*.<sup>112,113</sup> Treatment of *M. pneumoniae* infection with erythromycin, as with tetracycline, shortens the clinical course of the infection, even if started late in the course of illness; radiographic clearing of pulmonary lesions occurs earlier with erythromycin.<sup>164,165</sup> Clinical experience and studies in vitro and in guinea pigs suggest that erythromycin is effective in treating pneumonia caused by *L. pneumophila* or *Legionella micdadei*<sup>166</sup>; however, its use has been superseded by the newer macrolides, azithromycin and clarithromycin.<sup>167</sup> The US Food and Drug Administration has approved azithromycin and levofloxacin for the treatment of legionellosis, and they are now considered preferable to erythromycin, especially when given orally.<sup>155,168</sup>

Early treatment of pertussis with erythromycin is associated with clinical improvement, a rapid clearance of *B. pertussis* from the nasopharynx, and a reduction in secondary transmission in households. Erythromycin is also recommended for postexposure prophylaxis of pertussis.<sup>169,170</sup> Treatment of infants with erythromycin for pneumonia or conjunctivitis caused by *C. trachomatis* is approximately 80% effective, although a second course of antimicrobial therapy may be required.<sup>171</sup> Because of reports of an association between oral erythromycin and infantile hypertrophic pyloric stenosis, monitoring for its signs and symptoms should be implemented in treated infants who are younger than 6 weeks of age.<sup>147</sup>

TABLE 28.3

Potentially Clinically Significant Drug Interactions Produced by Macrolides \*

| DRUG           | INTERACTIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Erythromycin   | Alfentanil, alprazolam, amiodarone, bromocriptine, buspirone, carbamazepine, clobazam, cisapride, clomipramine plus risperidone, clozapine, colchicine, cyclosporine, diazepam, digoxin, disopyramide, dofetilide, ergot alkaloids, felodipine, levofloxacin, lidocaine, loratadine, lovastatin, methylprednisolone, midazolam, moxifloxacin, phenytoin, pimozide, quinidine, repaglinide, rifabutin, rifampin, ropivacaine, saquinavir, sertraline, sildenafil, simvastatin, sirolimus, sotalol, tacrolimus, terfenadine, theophylline, triazolam, valproate, verapamil, warfarin |
| Clarithromycin | Alprazolam, amiodarone, amprenavir, apixaban, atazanavir, atorvastatin, carbamazepine, cisapride, colchicine, cyclosporine, dabigatran, darunavir, diazepam, digoxin, disopyramide, disulfiram, dofetilide, efavirenz, ergot alkaloids, fluoxetine, itraconazole, lidocaine, loratadine, lovastatin, methylprednisolone, midazolam, phenytoin, pimozide, quinidine, repaglinide, rifabutin, rifampin, ritonavir, rivaroxaban, saquinavir, sildenafil, simvastatin, sirolimus, tacrolimus, terfenadine, theophylline, verapamil, warfarin, zidovudine                               |
| Azithromycin   | Cyclosporine (one case), digoxin, phenytoin, rifampin, tacrolimus, warfarin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

\* Interactions generally lead to increased levels of the listed drugs caused by interaction of the macrolide with cytochrome P-450 hepatic metabolism. Exceptions are digoxin, in which the raised levels are caused by interference with digoxin inactivation by gut flora, and zidovudine, in which serum concentrations may be decreased by unknown mechanisms. Rifampin and rifabutin may lower the levels of clarithromycin, and clarithromycin may raise the levels of rifampin and rifabutin. Ritonavir may raise clarithromycin levels. Clarithromycin may produce modest reduction of zidovudine levels.

TABLE 28.4

Major Indications for Use of Macrolides

| INFECTIONS IN WHICH MACROLIDES ARE THE DRUGS OF CHOICE                                             |                                                                                         |                                                                                                      |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Infection                                                                                          | Macrolide                                                                               | Adult Dosages <sup>a</sup>                                                                           |
| <i>Bartonella henselae</i> (cat-scratch disease bacillus)                                          | Azithromycin                                                                            | 500 mg PO on day 1, then 250 mg PO on days 2–5                                                       |
|                                                                                                    | Clarithromycin <sup>b</sup>                                                             | 500 mg PO bid for 7–10 d                                                                             |
| <i>Bartonella henselae</i> or <i>B. quintana</i> (bacillary angiomatosis, trench fever)            | Erythromycin                                                                            | 500 mg qid PO for 12 weeks                                                                           |
|                                                                                                    | Azithromycin                                                                            | 500 mg qd PO for 4–6 weeks                                                                           |
| <i>Bordetella pertussis</i>                                                                        | Erythromycin                                                                            | 250 mg qid PO for 14 d                                                                               |
|                                                                                                    | Azithromycin                                                                            | 500 mg PO on day 1, then 250 mg PO on days 2–5                                                       |
|                                                                                                    | Clarithromycin <sup>b</sup>                                                             | 500 mg PO bid for 7 d                                                                                |
| <i>Campylobacter jejuni</i> <sup>c</sup>                                                           | Azithromycin                                                                            | 500 mg daily PO for 3–7 d                                                                            |
|                                                                                                    | Erythromycin                                                                            | 500 mg PO bid for 5–7 d                                                                              |
| <i>Chlamydia pneumoniae</i> (TWAR strain)                                                          | Azithromycin                                                                            | 500 mg qd PO/IV for 1–2 d, then 500 mg PO qd to complete 7–10 d <sup>d</sup>                         |
|                                                                                                    | Clarithromycin <sup>b</sup>                                                             | 500 mg bid PO for 7–10 d                                                                             |
|                                                                                                    | Erythromycin                                                                            | 250–500 mg qid PO for 7–10 d                                                                         |
| <i>Chlamydia trachomatis</i> (inclusion conjunctivitis) <sup>e</sup>                               | Erythromycin                                                                            | Erythromycin base or ethylsuccinate 50 mg/kg/d PO divided into 4 doses daily for 14 d                |
|                                                                                                    | Azithromycin                                                                            | Single dose of 20 mg/kg up to a maximum dose of 1 g                                                  |
| <i>Chlamydia trachomatis</i> (pneumonia) <sup>e</sup>                                              | Erythromycin                                                                            | Erythromycin base or ethylsuccinate 50 mg/kg/d PO divided into 4 doses daily for 14 d                |
| <i>Chlamydia trachomatis</i> (trachoma)                                                            | Azithromycin                                                                            | 1 g PO, single dose                                                                                  |
| Prevention of endometritis and chorioamnionitis in patients with PPRPOM <34 weeks' gestational age | Erythromycin (+ ampicillin followed by amoxicillin)                                     | 250 mg every 6 hours for 48 h                                                                        |
|                                                                                                    | Azithromycin (+ ampicillin followed by amoxicillin)                                     | 1 g PO on admission                                                                                  |
| Diphtheria <sup>f</sup>                                                                            | Infection: erythromycin                                                                 | Erythromycin base or ethylsuccinate 40 mg/kg IV or PO in divided doses daily for 14 d, maximum 2 g/d |
|                                                                                                    | Carrier: erythromycin                                                                   | 250–500 mg qid PO for 7–10 d                                                                         |
| <i>Haemophilus ducreyi</i> (chancroid)                                                             | Azithromycin                                                                            | 1 g, single dose                                                                                     |
| <i>Helicobacter pylori</i>                                                                         | Clarithromycin <sup>b</sup> (+amoxicillin, or metronidazole + proton pump inhibitor)    | 500 mg bid PO for 10–14 d                                                                            |
| <i>Legionella</i> spp. pneumonia                                                                   | Azithromycin ± rifampin (or a fluoroquinolone ± rifampin)                               | 1 g IV or PO on day 1, then 500 mg qd for 7–10 d total duration                                      |
| <i>Mycobacterium abscessus</i>                                                                     | Azithromycin (+ 1–4 additional active agents guided by susceptibilities) <sup>g</sup>   | 250–500 mg PO qd OR 500 mg PO qd three times weekly                                                  |
|                                                                                                    | Clarithromycin (+ 1–4 additional active agents guided by susceptibilities) <sup>g</sup> | 500 mg PO bid qd OR 1000 mg PO in 2 divided doses three times weekly                                 |
| <i>Mycobacterium avium</i> complex disseminated disease                                            | Clarithromycin <sup>b</sup> (+ ethambutol ± rifabutin)                                  | 500 mg PO bid for variable periods <sup>h</sup>                                                      |
|                                                                                                    | Azithromycin (+ ethambutol ± rifabutin)                                                 | 500–600 mg qd PO for variable periods <sup>h</sup>                                                   |
| <i>Mycobacterium avium</i> complex prophylaxis                                                     | Azithromycin                                                                            | 1200 mg once weekly PO until initiation of effective antiretroviral therapy                          |
|                                                                                                    | Clarithromycin <sup>b</sup>                                                             | 500 mg bid PO until initiation of effective antiretroviral therapy                                   |
| <i>Mycobacterium avium</i> complex pulmonary infiltrative disease                                  | Clarithromycin <sup>b</sup> (+ ethambutol ± rifabutin)                                  | 500 mg bid PO for 1 year after sputum cultures are negative                                          |
|                                                                                                    | Azithromycin (+ ethambutol ± rifabutin)                                                 | 500–600 mg qd PO for 1 year after sputum cultures are negative                                       |

|                                                 |                                          |                          |
|-------------------------------------------------|------------------------------------------|--------------------------|
| <i>Mycobacterium fortuitum/chelonae</i> complex | Clarithromycin <sup>b</sup> (+ amikacin) | 500 mg bid PO for 4–6 mo |
| <i>Mycoplasma pneumoniae</i>                    | Azithromycin                             | 500 mg PO qd for 5–10 d  |
|                                                 | Clarithromycin <sup>b</sup>              | 250 mg PO bid for 14 d   |
|                                                 | Erythromycin                             | 500 mg qd PO for 14–21 d |

| INFECTIONS IN WHICH MACROLIDES ARE AN IMPORTANT ALTERNATIVE DRUG |                                                             |                                          |
|------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------|
| Infection                                                        | Macrolides and Adult Dosages <sup>a</sup>                   | Drug of Choice                           |
| Groups A, C, G streptococcal infection                           | Erythromycin 250–500 mg qid PO                              | Penicillin G or V                        |
|                                                                  | Azithromycin 500 mg PO on day 1, then 250 mg PO on days 2–5 |                                          |
|                                                                  | Clarithromycin <sup>b</sup> 250 mg bid PO                   |                                          |
| <i>Streptococcus pneumoniae</i> infection                        | Erythromycin 250–500 mg qid PO <sup>i</sup>                 | Penicillin G, ceftriaxone, or cefotaxime |
|                                                                  | Azithromycin 500 mg qd <sup>i</sup>                         |                                          |
|                                                                  | Clarithromycin <sup>b</sup> 250–500 mg bid <sup>i</sup>     |                                          |
| <i>Moraxella catarrhalis</i>                                     | Azithromycin 500 mg PO qd for 3 d                           | Cefuroxime; a fluoroquinolone            |
|                                                                  | Azithromycin 500 mg PO on day 1, then 250 mg PO on days 2–5 |                                          |
|                                                                  | Erythromycin 250–500 mg qid PO                              |                                          |
|                                                                  | Clarithromycin <sup>b</sup> 250–500 mg bid PO               |                                          |
| Table Continued                                                  |                                                             |                                          |

| INFECTIONS IN WHICH MACROLIDES ARE AN IMPORTANT ALTERNATIVE DRUG                           |                                                                                                                                                                   |                                                             |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Infection                                                                                  | Macrolides and Adult Dosages <sup>a</sup>                                                                                                                         | Drug of Choice                                              |
| <i>Haemophilus influenzae</i> (upper respiratory infection and bronchitis)                 | Azithromycin 500 mg PO qd for 3 d                                                                                                                                 | Trimethoprim-sulfamethoxazole                               |
|                                                                                            | Azithromycin 500 mg PO on day 1, then 250 mg PO on days 2–5                                                                                                       |                                                             |
|                                                                                            | Clarithromycin <sup>b</sup> 500 mg bid PO                                                                                                                         |                                                             |
| <i>Shigella</i>                                                                            | Azithromycin 500 mg PO daily for 3 days                                                                                                                           | Fluoroquinolone                                             |
| Prevention of infection after colorectal surgery                                           | 1 g PO each of neomycin and erythromycin base at 1, 2, and 11 PM on the day before 8 AM surgery (combined with vigorous purgation over second day before surgery) | Cefazolin plus metronidazole<br>Cefoxitin or cefotetan      |
| Rheumatic fever prophylaxis                                                                | Erythromycin 250 mg bid PO                                                                                                                                        | Penicillin G                                                |
| Anthrax                                                                                    | Erythromycin 500 mg qid PO for 10 d <sup>j</sup>                                                                                                                  | Ciprofloxacin, doxycycline                                  |
| <i>Lymphogranuloma venereum</i>                                                            | Erythromycin 500 mg qid PO for 21 d                                                                                                                               | A tetracycline                                              |
| <i>Acne vulgaris</i>                                                                       | Erythromycin 250 mg qid PO or topical preparation                                                                                                                 | A tetracycline PO and a number of topical drugs             |
| <i>Borrelia burgdorferi</i> (Lyme disease)                                                 | Azithromycin 500 mg PO for 7–10 d <sup>k</sup>                                                                                                                    | Doxycycline, amoxicillin, cefuroxime axetil PO <sup>j</sup> |
| <i>Babesia microti</i>                                                                     | Azithromycin 500 mg on day 1 and 250 mg on days 2–7 + atovaquone 750 mg q12h                                                                                      | Clindamycin + quinine                                       |
| <i>Chlamydia trachomatis</i> (urethritis or cervicitis)                                    | Azithromycin 1000mg PO single dose                                                                                                                                | Doxycycline                                                 |
| Nongonococcal urethritis in men ( <i>C. trachomatis</i> or <i>Ureaplasma urealyticum</i> ) | Azithromycin 1000 mg PO single dose                                                                                                                               | Doxycycline                                                 |

| INFECTIONS IN WHICH MACROLIDES ARE AN IMPORTANT ALTERNATIVE DRUG                           |                                                                                                                                                                   |                                                                |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Infection                                                                                  | Macrolides and Adult Dosages <sup>a</sup>                                                                                                                         | Drug of Choice                                                 |
| <i>Haemophilus influenzae</i> (upper respiratory infection and bronchitis)                 | Azithromycin 500 mg PO qd for 3 d                                                                                                                                 | Trimethoprim-sulfamethoxazole                                  |
|                                                                                            | Azithromycin 500 mg PO on day 1, then 250 mg PO on days 2–5                                                                                                       |                                                                |
|                                                                                            | Clarithromycin <sup>b</sup> 500 mg bid PO                                                                                                                         |                                                                |
| <i>Shigella</i>                                                                            | Azithromycin 500 mg PO daily for 3 days                                                                                                                           | Fluoroquinolone                                                |
| Prevention of infection after colorectal surgery                                           | 1 g PO each of neomycin and erythromycin base at 1, 2, and 11 PM on the day before 8 AM surgery (combined with vigorous purgation over second day before surgery) | Cefazolin plus metronidazole<br>Cefoxitin or cefotetan         |
| Rheumatic fever prophylaxis                                                                | Erythromycin 250 mg bid PO                                                                                                                                        | Penicillin G                                                   |
| Anthrax                                                                                    | Erythromycin 500 mg qid PO for 10 d <sup>l</sup>                                                                                                                  | Ciprofloxacin, doxycycline                                     |
| <i>Lymphogranuloma venereum</i>                                                            | Erythromycin 500 mg qid PO for 21 d                                                                                                                               | A tetracycline                                                 |
| <i>Acne vulgaris</i>                                                                       | Erythromycin 250 mg qid PO or topical preparation                                                                                                                 | A tetracycline PO and a number of topical drugs                |
| <i>Borrelia burgdorferi</i> (Lyme disease)                                                 | Azithromycin 500 mg PO for 7–10 d <sup>k</sup>                                                                                                                    | Doxycycline, amoxicillin, cefuroxime axetil<br>PO <sup>l</sup> |
| <i>Babesia microti</i>                                                                     | Azithromycin 500 mg on day 1 and 250 mg on days 2–7 + atovaquone 750 mg q12h                                                                                      | Clindamycin + quinine                                          |
| <i>Chlamydia trachomatis</i> (urethritis or cervicitis)                                    | Azithromycin 1000mg PO single dose                                                                                                                                | Doxycycline                                                    |
| Nongonococcal urethritis in men ( <i>C. trachomatis</i> or <i>Ureaplasma urealyticum</i> ) | Azithromycin 1000 mg PO single dose                                                                                                                               | Doxycycline                                                    |

<sup>a</sup> Intravenous therapy should be used in serious illness or when oral therapy is not possible or reliable.

<sup>b</sup> Not recommended for use in pregnancy.

<sup>c</sup> In some areas, such as in Thailand, macrolide- and fluoroquinolone-resistant strains have become common.

<sup>d</sup> Mild-to-moderate severity: azithromycin 500 mg PO on day 1, then 250 mg PO on days 2–5.

Erythromycin treatment of patients with gastroenteritis caused by *C. jejuni* hastens the eradication of the organism from the feces but does not appear to alter the clinical course of uncomplicated infection when therapy begins 4 days or more after the onset of symptoms.<sup>172</sup> However, earlier treatment of young children with acute dysentery Page 381 associated with *C. jejuni* has been shown to shorten the course of diarrhea and fecal excretion of susceptible organisms.<sup>173,174</sup>

Erythromycin base given orally together with neomycin on the day before colorectal surgery and combined with vigorous purgation is about as effective as parenteral cephalosporin administration just before surgery in decreasing the incidence of septic complications.<sup>175</sup> No advantage has been demonstrated for the use of a combination of oral and intravenous antibiotics.<sup>176</sup> In the presence of bowel obstruction or when there is need for emergency surgery, the parenteral antibiotic regimen should be used. Erythromycin is also currently recommended along with ampicillin for antimicrobial prophylaxis for prevention of endometritis for patients with preterm prelabor rupture of membranes (PPROM) before 34 weeks' gestation, but recent research suggests that azithromycin may be noninferior to erythromycin and result in fewer adverse effects.<sup>177–179</sup>

Erythromycin given orally for up to 3 months remains the drug of choice in treating certain visceral or angioproliferative *Bartonella* infections (bacillary angiomatosis and bacillary peliosis hepatis in immunocompromised patients), but it might be difficult to tolerate.<sup>180,181</sup> Relapses have been described, especially of lesions in bone and skin and when antibiotics are given for a shorter duration (<3 months), particularly in severely immunosuppressed patients.<sup>182</sup>

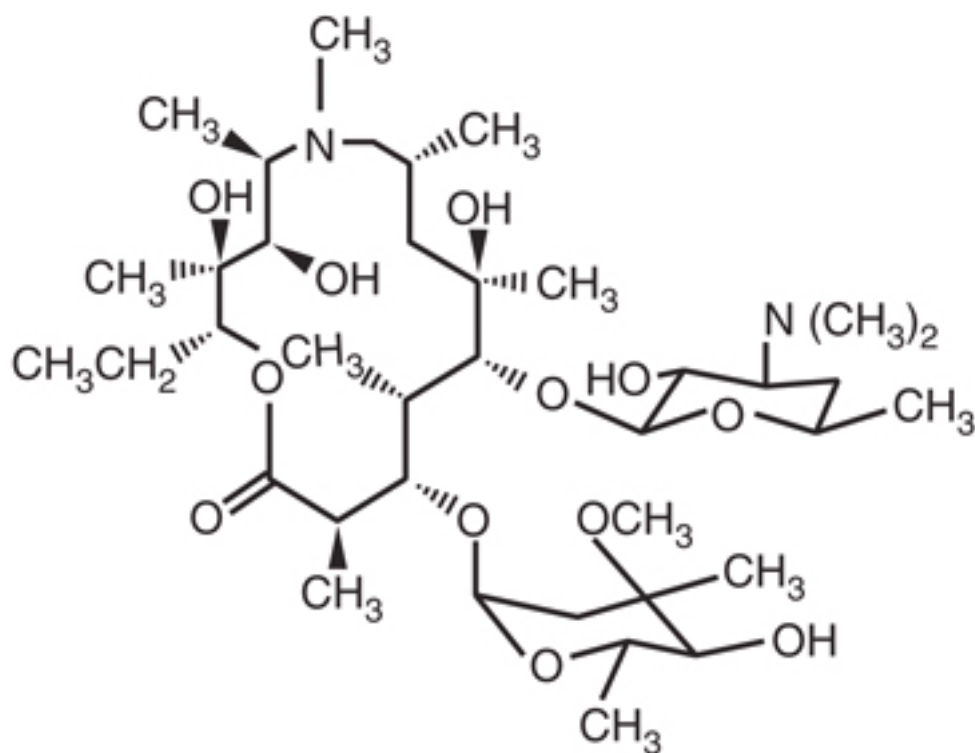
A comparative study involving a small number of children with cholera, who were all treated with rehydration solutions, showed that erythromycin or trimethoprim-sulfamethoxazole was effective and superior to treatment without an antimicrobial agent.<sup>183</sup> Erythromycin may be used as an alternative antibiotic in the treatment of anthrax. In view of the availability of more effective alternative drugs, erythromycin should not be used alone in the treatment of deep-seated staphylococcal infections because of the potential for the emergence of resistant strains during therapy.<sup>67,184</sup>

Erythromycin and other 14-member ring macrolides have a gastrointestinal motility–stimulating effect. In this regard, erythromycin acts as a motilin receptor agonist in the gut and gallbladder.<sup>185</sup> Erythromycin lactobionate 3 mg/kg IV every 8 hours has been effective for the treatment of gastroparesis in hospitalized diabetic patients.<sup>186</sup> When given orally, erythromycin may improve gastric emptying for several weeks but is often associated with tachyphylaxis due to downregulation of the motilin receptor when given for longer than 4 weeks.<sup>187</sup> These prokinetic effects have also been studied for the treatment of postvagotomy gastroparesis,<sup>188</sup> gastroparesis in critically ill patients receiving mechanical ventilation,<sup>189</sup> and intestinal dysmotility in young infants, and to aid in the placement of nasoenteric postpyloric feeding tubes.<sup>190,191</sup>

There has been interest in the antiinflammatory activities of low-dose erythromycin (600 mg/day) and other macrolides (reviewed by Kricker et al.<sup>192</sup>), first suggested because of the beneficial effect of erythromycin treatment of patients with diffuse panbronchiolitis.<sup>193,194</sup> Those activities include interference with oxidant production by neutrophils (in which the cladinoside moiety of



erythromycin was found to be the key structure<sup>195</sup>), acceleration of neutrophil apoptosis, suppression of the release of proinflammatory cytokines, and promotion of the release of nitric oxide from endothelial cells.<sup>195,196</sup> At the molecular level, macrolides appear to modulate inflammation in some cells, such as human bronchial epithelial cells, by inhibiting transcription factors,<sup>197,198</sup> which regulate the expression of interleukin-8, among others, a chemokine that acts as a major recruiter of neutrophils in chronic airway disease.<sup>199</sup> The antiinflammatory protective effects of erythromycin appear to be a slow process as demonstrated by the requirement of at least a 28-day pretreatment with erythromycin to suppress an inflammatory response in zymosan-induced peritonitis in rats.<sup>200</sup> All of this said, it is challenging to attribute effects of macrolides in vivo to antiinflammatory actions on host cells as opposed to their known antimicrobial actions, which can reduce inflammation indirectly. The 14-member ring macrolides have been effective in animal models in preventing the acute exacerbation of interstitial pneumonia Page 383 and acute lung injury, such as after the use of bleomycin.<sup>201,202</sup> One study evaluating the effectiveness of erythromycin compared with azithromycin in preventing exacerbations of non-cystic fibrosis (non-CF) bronchiectasis showed that azithromycin was more effective than erythromycin (RR, 0.35; 95% CI, 0.403 to 0.947) and resulted in fewer adverse effects.<sup>203</sup> In vitro studies have demonstrated that erythromycin and other macrolides at subinhibitory concentrations reduce the adherence of various pathogenic bacterial species, such as *Pseudomonas aeruginosa*, to host cells.<sup>204</sup>

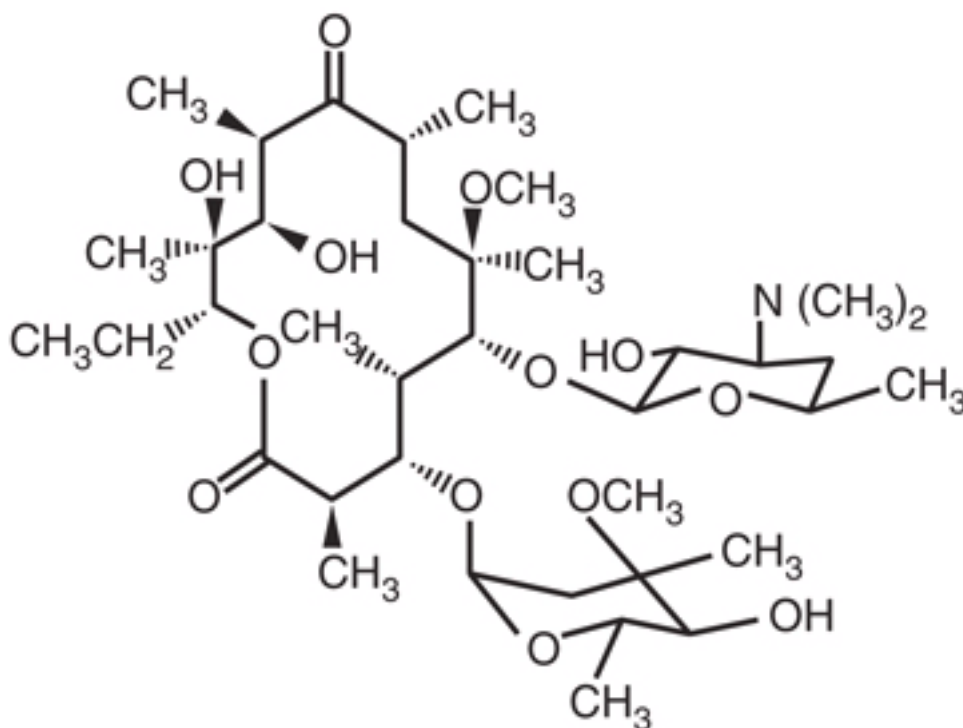


#### Azithromycin and Clarithromycin

Azithromycin and clarithromycin were developed to improve the qualities of erythromycin. They have better oral absorption, longer half-lives, fewer gastrointestinal side effects, and a greater antimicrobial spectrum of activity than erythromycin.

#### Derivation, Chemistry, and Preparations

Azithromycin is derived from erythromycin, differing in having a methyl-substituted nitrogen in its 15-member lactone ring (Fig. 28.2). It is therefore an azalide antibiotic. Clarithromycin, having a 14-member ring structure, is produced by modifying position C6 of the lactone ring of



in *M. avium* complex have been selected in patients undergoing clarithromycin or azithromycin monotherapy for disseminated infections.<sup>213</sup>

#### Antimicrobial Activity

Although one of the most widely prescribed antibiotics in the United States, azithromycin is about two- to fourfold less active than erythromycin against *S. pneumoniae* and *S. pyogenes*.<sup>205,206,214</sup> Clarithromycin is highly active against gram-positive bacteria, being two- to fourfold more active than erythromycin against most streptococci, including *S. pneumoniae* and *S. pyogenes*, and methicillin-sensitive *S. aureus*.<sup>47,205,206</sup> Streptococci and staphylococci that are resistant to erythromycin are resistant to clarithromycin and azithromycin.<sup>51,52,206,215</sup> Selection for penicillin and multidrug-resistant *S. pneumoniae* appears to be associated with use of macrolides and cephalosporins.<sup>214</sup> (Additional information regarding the emergence of macrolide resistance in clinical isolates of *S. pneumoniae* and *S. pyogenes* was discussed earlier under "Erythromycin.") Most methicillin-resistant staphylococci are resistant to the newer macrolides.<sup>63,64</sup> An active metabolite of clarithromycin, 14-hydroxyclearithromycin, has slightly greater activity than the parent compound against *S. aureus* and *S. pneumoniae* and is additive in vitro to the activity of clarithromycin.<sup>14,215</sup>

The activity of clarithromycin against many gram-negative bacteria is similar to that of erythromycin,<sup>14,206</sup> although it is slightly more active against *M. catarrhalis*. 14-Hydroxyclearithromycin also has slightly greater activity for *M. catarrhalis* and *H. influenzae* than the parent compound.<sup>215</sup> Azithromycin is more active than erythromycin or clarithromycin against gram-negative bacteria, especially against *H. influenzae* and *M. catarrhalis*, likely because of its higher penetration into the bacteria because of higher lipophilicity.<sup>205,216,217</sup> The greater activity of azithromycin against the Enterobacterales is of questionable clinical significance. However, it is of interest that prolonged incubation of *P. aeruginosa* strains with macrolides (especially azithromycin) at clinically achievable concentrations is associated with decreased viability and diminished protein synthesis.<sup>218</sup> That said, ribosomal mutations appear to confer reduced macrolide susceptibility in *P. aeruginosa* clinical isolates, which has been observed in patients with CF.<sup>219</sup> Macrolides have also been shown to inhibit the expression of quorum sensing of *P. aeruginosa*, thereby downregulating quorum sensing-regulated genes that encode virulence factors such as adhesion molecules and those that bring about cytotoxicity and inflammation.<sup>220</sup> Azithromycin is more active than erythromycin against *N. meningitidis* and *N. gonorrhoeae*; however, azithromycin resistance among *N. gonorrhoeae* isolates has increased up to 5% to 8% as of 2018 in the United States and European countries.<sup>221</sup>Page 384

Azithromycin and clarithromycin have equal or slightly better in vitro activities than erythromycin against *L. pneumophila*.<sup>42,92</sup> All three macrolides generally have good activity against *M. pneumoniae* and *C. pneumoniae*,<sup>42,90,215</sup> although macrolide-resistant *M. pneumoniae* isolates have been increasingly recognized globally, exhibiting marked geographic variability.<sup>222</sup> In the United States macrolide resistance among *M. pneumoniae* isolates hovers in the 5% to 10% range but is >50% in parts of the Western Pacific.<sup>222</sup> Azithromycin and clarithromycin have significantly greater activity than erythromycin against *C. trachomatis* and *U. urealyticum*.<sup>40,90</sup> and somewhat greater activity against *Borrelia burgdorferi*.<sup>206,207</sup> The small in vitro differences in potency among these macrolides may not have any clinical significance with regard to efficacy.<sup>223</sup> Both azithromycin and clarithromycin also have significant and approximately equal activity against *Toxoplasma gondii* in tissue culture systems.<sup>224,225</sup> Azithromycin or clarithromycin plus pyrimethamine plus leucovorin have demonstrated activity, with mixed clinical outcomes, against *T. gondii* in small cohorts of patients or case reports including few patients.<sup>226</sup> However, the dose of clarithromycin used in the case report is one that should not be used in patients with human immunodeficiency virus because of an association with increased mortality at doses higher than 500 mg/day.<sup>227</sup>

The macrolides generally show little activity against *M. tuberculosis*.<sup>103,228</sup> Clarithromycin shows substantial activity against *Mycobacterium leprae* and is superior in this respect to erythromycin and azithromycin.<sup>103,215</sup> Clarithromycin and azithromycin have appreciable activities against *M. avium* complex. Clarithromycin is about fourfold more active than azithromycin against this organism in vitro<sup>229</sup> and is somewhat more active in slowing its replication in infected human macrophages.<sup>230</sup>



Azithromycin and clarithromycin demonstrate activity against *Mycobacterium abscessus* complex species; however, this organism produces an inducible 23S rRNA methylase encoded by *erm*(41), which promotes macrolide resistance. Overall, clarithromycin induces greater *erm*(41) expression leading to higher macrolide resistance compared with azithromycin, but levels of inducible clarithromycin resistance are distinguishable among the subspecies of *M. abscessus*.<sup>231</sup> *M. abscessus* subsp. *bolletii* demonstrate inducible resistance, as does *M. abscessus* subsp. *abscessus*, if the *erm*(41) sequevar T28 is present. Conversely, *erm*(41) sequevar C28 is associated with clarithromycin susceptibility among *M. abscessus* subsp. *abscessus*. *M. abscessus* subsp. *massiliense* may also harbor *erm*(41) but remain susceptible to clarithromycin.<sup>232,233</sup>

#### Clinical Pharmacology

The oral bioavailability of azithromycin after a single 250 mg and 500 mg dose is 38% and 37%, respectively.<sup>234</sup> Food increases the maximum serum concentration by approximately 50%,<sup>235</sup> but absorption is unaffected by magnesium- or aluminum-containing antacids.<sup>236</sup> The maximum serum concentration achieved after a single 500-mg oral dose was 0.41 µg/mL within 2 to 4 hours.<sup>206</sup> The maximum concentration and 24-hour area under the curve (AUC) after a single 2-g dose of the sustained-release formulation are two and three times higher than those achieved with a total treatment dose of 1.5 g of conventional, immediate-release azithromycin.<sup>237</sup>

Protein binding of azithromycin in serum varies between 7% and 50% depending on the drug concentration.<sup>206</sup> Azithromycin is widely distributed in tissues, and for most tissues the drug concentration exceeds that in serum by 10- to 100-fold,<sup>234</sup> particularly in sputum and lung. Very high concentrations were found in alveolar macrophages and neutrophils.<sup>238</sup> The extensive tissue uptake of azithromycin has been attributed to cell uptake of this basic compound into relatively acidic lysosomes because of ionic trapping.<sup>238</sup> Very low concentrations were noted in CSF in patients without meningitis and in the aqueous humor of the uninflamed eye.<sup>239</sup> However, appreciable concentrations of azithromycin have been detected in the brains of patients undergoing resections of brain tumors after they received 500 mg orally.<sup>239</sup> The average half-life in many tissues is between 2 and 4 days,<sup>234</sup> so it is estimated that significant antibacterial activity against many pathogens persists in tissue for at least 5 days after a 5-day course of treatment.<sup>234</sup> The average terminal half-life is 68 hours, consistent with a slow release of drug from tissues, followed by elimination from the vascular compartment. About 6% of an oral dose appears as unchanged drug in the urine within 1 week of administration, and another small proportion is metabolized to inactive compounds, particularly by demethylation.<sup>234</sup> Most of the drug that is absorbed remains unmetabolized and is probably eliminated in feces by way of biliary excretion and possibly transintestinal elimination.<sup>234</sup> Biliary concentrations of azithromycin are higher than in the serum, and most of the drug in the bile is unchanged.<sup>234</sup> There are no data available on dose adjustments required with severe renal or hepatic failure.

Clarithromycin is well absorbed after oral administration, with or without food, and is approximately 55% bioavailable.<sup>206</sup> Mean peak serum concentrations at steady state with oral doses of 250 and 500 mg every 12 hours are 1 and 2 to 3 µg/mL, respectively. The elimination half-lives for those two regimens are 3 to 4 and 5 to 7 hours, respectively. Clarithromycin is appreciably metabolized in the liver by oxidation and hydrolysis to a number of compounds, accounting for a recovery of 78%. The major metabolite, 14-hydroxycarithromycin, has antibacterial activity and accounts for 20% of the metabolites.<sup>206</sup> With the 250-mg and 500-mg oral doses given every 12 hours, about 20% to 30%, respectively, of the drug is excreted into the urine unchanged and 10% to 15% as the hydroxy metabolite. At higher doses, there is some nonlinearity of half-life, apparently because of saturation of metabolic mechanisms with a higher proportion of unchanged drug eliminated in the urine.<sup>206</sup> About 65% to 70% of the drug is bound to protein in the serum. With renal insufficiency involving creatinine clearances of less than 30 mL/min, there is a marked increase in the half-life of clarithromycin.<sup>215</sup> Dose adjustment is suggested in patients with severe renal failure, including recommendations for a 500-mg loading dose, followed by 250 mg once or twice daily depending on the type of infection.<sup>215</sup> No dosage adjustment is recommended at present for severe hepatic dysfunction.<sup>215</sup>

Clarithromycin is widely distributed and penetrates well into various tissues, generally exceeding peak maximum serum levels by severalfold.<sup>215</sup> Concentrations of clarithromycin and its 14-hydroxy metabolite in middle ear fluids of children with acute otitis media exceeded the plasma

concentrations by approximately ninefold and fourfold, respectively, 12 hours after the sixth dose when the drug was given every 12 hours.<sup>240</sup> Concentrations generally exceed the MIC of most strains of middle ear pathogens, except for highly penicillin-resistant *S. pneumoniae*. Data indicate minimal penetration of clarithromycin and its 14-hydroxy metabolite into CSF in patients without meningitis.<sup>241</sup> The levels achieved were below the MIC for the usual pathogens associated with bacterial meningitis and were 1% to 2% of the corresponding plasma levels. However, CSF penetration appears to increase in the setting of infection and might achieve therapeutic levels.<sup>242</sup> Clarithromycin, like the other macrolides, penetrates well into phagocytic cells.<sup>243</sup>

#### Adverse Reactions

Adverse reactions to clarithromycin and azithromycin at the usual doses have been rare.<sup>205,206,244</sup> The most common complaints are gastrointestinal (diarrhea, nausea, abdominal pain), and discontinuance of therapy is rarely required. One retrospective study noted a small increased risk for hypertrophic pyloric stenosis in neonates exposed to azithromycin in the first 42 days of life, but this effect is more pronounced with exposure to other macrolides.<sup>245</sup> A proposed mechanism to explain this association is the capacity of macrolides to act as motilin agonists, interacting with motilin receptors and thereby directly stimulating contraction of the gastrointestinal smooth muscles, which could produce hypertrophy of the pylorus.<sup>246</sup> Azithromycin has also been shown to have a profound effect on the gastrointestinal microbiome. When compared with household contacts who did not receive antibiotics, patients who received either azithromycin or amoxicillin experienced substantial decreases in microbiota diversity that were sustained throughout the 6-month study period.<sup>247</sup> Studies in mice show long-term effects on the microbiome from macrolides, especially in young animals,<sup>248,249</sup> and studies in schoolchildren in Finland confirm the more profound and prolonged perturbation compared with  $\beta$ -lactam antibiotics.<sup>250</sup>Page 385Azithromycin and Clarithromycin  
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Azithromycin is derived from erythromycin, differing in having a methyl-substituted nitrogen in its 15-member lactone ring (Fig. 28.2). It is therefore an azalide antibiotic. Clarithromycin, having a 14-member ring structure, is produced by modifying position C6 of the lactone ring of erythromycin to possess a methoxy group (Fig. 28.3). These changes increase the stability of these compounds in gastric acid, improving absorption by the oral route.<sup>205</sup>

Azithromycin is available in capsules for oral use as azithromycin dihydrate equivalent to 250 mg of azithromycin; in film-coated tablets of 250 mg, 500 mg, and 600 mg; as a powder for oral suspension (1 g/packet, 100 mg/5 mL, and 200 mg/5 mL); and as intravenous powder for solution (500 mg). Azithromycin is also available as a 1% ophthalmic solution for bacterial conjunctivitis.

Clarithromycin is provided in 250-mg or 500-mg film-coated tablets, extended-release 250-mg and 500-mg tablets, and as granules for oral suspension (125 mg/5 mL and 250 mg/5 mL). Clarithromycin 500 mg is also available for intravenous administration in some countries.

#### Mechanisms of Action and Resistance

Limited studies suggest that azithromycin, clarithromycin, and erythromycin bind to the same receptor on the bacterial 50S ribosomal subunit and inhibit RNA-dependent protein synthesis by the same mechanism.<sup>11,206</sup> Azithromycin has greater activity than the 14-member macrolides erythromycin and clarithromycin against gram-negative bacteria (especially for *M. catarrhalis* and *H. influenzae*) and therefore appears to better penetrate the outer envelope of those organisms.<sup>207</sup> Like other macrolides, azithromycin and clarithromycin are generally considered to be bacteriostatic agents; however, bactericidal activity is demonstrated in vitro against such species as *S. pyogenes*, *S. pneumoniae*, and *H. influenzae*.<sup>206,207</sup> In vitro activity of the newer macrolides increases with rising pH, as with erythromycin.

Mechanisms of resistance to azithromycin or clarithromycin are the same as or similar to those for erythromycin.<sup>10,13,14</sup> There is complete cross-resistance among erythromycin, azithromycin, and clarithromycin for gram-positive organisms showing resistance to erythromycin by the MLSB

phenotype because the methylation mechanism already described operates for all of the 14- and 15-member macrolides.<sup>26</sup> Cross-resistance among the 14- and 15-member macrolides is also characteristic of the M phenotype and is the result of an efflux system for the drugs. Macrolide-resistant strains of *H. pylori* with point mutations in the 23S rRNA genes are increasing and are thought to be due to prior macrolide exposure.<sup>208,209</sup> This is an emerging mechanism of resistance that has been identified in other bacterial species including mycobacteria.<sup>210,211</sup> Further studies with such strains suggest that horizontal transfer of the mutated gene can occur.<sup>211,212</sup> Similar point mutations in the 23S rRNA gene associated with macrolide resistance in *M. avium* complex have been selected in patients undergoing clarithromycin or azithromycin monotherapy for disseminated infections.<sup>213</sup>

#### FIG. 28.3 Clarithromycin base.

Follow for extended description

##### Antimicrobial Activity

Although one of the most widely prescribed antibiotics in the United States, azithromycin is about two- to fourfold less active than erythromycin against *S. pneumoniae* and *S. pyogenes*.<sup>205,206,214</sup> Clarithromycin is highly active against gram-positive bacteria, being two- to fourfold more active than erythromycin against most streptococci, including *S. pneumoniae* and *S. pyogenes*, and methicillin-sensitive *S. aureus*.<sup>47,205,206</sup> Streptococci and staphylococci that are resistant to erythromycin are resistant to clarithromycin and azithromycin.<sup>51,52,206,215</sup> Selection for penicillin and multidrug-resistant *S. pneumoniae* appears to be associated with use of macrolides and cephalosporins.<sup>214</sup> (Additional information regarding the emergence of macrolide resistance in clinical isolates of *S. pneumoniae* and *S. pyogenes* was discussed earlier under “Erythromycin.”) Most methicillin-resistant staphylococci are resistant to the newer macrolides.<sup>63,64</sup> An active metabolite of clarithromycin, 14-hydroxyclearithromycin, has slightly greater activity than the parent compound against *S. aureus* and *S. pneumoniae* and is additive in vitro to the activity of clarithromycin.<sup>14,215</sup>

The activity of clarithromycin against many gram-negative bacteria is similar to that of erythromycin,<sup>14,206</sup> although it is slightly more active against *M. catarrhalis*. 14-Hydroxyclearithromycin also has slightly greater activity for *M. catarrhalis* and *H. influenzae* than the parent compound.<sup>215</sup> Azithromycin is more active than erythromycin or clarithromycin against gram-negative bacteria, especially against *H. influenzae* and *M. catarrhalis*, likely because of its higher penetration into the bacteria because of higher lipophilicity.<sup>205,216,217</sup> The greater activity of azithromycin against the Enterobacterales is of questionable clinical significance. However, it is of interest that prolonged incubation of *P. aeruginosa* strains with macrolides (especially azithromycin) at clinically achievable concentrations is associated with decreased viability and diminished protein synthesis.<sup>218</sup> That said, ribosomal mutations appear to confer reduced macrolide susceptibility in *P. aeruginosa* clinical isolates, which has been observed in patients with CF.<sup>219</sup> Macrolides have also been shown to inhibit the expression of quorum sensing of *P. aeruginosa*, thereby downregulating quorum sensing-regulated genes that encode virulence factors such as adhesion molecules and those that bring about cytotoxicity and inflammation.<sup>220</sup> Azithromycin is more active than erythromycin against *N. meningitidis* and *N. gonorrhoeae*; however, azithromycin resistance among *N. gonorrhoeae* isolates has increased up to 5% to 8% as of 2018 in the United States and European countries.<sup>221</sup>Page 384

Azithromycin and clarithromycin have equal or slightly better in vitro activities than erythromycin against *L. pneumophila*.<sup>42,92</sup> All three macrolides generally have good activity against *M. pneumoniae* and *C. pneumoniae*,<sup>42,90,215</sup> although macrolide-resistant *M. pneumoniae* isolates have been increasingly recognized globally, exhibiting marked geographic variability.<sup>222</sup> In the United States macrolide resistance among *M. pneumoniae* isolates hovers in the 5% to 10% range but is >50% in parts of the Western Pacific.<sup>222</sup> Azithromycin and clarithromycin have significantly greater activity than erythromycin against *C. trachomatis* and *U. urealyticum*.<sup>40,90</sup> and somewhat greater activity against *Borrelia burgdorferi*.<sup>206,207</sup> The small in vitro differences in potency among these macrolides may not have any clinical significance with regard to efficacy.<sup>223</sup> Both azithromycin and clarithromycin also have significant and approximately equal activity against *Toxoplasma gondii* in tissue culture systems.<sup>224,225</sup> Azithromycin or clarithromycin plus pyrimethamine plus leucovorin have demonstrated activity, with mixed clinical outcomes, against *T. gondii* in small cohorts of patients or case reports including few patients.<sup>226</sup>

However, the dose of clarithromycin used in the case report is one that should not be used in patients with human immunodeficiency virus because of an association with increased mortality at doses higher than 500 mg/day.<sup>227</sup>

The macrolides generally show little activity against *M. tuberculosis*.<sup>103,228</sup> Clarithromycin shows substantial activity against *Mycobacterium leprae* and is superior in this respect to erythromycin and azithromycin.<sup>103,215</sup> Clarithromycin and azithromycin have appreciable activities against *M. avium* complex. Clarithromycin is about fourfold more active than azithromycin against this organism in vitro<sup>229</sup> and is somewhat more active in slowing its replication in infected human macrophages.<sup>230</sup>

Azithromycin and clarithromycin demonstrate activity against *Mycobacterium abscessus* complex species; however, this organism produces an inducible 23S rRNA methylase encoded by *erm*(41), which promotes macrolide resistance. Overall, clarithromycin induces greater *erm*(41) expression leading to higher macrolide resistance compared with azithromycin, but levels of inducible clarithromycin resistance are distinguishable among the subspecies of *M. abscessus*.<sup>231</sup> *M. abscessus* subsp. *bolletii* demonstrate inducible resistance, as does *M. abscessus* subsp. *abscessus*, if the *erm*(41) sequevar T28 is present. Conversely, *erm*(41) sequevar C28 is associated with clarithromycin susceptibility among *M. abscessus* subsp. *abscessus*. *M. abscessus* subsp. *massiliense* may also harbor *erm*(41) but remain susceptible to clarithromycin.<sup>232,233</sup>

#### Clinical Pharmacology

The oral bioavailability of azithromycin after a single 250 mg and 500 mg dose is 38% and 37%, respectively.<sup>234</sup> Food increases the maximum serum concentration by approximately 50%,<sup>235</sup> but absorption is unaffected by magnesium- or aluminum-containing antacids.<sup>236</sup> The maximum serum concentration achieved after a single 500-mg oral dose was 0.41 µg/mL within 2 to 4 hours.<sup>206</sup> The maximum concentration and 24-hour area under the curve (AUC) after a single 2-g dose of the sustained-release formulation are two and three times higher than those achieved with a total treatment dose of 1.5 g of conventional, immediate-release azithromycin.<sup>237</sup>

Protein binding of azithromycin in serum varies between 7% and 50% depending on the drug concentration.<sup>206</sup> Azithromycin is widely distributed in tissues, and for most tissues the drug concentration exceeds that in serum by 10- to 100-fold,<sup>234</sup> particularly in sputum and lung. Very high concentrations were found in alveolar macrophages and neutrophils.<sup>238</sup> The extensive tissue uptake of azithromycin has been attributed to cell uptake of this basic compound into relatively acidic lysosomes because of ionic trapping.<sup>238</sup> Very low concentrations were noted in CSF in patients without meningitis and in the aqueous humor of the uninflamed eye.<sup>239</sup> However, appreciable concentrations of azithromycin have been detected in the brains of patients undergoing resections of brain tumors after they received 500 mg orally.<sup>239</sup> The average half-life in many tissues is between 2 and 4 days,<sup>234</sup> so it is estimated that significant antibacterial activity against many pathogens persists in tissue for at least 5 days after a 5-day course of treatment.<sup>234</sup> The average terminal half-life is 68 hours, consistent with a slow release of drug from tissues, followed by elimination from the vascular compartment. About 6% of an oral dose appears as unchanged drug in the urine within 1 week of administration, and another small proportion is metabolized to inactive compounds, particularly by demethylation.<sup>234</sup> Most of the drug that is absorbed remains unmetabolized and is probably eliminated in feces by way of biliary excretion and possibly transintestinal elimination.<sup>234</sup> Biliary concentrations of azithromycin are higher than in the serum, and most of the drug in the bile is unchanged.<sup>234</sup> There are no data available on dose adjustments required with severe renal or hepatic failure.

Clarithromycin is well absorbed after oral administration, with or without food, and is approximately 55% bioavailable.<sup>206</sup> Mean peak serum concentrations at steady state with oral doses of 250 and 500 mg every 12 hours are 1 and 2 to 3 µg/mL, respectively. The elimination half-lives for those two regimens are 3 to 4 and 5 to 7 hours, respectively. Clarithromycin is appreciably metabolized in the liver by oxidation and hydrolysis to a number of compounds, accounting for a recovery of 78%. The major metabolite, 14-hydroxycarithromycin, has antibacterial activity and accounts for 20% of the metabolites.<sup>206</sup> With the 250-mg and 500-mg oral doses given every 12 hours, about 20% to 30%, respectively, of the drug is excreted into the urine unchanged and 10% to 15% as the hydroxy metabolite. At higher doses, there is some

nonlinearity of half-life, apparently because of saturation of metabolic mechanisms with a higher proportion of unchanged drug eliminated in the urine.<sup>206</sup> About 65% to 70% of the drug is bound to protein in the serum. With renal insufficiency involving creatinine clearances of less than 30 mL/min, there is a marked increase in the half-life of clarithromycin.<sup>215</sup> Dose adjustment is suggested in patients with severe renal failure, including recommendations for a 500-mg loading dose, followed by 250 mg once or twice daily depending on the type of infection.<sup>215</sup> No dosage adjustment is recommended at present for severe hepatic dysfunction.<sup>215</sup>

Clarithromycin is widely distributed and penetrates well into various tissues, generally exceeding peak maximum serum levels by severalfold.<sup>215</sup> Concentrations of clarithromycin and its 14-hydroxy metabolite in middle ear fluids of children with acute otitis media exceeded the plasma concentrations by approximately ninefold and fourfold, respectively, 12 hours after the sixth dose when the drug was given every 12 hours.<sup>240</sup> Concentrations generally exceed the MIC of most strains of middle ear pathogens, except for highly penicillin-resistant *S. pneumoniae*. Data indicate minimal penetration of clarithromycin and its 14-hydroxy metabolite into CSF in patients without meningitis.<sup>241</sup> The levels achieved were below the MIC for the usual pathogens associated with bacterial meningitis and were 1% to 2% of the corresponding plasma levels. However, CSF penetration appears to increase in the setting of infection and might achieve therapeutic levels.<sup>242</sup> Clarithromycin, like the other macrolides, penetrates well into phagocytic cells.<sup>243</sup>

### Adverse Reactions

Adverse reactions to clarithromycin and azithromycin at the usual doses have been rare.<sup>205,206,244</sup> The most common complaints are gastrointestinal (diarrhea, nausea, abdominal pain), and discontinuance of therapy is rarely required. One retrospective study noted a small increased risk for hypertrophic pyloric stenosis in neonates exposed to azithromycin in the first 42 days of life, but this effect is more pronounced with exposure to other macrolides.<sup>245</sup> A proposed mechanism to explain this association is the capacity of macrolides to act as motilin agonists, interacting with motilin receptors and thereby directly stimulating contraction of the gastrointestinal smooth muscles, which could produce hypertrophy of the pylorus.<sup>246</sup> Azithromycin has also been shown to have a profound effect on the gastrointestinal microbiome. When compared with household contacts who did not receive antibiotics, patients who received either azithromycin or amoxicillin experienced substantial decreases in microbiota diversity that were sustained throughout the 6-month study period.<sup>247</sup> Studies in mice show long-term effects on the microbiome from macrolides, especially in young animals,<sup>248,249</sup> and studies in schoolchildren in Finland confirm the more profound and prolonged perturbation compared with  $\beta$ -lactam antibiotics.<sup>250</sup>Page 385

Acute psychosis or “mania,” akathisia, and visual hallucinations have been noted in a few patients receiving clarithromycin or azithromycin.<sup>251-255</sup> High doses of clarithromycin in animals have been associated with teratogenic effects, and therefore that drug is not recommended for use in pregnancy but definitive teratogenic effects in humans have not been substantiated.<sup>256,257</sup> Azithromycin is classified as pregnancy category B.

Abnormalities in liver function are occasionally encountered in patients treated with these drugs, and reversible cholestatic hepatitis has been reported with azithromycin.<sup>258</sup> With the high doses of these drugs used in the treatment of *M. avium* complex, tinnitus, dizziness, and reversible hearing loss have been reported.<sup>259,260</sup> Rarely, severe allergic reactions have occurred with the use of azithromycin, including erythema multiforme, Stevens-Johnson syndrome, and acute generalized exanthematous pustulosis.<sup>261-263</sup>

The risk of macrolide-associated torsades de pointes (a polymorphic ventricular tachycardia) has been associated with increasing age, female sex, chronic use, hemodialysis-dependent end stage renal failure, and concomitant drug use, especially with cisapride.<sup>264-266</sup> Chronic exposure to azithromycin in an animal model promoted intracellular sodium loading, which could lead to proarrhythmia.<sup>265</sup> A retrospective study reported that patients taking a 5-day course of azithromycin had a 2.88 increased risk of cardiovascular death compared with those taking no antibiotics, with risk being most pronounced among patients at high risk for cardiovascular disease.<sup>267,268</sup> Additionally, in an observational case series of patients who received treatment for *H. pylori* with either clarithromycin or amoxicillin, there was an increased risk of myocardial

infarction, arrhythmia, and cardiac mortality during or shortly after clarithromycin therapy. However, clarithromycin use was not associated with cardiovascular risks long term.<sup>269</sup> After following patients for a year or longer in a large randomized, multicenter study, an increased number of deaths were observed among patients with underlying coronary artery disease who received 2 weeks of clarithromycin.<sup>270</sup> Conversely, a recent retrospective cohort study found that macrolide antibiotics were not associated with a higher 30-day risk of ventricular arrhythmia or all-cause mortality compared with nonmacrolide antibiotics.<sup>271</sup> Although these conflicting observations are of concern, definitive association of increased cardiovascular-related death and exposure to azithromycin or clarithromycin cannot be based on a retrospective cohort design, given residual confounding that may have influenced the outcome.<sup>272</sup> However, as a result of that study, the US Food and Drug Administration has required changes in the labeling of azithromycin warning about the risk of QT interval prolongation and cardiac arrhythmias, as well as changes in the labeling of clarithromycin warning about the increased risk of death among patients with underlying heart disease even in the setting of short courses of therapy based on the study by Jespersen and coauthors.<sup>270,273</sup> It is advised to avoid prescribing a macrolide antibiotic together with another drug known to prolong the QT interval and to exercise caution in the use of a macrolide antibiotic, even alone, in patients with underlying heart disease.

### Drug Interactions

Azithromycin minimally inhibits CYP3A4 and therefore appears to be the safest macrolide derivative from a drug interaction perspective.<sup>274</sup>

Clarithromycin has been reported to be associated with increased concentrations of several drugs that undergo hepatic metabolism by the CYP3A subclass of the cytochrome P-450 enzyme system (see Table 28.3).<sup>275</sup> As with erythromycin, these interactions can lead to serious toxicity. It is not yet clear whether the clarithromycin interactions can occur with all the drugs that interact with erythromycin, but a conservative clinical approach would be to consider that potential. These interactions with the hepatic metabolism of other drugs have not been documented with azithromycin, which does not appear to induce or bind and inactivate the cytochrome P-450 enzymes, probably because of its different azalide structure.<sup>149,276</sup>

As a potent P-glycoprotein inhibitor, clarithromycin coadministration with direct oral anticoagulants (DOACs) has led to twofold increases in AUCs of dabigatran and rivaroxaban.<sup>277</sup> To better understand the clinical ramifications of this interaction, a study demonstrated higher rates of hospitalization with hemorrhage associated with coprescribing clarithromycin (0.77%) versus azithromycin (0.43%) with DOACs (adjusted hazard ratio, 1.71; 95% CI, 1.20 to 2.45).<sup>278</sup> Clarithromycin may decrease the serum concentration of zidovudine by unknown mechanisms when the two drugs are taken at the same time.<sup>279</sup> However, in another study with a somewhat different design, there was no significant alteration of zidovudine bioavailability in volunteers with acquired immunodeficiency syndrome (AIDS) who took the two drugs 2 hours apart.<sup>280</sup>

Clarithromycin, like erythromycin, may occasionally lead to digoxin toxicity, possibly by diminishing the bacterial metabolism of digoxin in the gut.<sup>281</sup>

### Uses of Clarithromycin and Azithromycin

Clarithromycin and azithromycin have several indications for use as the drug of choice and some important applications as an alternative drug (see Table 28.4).<sup>14</sup> Clarithromycin and azithromycin were as effective as other commonly used antimicrobial agents when employed in randomized multicenter trials for the treatment of pharyngitis, sinusitis, community-acquired pneumonia (including *M. pneumoniae* and *C. pneumoniae* pneumonia), and skin infections.<sup>163,205,206,282</sup> Clarithromycin and azithromycin have been considered alternatives to penicillin in treating group A  $\beta$ -hemolytic streptococcal pharyngitis in patients with penicillin allergies.<sup>283</sup> Treatment with a 10- to 14-day course of either clarithromycin (250 mg twice daily) or penicillin VK (250 mg every 6 hours) is equally effective.<sup>206,284</sup> Azithromycin administered for 5 days (500 mg on day 1, followed by 250 mg daily for 4 days) is effective in eradicating group A streptococci from the pharynx,<sup>285</sup> although shorter courses of 3-day treatment with azithromycin were associated with lower levels of bacteriologic eradication when compared with a 10-day course of penicillin V.<sup>286</sup> One study demonstrated that a 10-day course of treatment with clarithromycin (250 mg twice daily) led to more bacteriologic eradication compared with a 5-day course with azithromycin (500 mg on day 1, followed by 250 mg daily).<sup>287</sup> Clinical studies have demonstrated equal or superior

tolerability and also clinical and bacterial efficacy of azithromycin (12 mg/kg, once daily for 5 days) compared with a 10-day therapy with penicillin V in the treatment of pediatric streptococcal pharyngitis.<sup>288,289</sup> However, the effectiveness of a 5-day course of azithromycin in preventing acute rheumatic fever is unknown.<sup>290</sup> Moreover, with the variable prevalence of macrolide resistance in group A streptococci, the effectiveness of these agents in treating such infections when the susceptibility of the organism is unknown should be questioned.

Azithromycin administered for 5 days or clarithromycin for 7 to 10 days for the treatment of acute otitis media in children has been effective,<sup>291,292</sup> but use is limited because of increasing macrolide resistance among *S. pneumoniae* isolates, and cases of meningitis have been reported during oral therapy.<sup>293</sup>

Guidelines available for the treatment of adult outpatients with community-acquired pneumonia include the use of a macrolide alone as a first-line agent in patients in areas where macrolide resistance in pneumococci is documented to be <25%, and the patient has no significant medical problems or risk factors for antibiotic resistant pathogens, or as combination therapy with a  $\beta$ -lactam agent in patients with comorbidities such as chronic heart, lung, or liver disease; diabetes; or immunosuppressing conditions.<sup>163,294</sup> The aforementioned guideline is now further complicated for patients with heart disease by the concern for potentiating cardiac arrhythmias with either a macrolide or a fluoroquinolone. All three macrolides have good activity against most pathogens that commonly cause community-acquired pneumonia, including many strains of *S. pneumoniae* (though resistance is emerging) and almost all strains of *M. pneumoniae*, *C. pneumoniae*, *Legionella* spp., and *M. catarrhalis* (see Table 28.1).<sup>47</sup> However, the rising prevalence, already discussed, of macrolide resistance by strains of *S. pneumoniae*, especially those that are penicillin resistant, requires caution in the use of any of the macrolides as a sole agent when that organism may be causing pneumonia. That concern would be greatest for older adults or persons with underlying medical conditions. A study by Martinez and colleagues<sup>295</sup> suggested lower mortality benefit with combination therapy including a macrolide in community-acquired pneumonia associated with *S. pneumoniae*, including penicillin-susceptible strains, although sample size was small and the study was Page 386retrospective in design. Macrolides, with their ability to concentrate intracellularly, are effective against intracellular pathogens, including *C. pneumoniae*, *Legionella* spp., and *Coxiella burnetii* (the agent of Q fever).<sup>296</sup> Treatment with macrolides of lower respiratory tract infection with *M. pneumoniae* and *C. pneumoniae* in children shortens the clinical course of infection.<sup>297</sup> In the treatment of patients with nonpneumococcal pneumonia, erythromycin and azithromycin were equally effective (76% and 79%, respectively) in achieving clinical and radiologic resolution.<sup>223</sup> The longer half-life of azithromycin allows a shorter duration of therapy, and in one retrospective study, a 3-day course of azithromycin (daily dose, 500 mg) was as effective as a 5-day course (500 mg in a single dose the first day, followed by 250 mg daily for 4 days) in the treatment of atypical pneumonia.<sup>298</sup>

#### Azithromycin and Clarithromycin

Azithromycin and clarithromycin were developed to improve the qualities of erythromycin. They have better oral absorption, longer half-lives, fewer gastrointestinal side effects, and a greater antimicrobial spectrum of activity than erythromycin.

#### Derivation, Chemistry, and Preparations

Azithromycin is derived from erythromycin, differing in having a methyl-substituted nitrogen in its 15-member lactone ring (Fig. 28.2). It is therefore an azalide antibiotic. Clarithromycin, having a 14-member ring structure, is produced by modifying position C6 of the lactone ring of erythromycin to possess a methoxy group (Fig. 28.3). These changes increase the stability of these compounds in gastric acid, improving absorption by the oral route.<sup>205</sup>

Azithromycin is available in capsules for oral use as azithromycin dihydrate equivalent to 250 mg of azithromycin; in film-coated tablets of 250 mg, 500 mg, and 600 mg; as a powder for oral suspension (1 g/packet, 100 mg/5 mL, and 200 mg/5 mL); and as intravenous powder for solution (500 mg). Azithromycin is also available as a 1% ophthalmic solution for bacterial conjunctivitis.

Clarithromycin is provided in 250-mg or 500-mg film-coated tablets, extended-release 250-mg and 500-mg tablets, and as granules for oral suspension (125 mg/5 mL and 250 mg/5 mL). Clarithromycin 500 mg is also available for intravenous administration in some countries.

### Mechanisms of Action and Resistance

Limited studies suggest that azithromycin, clarithromycin, and erythromycin bind to the same receptor on the bacterial 50S ribosomal subunit and inhibit RNA-dependent protein synthesis by the same mechanism.<sup>11,206</sup> Azithromycin has greater activity than the 14-member macrolides erythromycin and clarithromycin against gram-negative bacteria (especially for *M. catarrhalis* and *H. influenzae*) and therefore appears to better penetrate the outer envelope of those organisms.<sup>207</sup> Like other macrolides, azithromycin and clarithromycin are generally considered to be bacteriostatic agents; however, bactericidal activity is demonstrated in vitro against such species as *S. pyogenes*, *S. pneumoniae*, and *H. influenzae*.<sup>206,207</sup> In vitro activity of the newer macrolides increases with rising pH, as with erythromycin.

Mechanisms of resistance to azithromycin or clarithromycin are the same as or similar to those for erythromycin.<sup>10,13,14</sup> There is complete cross-resistance among erythromycin, azithromycin, and clarithromycin for gram-positive organisms showing resistance to erythromycin by the MLSB phenotype because the methylation mechanism already described operates for all of the 14- and 15-member macrolides.<sup>26</sup> Cross-resistance among the 14- and 15-member macrolides is also characteristic of the M phenotype and is the result of an efflux system for the drugs. Macrolide-resistant strains of *H. pylori* with point mutations in the 23S rRNA genes are increasing and are thought to be due to prior macrolide exposure.<sup>208,209</sup> This is an emerging mechanism of resistance that has been identified in other bacterial species including mycobacteria.<sup>210,211</sup> Further studies with such strains suggest that horizontal transfer of the mutated gene can occur.<sup>211,212</sup> Similar point mutations in the 23S rRNA gene associated with macrolide resistance in *M. avium* complex have been selected in patients undergoing clarithromycin or azithromycin monotherapy for disseminated infections.<sup>213</sup>

### FIG. 28.3 Clarithromycin base.

Follow for extended description

#### Antimicrobial Activity

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#### Clinical Pharmacology

The oral bioavailability of azithromycin after a single 250 mg and 500 mg dose is 38% and 37%, respectively.<sup>234</sup> Food increases the maximum serum concentration by approximately 50%,<sup>235</sup> but absorption is unaffected by magnesium- or aluminum-containing antacids.<sup>236</sup> The maximum serum concentration achieved after a single 500-mg oral dose was 0.41 µg/mL within 2 to 4 hours.<sup>206</sup> The maximum concentration and 24-hour area under the curve (AUC) after a single 2-g dose of the sustained-release formulation are two and three times higher than those achieved with a total treatment dose of 1.5 g of conventional, immediate-release azithromycin.<sup>237</sup>

Protein binding of azithromycin in serum varies between 7% and 50% depending on the drug concentration.<sup>206</sup> Azithromycin is widely distributed in tissues, and for most tissues the drug concentration exceeds that in serum by 10- to 100-fold,<sup>234</sup> particularly in sputum and lung. Very high concentrations were found in alveolar macrophages and neutrophils.<sup>238</sup> The extensive tissue uptake of azithromycin has been attributed to cell uptake of this basic compound into relatively acidic lysosomes because of ionic trapping.<sup>238</sup> Very low concentrations were noted in CSF in patients without meningitis and in the aqueous humor of the uninflamed eye.<sup>239</sup> However, appreciable concentrations of azithromycin have been detected in the brains of patients undergoing resections of brain tumors after they received 500 mg orally.<sup>239</sup> The average half-life in many tissues is between 2 and 4 days,<sup>234</sup> so it is estimated that significant antibacterial activity against many pathogens persists in tissue for at least 5 days after a 5-day course of treatment.<sup>234</sup> The average terminal half-life is 68 hours, consistent with a slow release of drug from tissues, followed by elimination from the vascular compartment. About 6% of an oral dose appears as unchanged drug in the urine within 1 week of administration, and another small proportion is metabolized to inactive compounds, particularly by demethylation.<sup>234</sup> Most of the

drug that is absorbed remains unmetabolized and is probably eliminated in feces by way of biliary excretion and possibly transintestinal elimination.<sup>234</sup> Biliary concentrations of azithromycin are higher than in the serum, and most of the drug in the bile is unchanged.<sup>234</sup> There are no data available on dose adjustments required with severe renal or hepatic failure.

Clarithromycin is well absorbed after oral administration, with or without food, and is approximately 55% bioavailable.<sup>206</sup> Mean peak serum concentrations at steady state with oral doses of 250 and 500 mg every 12 hours are 1 and 2 to 3 µg/mL, respectively. The elimination half-lives for those two regimens are 3 to 4 and 5 to 7 hours, respectively. Clarithromycin is appreciably metabolized in the liver by oxidation and hydrolysis to a number of compounds, accounting for a recovery of 78%. The major metabolite, 14-hydroxyclearithromycin, has antibacterial activity and accounts for 20% of the metabolites.<sup>206</sup> With the 250-mg and 500-mg oral doses given every 12 hours, about 20% to 30%, respectively, of the drug is excreted into the urine unchanged and 10% to 15% as the hydroxy metabolite. At higher doses, there is some nonlinearity of half-life, apparently because of saturation of metabolic mechanisms with a higher proportion of unchanged drug eliminated in the urine.<sup>206</sup> About 65% to 70% of the drug is bound to protein in the serum. With renal insufficiency involving creatinine clearances of less than 30 mL/min, there is a marked increase in the half-life of clarithromycin.<sup>215</sup> Dose adjustment is suggested in patients with severe renal failure, including recommendations for a 500-mg loading dose, followed by 250 mg once or twice daily depending on the type of infection.<sup>215</sup> No dosage adjustment is recommended at present for severe hepatic dysfunction.<sup>215</sup>

Clarithromycin is widely distributed and penetrates well into various tissues, generally exceeding peak maximum serum levels by severalfold.<sup>215</sup> Concentrations of clarithromycin and its 14-hydroxy metabolite in middle ear fluids of children with acute otitis media exceeded the plasma concentrations by approximately ninefold and fourfold, respectively, 12 hours after the sixth dose when the drug was given every 12 hours.<sup>240</sup> Concentrations generally exceed the MIC of most strains of middle ear pathogens, except for highly penicillin-resistant *S. pneumoniae*. Data indicate minimal penetration of clarithromycin and its 14-hydroxy metabolite into CSF in patients without meningitis.<sup>241</sup> The levels achieved were below the MIC for the usual pathogens associated with bacterial meningitis and were 1% to 2% of the corresponding plasma levels. However, CSF penetration appears to increase in the setting of infection and might achieve therapeutic levels.<sup>242</sup> Clarithromycin, like the other macrolides, penetrates well into phagocytic cells.<sup>243</sup>

#### Adverse Reactions

Adverse reactions to clarithromycin and azithromycin at the usual doses have been rare.<sup>205,206,244</sup> The most common complaints are gastrointestinal (diarrhea, nausea, abdominal pain), and discontinuance of therapy is rarely required. One retrospective study noted a small increased risk for hypertrophic pyloric stenosis in neonates exposed to azithromycin in the first 42 days of life, but this effect is more pronounced with exposure to other macrolides.<sup>245</sup> A proposed mechanism to explain this association is the capacity of macrolides to act as motilin agonists, interacting with motilin receptors and thereby directly stimulating contraction of the gastrointestinal smooth muscles, which could produce hypertrophy of the pylorus.<sup>246</sup> Azithromycin has also been shown to have a profound effect on the gastrointestinal microbiome. When compared with household contacts who did not receive antibiotics, patients who received either azithromycin or amoxicillin experienced substantial decreases in microbiota diversity that were sustained throughout the 6-month study period.<sup>247</sup> Studies in mice show long-term effects on the microbiome from macrolides, especially in young animals,<sup>248,249</sup> and studies in schoolchildren in Finland confirm the more profound and prolonged perturbation compared with β-lactam antibiotics.<sup>250</sup>Page 385

Acute psychosis or “mania,” akathisia, and visual hallucinations have been noted in a few patients receiving clarithromycin or azithromycin.<sup>251–255</sup> High doses of clarithromycin in animals have been associated with teratogenic effects, and therefore that drug is not recommended for use in pregnancy but definitive teratogenic effects in humans have not been substantiated.<sup>256,257</sup> Azithromycin is classified as pregnancy category B.

Abnormalities in liver function are occasionally encountered in patients treated with these drugs, and reversible cholestatic hepatitis has been reported with azithromycin.<sup>258</sup> With the high doses

of these drugs used in the treatment of *M. avium* complex, tinnitus, dizziness, and reversible hearing loss have been reported.<sup>259,260</sup> Rarely, severe allergic reactions have occurred with the use of azithromycin, including erythema multiforme, Stevens-Johnson syndrome, and acute generalized exanthematous pustulosis.<sup>261-263</sup>

The risk of macrolide-associated torsades de pointes (a polymorphic ventricular tachycardia) has been associated with increasing age, female sex, chronic use, hemodialysis-dependent end stage renal failure, and concomitant drug use, especially with cisapride.<sup>264-266</sup> Chronic exposure to azithromycin in an animal model promoted intracellular sodium loading, which could lead to proarrhythmia.<sup>265</sup> A retrospective study reported that patients taking a 5-day course of azithromycin had a 2.88 increased risk of cardiovascular death compared with those taking no antibiotics, with risk being most pronounced among patients at high risk for cardiovascular disease.<sup>267,268</sup> Additionally, in an observational case series of patients who received treatment for *H. pylori* with either clarithromycin or amoxicillin, there was an increased risk of myocardial infarction, arrhythmia, and cardiac mortality during or shortly after clarithromycin therapy. However, clarithromycin use was not associated with cardiovascular risks long term.<sup>269</sup> After following patients for a year or longer in a large randomized, multicenter study, an increased number of deaths were observed among patients with underlying coronary artery disease who received 2 weeks of clarithromycin.<sup>270</sup> Conversely, a recent retrospective cohort study found that macrolide antibiotics were not associated with a higher 30-day risk of ventricular arrhythmia or all-cause mortality compared with nonmacrolide antibiotics.<sup>271</sup> Although these conflicting observations are of concern, definitive association of increased cardiovascular-related death and exposure to azithromycin or clarithromycin cannot be based on a retrospective cohort design, given residual confounding that may have influenced the outcome.<sup>272</sup> However, as a result of that study, the US Food and Drug Administration has required changes in the labeling of azithromycin warning about the risk of QT interval prolongation and cardiac arrhythmias, as well as changes in the labeling of clarithromycin warning about the increased risk of death among patients with underlying heart disease even in the setting of short courses of therapy based on the study by Jespersen and coauthors.<sup>270,273</sup> It is advised to avoid prescribing a macrolide antibiotic together with another drug known to prolong the QT interval and to exercise caution in the use of a macrolide antibiotic, even alone, in patients with underlying heart disease.

#### Drug Interactions

Azithromycin minimally inhibits CYP3A4 and therefore appears to be the safest macrolide derivative from a drug interaction perspective.<sup>274</sup>

Clarithromycin has been reported to be associated with increased concentrations of several drugs that undergo hepatic metabolism by the CYP3A subclass of the cytochrome P-450 enzyme system (see Table 28.3).<sup>275</sup> As with erythromycin, these interactions can lead to serious toxicity. It is not yet clear whether the clarithromycin interactions can occur with all the drugs that interact with erythromycin, but a conservative clinical approach would be to consider that potential. These interactions with the hepatic metabolism of other drugs have not been documented with azithromycin, which does not appear to induce or bind and inactivate the cytochrome P-450 enzymes, probably because of its different azalide structure.<sup>149,276</sup>

As a potent P-glycoprotein inhibitor, clarithromycin coadministration with direct oral anticoagulants (DOACs) has led to twofold increases in AUCs of dabigatran and rivaroxaban.<sup>277</sup> To better understand the clinical ramifications of this interaction, a study demonstrated higher rates of hospitalization with hemorrhage associated with coprescribing clarithromycin (0.77%) versus azithromycin (0.43%) with DOACs (adjusted hazard ratio, 1.71; 95% CI, 1.20 to 2.45).<sup>278</sup> Clarithromycin may decrease the serum concentration of zidovudine by unknown mechanisms when the two drugs are taken at the same time.<sup>279</sup> However, in another study with a somewhat different design, there was no significant alteration of zidovudine bioavailability in volunteers with acquired immunodeficiency syndrome (AIDS) who took the two drugs 2 hours apart.<sup>280</sup>

Clarithromycin, like erythromycin, may occasionally lead to digoxin toxicity, possibly by diminishing the bacterial metabolism of digoxin in the gut.<sup>281</sup>

#### Uses of Clarithromycin and Azithromycin

Clarithromycin and azithromycin have several indications for use as the drug of choice and some important applications as an alternative drug (see Table 28.4).<sup>14</sup> Clarithromycin and azithromycin were as effective as other commonly used antimicrobial agents when employed in randomized multicenter trials for the treatment of pharyngitis, sinusitis, community-acquired pneumonia (including *M. pneumoniae* and *C. pneumoniae* pneumonia), and skin infections.<sup>163,205,206,282</sup> Clarithromycin and azithromycin have been considered alternatives to penicillin in treating group A  $\beta$ -hemolytic streptococcal pharyngitis in patients with penicillin allergies.<sup>283</sup> Treatment with a 10- to 14-day course of either clarithromycin (250 mg twice daily) or penicillin VK (250 mg every 6 hours) is equally effective.<sup>206,284</sup> Azithromycin administered for 5 days (500 mg on day 1, followed by 250 mg daily for 4 days) is effective in eradicating group A streptococci from the pharynx,<sup>285</sup> although shorter courses of 3-day treatment with azithromycin were associated with lower levels of bacteriologic eradication when compared with a 10-day course of penicillin V.<sup>286</sup> One study demonstrated that a 10-day course of treatment with clarithromycin (250 mg twice daily) led to more bacteriologic eradication compared with a 5-day course with azithromycin (500 mg on day 1, followed by 250 mg daily).<sup>287</sup> Clinical studies have demonstrated equal or superior tolerability and also clinical and bacterial efficacy of azithromycin (12 mg/kg, once daily for 5 days) compared with a 10-day therapy with penicillin V in the treatment of pediatric streptococcal pharyngitis.<sup>288,289</sup> However, the effectiveness of a 5-day course of azithromycin in preventing acute rheumatic fever is unknown.<sup>290</sup> Moreover, with the variable prevalence of macrolide resistance in group A streptococci, the effectiveness of these agents in treating such infections when the susceptibility of the organism is unknown should be questioned.

Azithromycin administered for 5 days or clarithromycin for 7 to 10 days for the treatment of acute otitis media in children has been effective,<sup>291,292</sup> but use is limited because of increasing macrolide resistance among *S. pneumoniae* isolates, and cases of meningitis have been reported during oral therapy.<sup>293</sup>

Guidelines available for the treatment of adult outpatients with community-acquired pneumonia include the use of a macrolide alone as a first-line agent in patients in areas where macrolide resistance in pneumococci is documented to be <25%, and the patient has no significant medical problems or risk factors for antibiotic resistant pathogens, or as combination therapy with a  $\beta$ -lactam agent in patients with comorbidities such as chronic heart, lung, or liver disease; diabetes; or immunosuppressing conditions.<sup>163,294</sup> The aforementioned guideline is now further complicated for patients with heart disease by the concern for potentiating cardiac arrhythmias with either a macrolide or a fluoroquinolone. All three macrolides have good activity against most pathogens that commonly cause community-acquired pneumonia, including many strains of *S. pneumoniae* (though resistance is emerging) and almost all strains of *M. pneumoniae*, *C. pneumoniae*, *Legionella* spp., and *M. catarrhalis* (see Table 28.1).<sup>47</sup> However, the rising prevalence, already discussed, of macrolide resistance by strains of *S. pneumoniae*, especially those that are penicillin resistant, requires caution in the use of any of the macrolides as a sole agent when that organism may be causing pneumonia. That concern would be greatest for older adults or persons with underlying medical conditions. A study by Martinez and colleagues<sup>295</sup> suggested lower mortality benefit with combination therapy including a macrolide in community-acquired pneumonia associated with *S. pneumoniae*, including penicillin-susceptible strains, although sample size was small and the study was Page 386retrospective in design. Macrolides, with their ability to concentrate intracellularly, are effective against intracellular pathogens, including *C. pneumoniae*, *Legionella* spp., and *Coxiella burnetii* (the agent of Q fever).<sup>296</sup> Treatment with macrolides of lower respiratory tract infection with *M. pneumoniae* and *C. pneumoniae* in children shortens the clinical course of infection.<sup>297</sup> In the treatment of patients with nonpneumococcal pneumonia, erythromycin and azithromycin were equally effective (76% and 79%, respectively) in achieving clinical and radiologic resolution.<sup>223</sup> The longer half-life of azithromycin allows a shorter duration of therapy, and in one retrospective study, a 3-day course of azithromycin (daily dose, 500 mg) was as effective as a 5-day course (500 mg in a single dose the first day, followed by 250 mg daily for 4 days) in the treatment of atypical pneumonia.<sup>298</sup>

The generally recommended empirical therapy for community-acquired pneumonia requiring hospitalization includes the combination of a macrolide and a  $\beta$ -lactam,<sup>294</sup> which was associated with briefer hospital stays and a lower mortality rate than treatment with a cephalosporin alone.<sup>299,300</sup> In addition, two retrospective studies found that in cases of bacteremic

pneumococcal pneumonia, dual antimicrobial therapy, including a macrolide, reduced mortality.<sup>301,302</sup>

*B. pertussis* is susceptible in vitro to azithromycin and clarithromycin. Furthermore, data from a multicenter randomized trial in North America demonstrated that azithromycin was as effective as erythromycin, better tolerated, and associated with improved compliance.<sup>303</sup> The Centers for Disease Control and Prevention (CDC) and others recommend that azithromycin should be used for all neonates younger than 1 month, and erythromycin, clarithromycin, or azithromycin are noted to be acceptable for the treatment of pertussis in persons age 1 month or older.<sup>304,305</sup> Resistance to azithromycin has been rarely reported, but limited studies are available on its clinical effectiveness.<sup>306</sup> In a study involving a small number of children with pertussis in Japan, a 5-day course of azithromycin and a 7-day course of clarithromycin were as effective in eradicating *B. pertussis* from cultures 1 week after treatment as in historical control subjects who had been treated with erythromycin for 2 weeks.<sup>152</sup>

Clarithromycin (500 mg orally twice daily) or azithromycin (500–600 mg orally once daily) in addition to ethambutol with or without rifabutin are now considered the drugs of choice in the treatment of disseminated *M. avium* complex infections in persons living with AIDS. However, recent US National Institutes of Health guidelines no longer recommend primary prophylaxis for *M. avium* complex infections in persons living with AIDS who immediately initiate antiretroviral therapy (ART).<sup>226</sup> Relatively few mutations are associated with macrolide resistance in mycobacteria, and all are in genes encoding the peptidyl transferase region of the 23S rRNA.<sup>307</sup> Therefore the use of macrolides alone is often associated with clinical relapse and the emergence of macrolide-resistant organisms.<sup>308</sup> In persons living with AIDS who are not receiving ART or who remain viremic on ART but have no current options for a fully suppressive ART regimen and have CD4 counts <50 cells/mm<sup>3</sup>, prophylaxis of disseminated *M. avium* complex infection with clarithromycin (500 mg orally twice daily) or azithromycin (1200 mg orally once weekly) is effective.<sup>226,309,310</sup> In patients who develop disseminated infection despite prophylaxis, macrolide-resistant isolates are frequently found.<sup>309</sup>

While some studies have shown that mass administration of a single weight-based dose of azithromycin every 6 months improved childhood mortality in sub-Saharan Africa, such approaches require further validation and careful attention to potential adverse effects of mass antibacterial use.<sup>311</sup> The MORDOR I trial randomized children aged 1 to 59 months to receive biennial single doses of azithromycin or placebo, and found that the groups that received azithromycin resulted in approximately a third fewer deaths due to meningitis and dysentery, and a fifth fewer deaths due to malaria and pneumonia.<sup>311</sup> The MORDOR II trial showed that this benefit to childhood mortality persisted after 3 years of biennial azithromycin administration.<sup>312</sup> In a recent study of azithromycin mass administration, childhood mortality was incidentally found to be 13.5% lower ( $P < .001$ ) among children in communities who received azithromycin than in those who received placebo.<sup>313</sup> A single oral dose of azithromycin (1-g single dose oral dose for adults or 20 mg/kg for children) is a highly effective treatment for trachoma caused by *C. trachomatis*, and studies have been performed assessing the value of mass administration on childhood trachoma rates as well.<sup>314,315</sup> As a result, the WHO recommends mass azithromycin administration campaigns and hygiene interventions in areas with high rates of trachoma in children.<sup>316,317</sup> Additionally, biennial mass administration of azithromycin has been shown to decrease rates of yaws (*Treponema pallidum* subsp. *pertenue*) in children in yaws-endemic areas.<sup>318,319</sup> Biennial mass administration of azithromycin has been shown to have a low incidence of side effects, but concerns exist about the effect of mass antibiotic exposure on rates of antimicrobial resistance and effects on the human microbiome in developing nations.<sup>315,320–323</sup>

As mentioned in “Erythromycin,” erythromycin has been recommended in addition to ampicillin for prevention of endometritis in PPRM for pregnant patients before 34 weeks’ gestational age.<sup>177–179</sup> Azithromycin has also been studied and is an acceptable alternative and may in fact result in a lower rate of endometritis and chorioamnionitis in these patients compared with erythromycin.<sup>179</sup> Several studies suggest that the addition of azithromycin to standard perioperative antimicrobial prophylaxis may reduce the risk of infection after cesarean delivery.<sup>324–327</sup> One randomized controlled trial added azithromycin or placebo to cefuroxime for

perioperative prophylaxis for cesarean deliveries and showed that patients who received azithromycin had a lower incidence of postoperative cesarean scar defect.<sup>328</sup>

Azithromycin is no longer recommended as first line therapy for *C. trachomatis* urethritis, cervicitis, proctitis, or pharyngitis due to increasing rates of macrolide resistance.<sup>329,330</sup> Doxycycline 100 mg by mouth twice per day for 7 days is now recommended.<sup>329,330</sup> One randomized controlled trial showed that doxycycline 100 mg by mouth twice per day for 7 days resulted in a 26% (95% CI, 16% to 36%;  $P < .001$ ) greater rate of microbiologic cure than a single oral 1-g dose of azithromycin.<sup>331</sup> Another study in adolescents with urogenital *C. trachomatis* infection showed that azithromycin single oral 1-g dose was associated with a 97% rate of treatment success compared with a 100% treatment success rate with doxycycline 100 mg by mouth twice per day for 7 days.<sup>332</sup> Doxycycline 100 mg by mouth twice a day for 7 days was also showed to be superior to azithromycin 1 g as a single oral dose for rectal *C. trachomatis*.<sup>333</sup> Azithromycin is also no longer considered first-line treatment for lymphogranuloma venereum cause by *C. trachomatis* serovars L1, L2, or L3, but it is still considered an alternate treatment.<sup>334</sup> Azithromycin is no longer recommended for *M. genitalium* infection because of rising rates of resistance in North America, Europe, and Australia.<sup>335</sup> Azithromycin (1-g single oral dose) is as effective as ceftriaxone (250-mg IM dose) in treating chancroid (*Haemophilus ducreyi*) and is considered first-line treatment.<sup>336</sup> A single 2-g dose of azithromycin is not recommended as treatment for *N. gonorrhoeae* infection because of gastrointestinal side effects, expense, and emerging antibiotic resistance.<sup>336</sup>

Clarithromycin (500 mg orally twice daily) combined with amoxicillin (1 g twice daily) or metronidazole (500 mg two to three times daily) and a proton pump inhibitor for 14 days is recommended as a first-line therapy for the treatment of *H. pylori* in areas where clarithromycin resistance for this pathogen is  $<15\%$  and in patients without a history of macrolide exposure.<sup>337</sup> Failure is often associated with primary resistance by *H. pylori* strains to one of these antimicrobials<sup>338</sup> or with the emergence of secondary resistance, which occurs more frequently if only one antimicrobial agent is used or if two are used in the presence of primary resistance to one of them.<sup>339,340</sup> Sequential therapy involving 5 days of use of a proton pump inhibitor together with amoxicillin, followed by another 5 days of the proton pump inhibitor plus clarithromycin and a nitroimidazole, is associated with high cure rates (approximately 90%)<sup>338</sup> and is a suggested first-line therapy.<sup>337</sup>

Azithromycin is effective in the treatment of *Campylobacter* enteritis, but in some areas macrolide-resistant strains are common.<sup>153,341</sup> Because of its lower toxicity, azithromycin should be used instead of erythromycin when therapy is indicated and cost not a major factor. Azithromycin treatment for 5 days was found to be as effective as Page 387ciprofloxacin in the treatment of shigellosis in a randomized controlled trial and can be considered an alternative therapy for that condition, especially for children and pregnant women.<sup>158</sup>

Clarithromycin and azithromycin have been used effectively as the central drug or in combination with common antituberculous drugs in a variety of mycobacterial infections, including *M. tuberculosis*, *M. avium* complex, *M. chelonae*, *M. fortuitum*, *M. genavense*, and *M. kansasii*.<sup>342</sup> In deep-seated infections and in immunocompromised patients with these infections, combination therapy with two or more active agents is suggested to decrease the chance for the emergence of resistant strains.<sup>343,344</sup> Antimicrobial resistance testing, including testing for the presence of genes conferring macrolide resistance, is recommended to guide individual treatment regimens for nontuberculous mycobacterial infections.<sup>344</sup> Limited data suggest the effectiveness of clarithromycin in treating leprosy, for which it can serve as an alternative drug.<sup>342,345</sup>

Clarithromycin (500 mg twice daily for 14–21 days) or azithromycin (500 mg for 7–10 days) is an alternative treatment of early Lyme disease<sup>346,347</sup>; however, it is less effective than doxycycline, amoxicillin,<sup>347</sup> and cefuroxime.<sup>348</sup> In some situations, such as allergy to  $\beta$ -lactam antibiotics in pregnant women or allergy to both  $\beta$ -lactam and tetracycline antibiotics, azithromycin may be an effective alternative treatment for Lyme disease. However, present guidelines do not recommend macrolides as first-line therapy for Lyme disease.<sup>3</sup> Several studies have shown favorable outcomes and safety profiles when azithromycin has been used to treat scrub typhus compared with rifamycins, tetracyclines, and Azithromycin and Clarithromycin

Azithromycin and clarithromycin were developed to improve the qualities of erythromycin. They have better oral absorption, longer half-lives, fewer gastrointestinal side effects, and a greater antimicrobial spectrum of activity than erythromycin.

#### Derivation, Chemistry, and Preparations

Azithromycin is derived from erythromycin, differing in having a methyl-substituted nitrogen in its 15-member lactone ring (Fig. 28.2). It is therefore an azalide antibiotic. Clarithromycin, having a 14-member ring structure, is produced by modifying position C6 of the lactone ring of erythromycin to possess a methoxy group (Fig. 28.3). These changes increase the stability of these compounds in gastric acid, improving absorption by the oral route.<sup>205</sup>

Azithromycin is available in capsules for oral use as azithromycin dihydrate equivalent to 250 mg of azithromycin; in film-coated tablets of 250 mg, 500 mg, and 600 mg; as a powder for oral suspension (1 g/packet, 100 mg/5 mL, and 200 mg/5 mL); and as intravenous powder for solution (500 mg). Azithromycin is also available as a 1% ophthalmic solution for bacterial conjunctivitis.

Clarithromycin is provided in 250-mg or 500-mg film-coated tablets, extended-release 250-mg and 500-mg tablets, and as granules for oral suspension (125 mg/5 mL and 250 mg/5 mL). Clarithromycin 500 mg is also available for intravenous administration in some countries.

#### Mechanisms of Action and Resistance

Limited studies suggest that azithromycin, clarithromycin, and erythromycin bind to the same receptor on the bacterial 50S ribosomal subunit and inhibit RNA-dependent protein synthesis by the same mechanism.<sup>11,206</sup> Azithromycin has greater activity than the 14-member macrolides erythromycin and clarithromycin against gram-negative bacteria (especially for *M. catarrhalis* and *H. influenzae*) and therefore appears to better penetrate the outer envelope of those organisms.<sup>207</sup> Like other macrolides, azithromycin and clarithromycin are generally considered to be bacteriostatic agents; however, bactericidal activity is demonstrated in vitro against such species as *S. pyogenes*, *S. pneumoniae*, and *H. influenzae*.<sup>206,207</sup> In vitro activity of the newer macrolides increases with rising pH, as with erythromycin.

Mechanisms of resistance to azithromycin or clarithromycin are the same as or similar to those for erythromycin.<sup>10,13,14</sup> There is complete cross-resistance among erythromycin, azithromycin, and clarithromycin for gram-positive organisms showing resistance to erythromycin by the MLSB phenotype because the methylation mechanism already described operates for all of the 14- and 15-member macrolides.<sup>26</sup> Cross-resistance among the 14- and 15-member macrolides is also characteristic of the M phenotype and is the result of an efflux system for the drugs. Macrolide-resistant strains of *H. pylori* with point mutations in the 23S rRNA genes are increasing and are thought to be due to prior macrolide exposure.<sup>208,209</sup> This is an emerging mechanism of resistance that has been identified in other bacterial species including mycobacteria.<sup>210,211</sup> Further studies with such strains suggest that horizontal transfer of the mutated gene can occur.<sup>211,212</sup> Similar point mutations in the 23S rRNA gene associated with macrolide resistance in *M. avium* complex have been selected in patients undergoing clarithromycin or azithromycin monotherapy for disseminated infections.<sup>213</sup>

#### FIG. 28.3 Clarithromycin base.

Follow for extended description

#### Antimicrobial Activity

Although one of the most widely prescribed antibiotics in the United States, azithromycin is about two- to fourfold less active than erythromycin against *S. pneumoniae* and *S. pyogenes*.<sup>205,206,214</sup> Clarithromycin is highly active against gram-positive bacteria, being two- to fourfold more active than erythromycin against most streptococci, including *S. pneumoniae* and *S. pyogenes*, and methicillin-sensitive *S. aureus*.<sup>47,205,206</sup> Streptococci and staphylococci that are resistant to erythromycin are resistant to clarithromycin and azithromycin.<sup>51,52,206,215</sup> Selection for penicillin and multidrug-resistant *S. pneumoniae* appears to be associated with use of macrolides and cephalosporins.<sup>214</sup> (Additional information regarding the emergence of macrolide resistance in clinical isolates of *S. pneumoniae* and *S. pyogenes* was discussed earlier under "Erythromycin.") Most methicillin-resistant staphylococci are resistant to the newer macrolides.<sup>63,64</sup> An active metabolite of clarithromycin, 14-hydroxylclarithromycin, has slightly

greater activity than the parent compound against *S. aureus* and *S. pneumoniae* and is additive in vitro to the activity of clarithromycin.<sup>14,215</sup>

The activity of clarithromycin against many gram-negative bacteria is similar to that of erythromycin,<sup>14,206</sup> although it is slightly more active against *M. catarrhalis*.<sup>14</sup> Hydroxyclearithromycin also has slightly greater activity for *M. catarrhalis* and *H. influenzae* than the parent compound.<sup>215</sup> Azithromycin is more active than erythromycin or clarithromycin against gram-negative bacteria, especially against *H. influenzae* and *M. catarrhalis*, likely because of its higher penetration into the bacteria because of higher lipophilicity.<sup>205,216,217</sup> The greater activity of azithromycin against the Enterobacterales is of questionable clinical significance. However, it is of interest that prolonged incubation of *P. aeruginosa* strains with macrolides (especially azithromycin) at clinically achievable concentrations is associated with decreased viability and diminished protein synthesis.<sup>218</sup> That said, ribosomal mutations appear to confer reduced macrolide susceptibility in *P. aeruginosa* clinical isolates, which has been observed in patients with CF.<sup>219</sup> Macrolides have also been shown to inhibit the expression of quorum sensing of *P. aeruginosa*, thereby downregulating quorum sensing-regulated genes that encode virulence factors such as adhesion molecules and those that bring about cytotoxicity and inflammation.<sup>220</sup> Azithromycin is more active than erythromycin against *N. meningitidis* and *N. gonorrhoeae*; however, azithromycin resistance among *N. gonorrhoeae* isolates has increased up to 5% to 8% as of 2018 in the United States and European countries.<sup>221</sup>Page 384

Azithromycin and clarithromycin have equal or slightly better in vitro activities than erythromycin against *L. pneumophila*.<sup>42,92</sup> All three macrolides generally have good activity against *M. pneumoniae* and *C. pneumoniae*,<sup>42,90,215</sup> although macrolide-resistant *M. pneumoniae* isolates have been increasingly recognized globally, exhibiting marked geographic variability.<sup>222</sup> In the United States macrolide resistance among *M. pneumoniae* isolates hovers in the 5% to 10% range but is >50% in parts of the Western Pacific.<sup>222</sup> Azithromycin and clarithromycin have significantly greater activity than erythromycin against *C. trachomatis* and *U. urealyticum*<sup>40,90</sup> and somewhat greater activity against *Borrelia burgdorferi*.<sup>206,207</sup> The small in vitro differences in potency among these macrolides may not have any clinical significance with regard to efficacy.<sup>223</sup> Both azithromycin and clarithromycin also have significant and approximately equal activity against *Toxoplasma gondii* in tissue culture systems.<sup>224,225</sup> Azithromycin or clarithromycin plus pyrimethamine plus leucovorin have demonstrated activity, with mixed clinical outcomes, against *T. gondii* in small cohorts of patients or case reports including few patients.<sup>226</sup> However, the dose of clarithromycin used in the case report is one that should not be used in patients with human immunodeficiency virus because of an association with increased mortality at doses higher than 500 mg/day.<sup>227</sup>

The macrolides generally show little activity against *M. tuberculosis*.<sup>103,228</sup> Clarithromycin shows substantial activity against *Mycobacterium leprae* and is superior in this respect to erythromycin and azithromycin.<sup>103,215</sup> Clarithromycin and azithromycin have appreciable activities against *M. avium* complex. Clarithromycin is about fourfold more active than azithromycin against this organism in vitro<sup>229</sup> and is somewhat more active in slowing its replication in infected human macrophages.<sup>230</sup>

Azithromycin and clarithromycin demonstrate activity against *Mycobacterium abscessus* complex species; however, this organism produces an inducible 23S rRNA methylase encoded by *erm*(41), which promotes macrolide resistance. Overall, clarithromycin induces greater *erm*(41) expression leading to higher macrolide resistance compared with azithromycin, but levels of inducible clarithromycin resistance are distinguishable among the subspecies of *M. abscessus*.<sup>231</sup> *M. abscessus* subsp. *bolletii* demonstrate inducible resistance, as does *M. abscessus* subsp. *abscessus*, if the *erm*(41) sequevar T28 is present. Conversely, *erm*(41) sequevar C28 is associated with clarithromycin susceptibility among *M. abscessus* subsp. *abscessus*. *M. abscessus* subsp. *massiliense* may also harbor *erm*(41) but remain susceptible to clarithromycin.<sup>232,233</sup>

#### Clinical Pharmacology

The oral bioavailability of azithromycin after a single 250 mg and 500 mg dose is 38% and 37%, respectively.<sup>234</sup> Food increases the maximum serum concentration by approximately 50%,<sup>235</sup> but absorption is unaffected by magnesium- or aluminum-containing antacids.<sup>236</sup> The maximum



serum concentration achieved after a single 500-mg oral dose was 0.41 µg/mL within 2 to 4 hours.<sup>206</sup> The maximum concentration and 24-hour area under the curve (AUC) after a single 2-g dose of the sustained-release formulation are two and three times higher than those achieved with a total treatment dose of 1.5 g of conventional, immediate-release azithromycin.<sup>237</sup>

Protein binding of azithromycin in serum varies between 7% and 50% depending on the drug concentration.<sup>206</sup> Azithromycin is widely distributed in tissues, and for most tissues the drug concentration exceeds that in serum by 10- to 100-fold,<sup>234</sup> particularly in sputum and lung. Very high concentrations were found in alveolar macrophages and neutrophils.<sup>238</sup> The extensive tissue uptake of azithromycin has been attributed to cell uptake of this basic compound into relatively acidic lysosomes because of ionic trapping.<sup>238</sup> Very low concentrations were noted in CSF in patients without meningitis and in the aqueous humor of the uninflamed eye.<sup>239</sup> However, appreciable concentrations of azithromycin have been detected in the brains of patients undergoing resections of brain tumors after they received 500 mg orally.<sup>239</sup> The average half-life in many tissues is between 2 and 4 days,<sup>234</sup> so it is estimated that significant antibacterial activity against many pathogens persists in tissue for at least 5 days after a 5-day course of treatment.<sup>234</sup> The average terminal half-life is 68 hours, consistent with a slow release of drug from tissues, followed by elimination from the vascular compartment. About 6% of an oral dose appears as unchanged drug in the urine within 1 week of administration, and another small proportion is metabolized to inactive compounds, particularly by demethylation.<sup>234</sup> Most of the drug that is absorbed remains unmetabolized and is probably eliminated in feces by way of biliary excretion and possibly transintestinal elimination.<sup>234</sup> Biliary concentrations of azithromycin are higher than in the serum, and most of the drug in the bile is unchanged.<sup>234</sup> There are no data available on dose adjustments required with severe renal or hepatic failure.

Clarithromycin is well absorbed after oral administration, with or without food, and is approximately 55% bioavailable.<sup>206</sup> Mean peak serum concentrations at steady state with oral doses of 250 and 500 mg every 12 hours are 1 and 2 to 3 µg/mL, respectively. The elimination half-lives for those two regimens are 3 to 4 and 5 to 7 hours, respectively. Clarithromycin is appreciably metabolized in the liver by oxidation and hydrolysis to a number of compounds, accounting for a recovery of 78%. The major metabolite, 14-hydroxyclearithromycin, has antibacterial activity and accounts for 20% of the metabolites.<sup>206</sup> With the 250-mg and 500-mg oral doses given every 12 hours, about 20% to 30%, respectively, of the drug is excreted into the urine unchanged and 10% to 15% as the hydroxy metabolite. At higher doses, there is some nonlinearity of half-life, apparently because of saturation of metabolic mechanisms with a higher proportion of unchanged drug eliminated in the urine.<sup>206</sup> About 65% to 70% of the drug is bound to protein in the serum. With renal insufficiency involving creatinine clearances of less than 30 mL/min, there is a marked increase in the half-life of clarithromycin.<sup>215</sup> Dose adjustment is suggested in patients with severe renal failure, including recommendations for a 500-mg loading dose, followed by 250 mg once or twice daily depending on the type of infection.<sup>215</sup> No dosage adjustment is recommended at present for severe hepatic dysfunction.<sup>215</sup>

Clarithromycin is widely distributed and penetrates well into various tissues, generally exceeding peak maximum serum levels by severalfold.<sup>215</sup> Concentrations of clarithromycin and its 14-hydroxy metabolite in middle ear fluids of children with acute otitis media exceeded the plasma concentrations by approximately ninefold and fourfold, respectively, 12 hours after the sixth dose when the drug was given every 12 hours.<sup>240</sup> Concentrations generally exceed the MIC of most strains of middle ear pathogens, except for highly penicillin-resistant *S. pneumoniae*. Data indicate minimal penetration of clarithromycin and its 14-hydroxy metabolite into CSF in patients without meningitis.<sup>241</sup> The levels achieved were below the MIC for the usual pathogens associated with bacterial meningitis and were 1% to 2% of the corresponding plasma levels. However, CSF penetration appears to increase in the setting of infection and might achieve therapeutic levels.<sup>242</sup> Clarithromycin, like the other macrolides, penetrates well into phagocytic cells.<sup>243</sup>

#### Adverse Reactions

Adverse reactions to clarithromycin and azithromycin at the usual doses have been rare.<sup>205,206,244</sup> The most common complaints are gastrointestinal (diarrhea, nausea, abdominal pain), and discontinuance of therapy is rarely required. One retrospective study noted a small increased risk for hypertrophic pyloric stenosis in neonates exposed to azithromycin in the first 42

days of life, but this effect is more pronounced with exposure to other macrolides.<sup>245</sup> A proposed mechanism to explain this association is the capacity of macrolides to act as motilin agonists, interacting with motilin receptors and thereby directly stimulating contraction of the gastrointestinal smooth muscles, which could produce hypertrophy of the pylorus.<sup>246</sup> Azithromycin has also been shown to have a profound effect on the gastrointestinal microbiome. When compared with household contacts who did not receive antibiotics, patients who received either azithromycin or amoxicillin experienced substantial decreases in microbiota diversity that were sustained throughout the 6-month study period.<sup>247</sup> Studies in mice show long-term effects on the microbiome from macrolides, especially in young animals,<sup>248,249</sup> and studies in schoolchildren in Finland confirm the more profound and prolonged perturbation compared with  $\beta$ -lactam antibiotics.<sup>250</sup>Page 385

Acute psychosis or “mania,” akathisia, and visual hallucinations have been noted in a few patients receiving clarithromycin or azithromycin.<sup>251-255</sup> High doses of clarithromycin in animals have been associated with teratogenic effects, and therefore that drug is not recommended for use in pregnancy but definitive teratogenic effects in humans have not been substantiated.<sup>256,257</sup> Azithromycin is classified as pregnancy category B.

Abnormalities in liver function are occasionally encountered in patients treated with these drugs, and reversible cholestatic hepatitis has been reported with azithromycin.<sup>258</sup> With the high doses of these drugs used in the treatment of *M. avium* complex, tinnitus, dizziness, and reversible hearing loss have been reported.<sup>259,260</sup> Rarely, severe allergic reactions have occurred with the use of azithromycin, including erythema multiforme, Stevens-Johnson syndrome, and acute generalized exanthematous pustulosis.<sup>261-263</sup>

The risk of macrolide-associated torsades de pointes (a polymorphic ventricular tachycardia) has been associated with increasing age, female sex, chronic use, hemodialysis-dependent end stage renal failure, and concomitant drug use, especially with cisapride.<sup>264-266</sup> Chronic exposure to azithromycin in an animal model promoted intracellular sodium loading, which could lead to proarrhythmia.<sup>265</sup> A retrospective study reported that patients taking a 5-day course of azithromycin had a 2.88 increased risk of cardiovascular death compared with those taking no antibiotics, with risk being most pronounced among patients at high risk for cardiovascular disease.<sup>267,268</sup> Additionally, in an observational case series of patients who received treatment for *H. pylori* with either clarithromycin or amoxicillin, there was an increased risk of myocardial infarction, arrhythmia, and cardiac mortality during or shortly after clarithromycin therapy. However, clarithromycin use was not associated with cardiovascular risks long term.<sup>269</sup> After following patients for a year or longer in a large randomized, multicenter study, an increased number of deaths were observed among patients with underlying coronary artery disease who received 2 weeks of clarithromycin.<sup>270</sup> Conversely, a recent retrospective cohort study found that macrolide antibiotics were not associated with a higher 30-day risk of ventricular arrhythmia or all-cause mortality compared with nonmacrolide antibiotics.<sup>271</sup> Although these conflicting observations are of concern, definitive association of increased cardiovascular-related death and exposure to azithromycin or clarithromycin cannot be based on a retrospective cohort design, given residual confounding that may have influenced the outcome.<sup>272</sup> However, as a result of that study, the US Food and Drug Administration has required changes in the labeling of azithromycin warning about the risk of QT interval prolongation and cardiac arrhythmias, as well as changes in the labeling of clarithromycin warning about the increased risk of death among patients with underlying heart disease even in the setting of short courses of therapy based on the study by Jespersen and coauthors.<sup>270,273</sup> It is advised to avoid prescribing a macrolide antibiotic together with another drug known to prolong the QT interval and to exercise caution in the use of a macrolide antibiotic, even alone, in patients with underlying heart disease.

#### Drug Interactions

Azithromycin minimally inhibits CYP3A4 and therefore appears to be the safest macrolide derivative from a drug interaction perspective.<sup>274</sup>

Clarithromycin has been reported to be associated with increased concentrations of several drugs that undergo hepatic metabolism by the CYP3A subclass of the cytochrome P-450 enzyme system (see Table 28.3).<sup>275</sup> As with erythromycin, these interactions can lead to serious toxicity. It is not yet clear whether the clarithromycin interactions can occur with all the drugs that interact

with erythromycin, but a conservative clinical approach would be to consider that potential. These interactions with the hepatic metabolism of other drugs have not been documented with azithromycin, which does not appear to induce or bind and inactivate the cytochrome P-450 enzymes, probably because of its different azalide structure.<sup>149,276</sup>

As a potent P-glycoprotein inhibitor, clarithromycin coadministration with direct oral anticoagulants (DOACs) has led to twofold increases in AUCs of dabigatran and rivaroxaban.<sup>277</sup> To better understand the clinical ramifications of this interaction, a study demonstrated higher rates of hospitalization with hemorrhage associated with coprescribing clarithromycin (0.77%) versus azithromycin (0.43%) with DOACs (adjusted hazard ratio, 1.71; 95% CI, 1.20 to 2.45).<sup>278</sup> Clarithromycin may decrease the serum concentration of zidovudine by unknown mechanisms when the two drugs are taken at the same time.<sup>279</sup> However, in another study with a somewhat different design, there was no significant alteration of zidovudine bioavailability in volunteers with acquired immunodeficiency syndrome (AIDS) who took the two drugs 2 hours apart.<sup>280</sup>

Clarithromycin, like erythromycin, may occasionally lead to digoxin toxicity, possibly by diminishing the bacterial metabolism of digoxin in the gut.<sup>281</sup>

#### Uses of Clarithromycin and Azithromycin

Clarithromycin and azithromycin have several indications for use as the drug of choice and some important applications as an alternative drug (see Table 28.4).<sup>14</sup> Clarithromycin and azithromycin were as effective as other commonly used antimicrobial agents when employed in randomized multicenter trials for the treatment of pharyngitis, sinusitis, community-acquired pneumonia (including *M. pneumoniae* and *C. pneumoniae* pneumonia), and skin infections.<sup>163,205,206,282</sup> Clarithromycin and azithromycin have been considered alternatives to penicillin in treating group A  $\beta$ -hemolytic streptococcal pharyngitis in patients with penicillin allergies.<sup>283</sup> Treatment with a 10- to 14-day course of either clarithromycin (250 mg twice daily) or penicillin VK (250 mg every 6 hours) is equally effective.<sup>206,284</sup> Azithromycin administered for 5 days (500 mg on day 1, followed by 250 mg daily for 4 days) is effective in eradicating group A streptococci from the pharynx,<sup>285</sup> although shorter courses of 3-day treatment with azithromycin were associated with lower levels of bacteriologic eradication when compared with a 10-day course of penicillin V.<sup>286</sup> One study demonstrated that a 10-day course of treatment with clarithromycin (250 mg twice daily) led to more bacteriologic eradication compared with a 5-day course with azithromycin (500 mg on day 1, followed by 250 mg daily).<sup>287</sup> Clinical studies have demonstrated equal or superior tolerability and also clinical and bacterial efficacy of azithromycin (12 mg/kg, once daily for 5 days) compared with a 10-day therapy with penicillin V in the treatment of pediatric streptococcal pharyngitis.<sup>288,289</sup> However, the effectiveness of a 5-day course of azithromycin in preventing acute rheumatic fever is unknown.<sup>290</sup> Moreover, with the variable prevalence of macrolide resistance in group A streptococci, the effectiveness of these agents in treating such infections when the susceptibility of the organism is unknown should be questioned.

Azithromycin administered for 5 days or clarithromycin for 7 to 10 days for the treatment of acute otitis media in children has been effective,<sup>291,292</sup> but use is limited because of increasing macrolide resistance among *S. pneumoniae* isolates, and cases of meningitis have been reported during oral therapy.<sup>293</sup>

Guidelines available for the treatment of adult outpatients with community-acquired pneumonia include the use of a macrolide alone as a first-line agent in patients in areas where macrolide resistance in pneumococci is documented to be <25%, and the patient has no significant medical problems or risk factors for antibiotic resistant pathogens, or as combination therapy with a  $\beta$ -lactam agent in patients with comorbidities such as chronic heart, lung, or liver disease; diabetes; or immunosuppressing conditions.<sup>163,294</sup> The aforementioned guideline is now further complicated for patients with heart disease by the concern for potentiating cardiac arrhythmias with either a macrolide or a fluoroquinolone. All three macrolides have good activity against most pathogens that commonly cause community-acquired pneumonia, including many strains of *S. pneumoniae* (though resistance is emerging) and almost all strains of *M. pneumoniae*, *C. pneumoniae*, *Legionella* spp., and *M. catarrhalis* (see Table 28.1).<sup>47</sup> However, the rising prevalence, already discussed, of macrolide resistance by strains of *S. pneumoniae*, especially those that are penicillin resistant, requires caution in the use of any of the macrolides as a sole agent when that organism may be causing pneumonia. That concern would be greatest for older

adults or persons with underlying medical conditions. A study by Martinez and colleagues<sup>295</sup> suggested lower mortality benefit with combination therapy including a macrolide in community-acquired pneumonia associated with *S. pneumoniae*, including penicillin-susceptible strains, although sample size was small and the study was Page 386retrospective in design. Macrolides, with their ability to concentrate intracellularly, are effective against intracellular pathogens, including *C. pneumoniae*, *Legionella* spp., and *Coxiella burnetii* (the agent of Q fever).<sup>296</sup> Treatment with macrolides of lower respiratory tract infection with *M. pneumoniae* and *C. pneumoniae* in children shortens the clinical course of infection.<sup>297</sup> In the treatment of patients with nonpneumococcal pneumonia, erythromycin and azithromycin were equally effective (76% and 79%, respectively) in achieving clinical and radiologic resolution.<sup>223</sup> The longer half-life of azithromycin allows a shorter duration of therapy, and in one retrospective study, a 3-day course of azithromycin (daily dose, 500 mg) was as effective as a 5-day course (500 mg in a single dose the first day, followed by 250 mg daily for 4 days) in the treatment of atypical pneumonia.<sup>298</sup>

The generally recommended empirical therapy for community-acquired pneumonia requiring hospitalization includes the combination of a macrolide and a  $\beta$ -lactam,<sup>294</sup> which was associated with briefer hospital stays and a lower mortality rate than treatment with a cephalosporin alone.<sup>299,300</sup> In addition, two retrospective studies found that in cases of bacteremic pneumococcal pneumonia, dual antimicrobial therapy, including a macrolide, reduced mortality.<sup>301,302</sup>

*B. pertussis* is susceptible in vitro to azithromycin and clarithromycin. Furthermore, data from a multicenter randomized trial in North America demonstrated that azithromycin was as effective as erythromycin, better tolerated, and associated with improved compliance.<sup>303</sup> The Centers for Disease Control and Prevention (CDC) and others recommend that azithromycin should be used for all neonates younger than 1 month, and erythromycin, clarithromycin, or azithromycin are noted to be acceptable for the treatment of pertussis in persons age 1 month or older.<sup>304,305</sup> Resistance to azithromycin has been rarely reported, but limited studies are available on its clinical effectiveness.<sup>306</sup> In a study involving a small number of children with pertussis in Japan, a 5-day course of azithromycin and a 7-day course of clarithromycin were as effective in eradicating *B. pertussis* from cultures 1 week after treatment as in historical control subjects who had been treated with erythromycin for 2 weeks.<sup>152</sup>

Clarithromycin (500 mg orally twice daily) or azithromycin (500–600 mg orally once daily) in addition to ethambutol with or without rifabutin are now considered the drugs of choice in the treatment of disseminated *M. avium* complex infections in persons living with AIDS. However, recent US National Institutes of Health guidelines no longer recommend primary prophylaxis for *M. avium* complex infections in persons living with AIDS who immediately initiate antiretroviral therapy (ART).<sup>226</sup> Relatively few mutations are associated with macrolide resistance in mycobacteria, and all are in genes encoding the peptidyl transferase region of the 23S rRNA.<sup>307</sup> Therefore the use of macrolides alone is often associated with clinical relapse and the emergence of macrolide-resistant organisms.<sup>308</sup> In persons living with AIDS who are not receiving ART or who remain viremic on ART but have no current options for a fully suppressive ART regimen and have CD4 counts <50 cells/mm<sup>3</sup>, prophylaxis of disseminated *M. avium* complex infection with clarithromycin (500 mg orally twice daily) or azithromycin (1200 mg orally once weekly) is effective.<sup>226,309,310</sup> In patients who develop disseminated infection despite prophylaxis, macrolide-resistant isolates are frequently found.<sup>309</sup>

While some studies have shown that mass administration of a single weight-based dose of azithromycin every 6 months improved childhood mortality in sub-Saharan Africa, such approaches require further validation and careful attention to potential adverse effects of mass antibacterial use.<sup>311</sup> The MORDOR I trial randomized children aged 1 to 59 months to receive biennial single doses of azithromycin or placebo, and found that the groups that received azithromycin resulted in approximately a third fewer deaths due to meningitis and dysentery, and a fifth fewer deaths due to malaria and pneumonia.<sup>311</sup> The MORDOR II trial showed that this benefit to childhood mortality persisted after 3 years of biennial azithromycin administration.<sup>312</sup> In a recent study of azithromycin mass administration, childhood mortality was incidentally found to be 13.5% lower ( $P < .001$ ) among children in communities who received azithromycin than in those who received placebo.<sup>313</sup> A single oral dose of azithromycin (1-g single dose oral dose for adults or 20 mg/kg for children) is a highly effective treatment for trachoma caused by *C.*

trachomatis, and studies have been performed assessing the value of mass administration on childhood trachoma rates as well.<sup>314,315</sup> As a result, the WHO recommends mass azithromycin administration campaigns and hygiene interventions in areas with high rates of trachoma in children.<sup>316,317</sup> Additionally, biennial mass administration of azithromycin has been shown to decrease rates of yaws (*Treponema pallidum* subsp. *pertenue*) in children in yaws-endemic areas.<sup>318,319</sup> Biennial mass administration of azithromycin has been shown to have a low incidence of side effects, but concerns exist about the effect of mass antibiotic exposure on rates of antimicrobial resistance and effects on the human microbiome in developing nations.<sup>315,320-323</sup>

As mentioned in “Erythromycin,” erythromycin has been recommended in addition to ampicillin for prevention of endometritis in PPROM for pregnant patients before 34 weeks’ gestational age.<sup>177-179</sup> Azithromycin has also been studied and is an acceptable alternative and may in fact result in a lower rate of endometritis and chorioamnionitis in these patients compared with erythromycin.<sup>179</sup> Several studies suggest that the addition of azithromycin to standard perioperative antimicrobial prophylaxis may reduce the risk of infection after cesarean delivery.<sup>324-327</sup> One randomized controlled trial added azithromycin or placebo to cefuroxime for perioperative prophylaxis for cesarean deliveries and showed that patients who received azithromycin had a lower incidence of postoperative cesarean scar defect.<sup>328</sup>

Azithromycin is no longer recommended as first line therapy for *C. trachomatis* urethritis, cervicitis, proctitis, or pharyngitis due to increasing rates of macrolide resistance.<sup>329,330</sup> Doxycycline 100 mg by mouth twice per day for 7 days is now recommended.<sup>329,330</sup> One randomized controlled trial showed that doxycycline 100 mg by mouth twice per day for 7 days resulted in a 26% (95% CI, 16% to 36%;  $P < .001$ ) greater rate of microbiologic cure than a single oral 1-g dose of azithromycin.<sup>331</sup> Another study in adolescents with urogenital *C. trachomatis* infection showed that azithromycin single oral 1-g dose was associated with a 97% rate of treatment success compared with a 100% treatment success rate with doxycycline 100 mg by mouth twice per day for 7 days.<sup>332</sup> Doxycycline 100 mg by mouth twice a day for 7 days was also showed to be superior to azithromycin 1 g as a single oral dose for rectal *C. trachomatis*.<sup>333</sup> Azithromycin is also no longer considered first-line treatment for lymphogranuloma venereum cause by *C. trachomatis* serovars L1, L2, or L3, but it is still considered an alternate treatment.<sup>334</sup> Azithromycin is no longer recommended for *M. genitalium* infection because of rising rates of resistance in North America, Europe, and Australia.<sup>335</sup> Azithromycin (1-g single oral dose) is as effective as ceftriaxone (250-mg IM dose) in treating chancroid (*Haemophilus ducreyi*) and is considered first-line treatment.<sup>336</sup> A single 2-g dose of azithromycin is not recommended as treatment for *N. gonorrhoeae* infection because of gastrointestinal side effects, expense, and emerging antibiotic resistance.<sup>336</sup>

Clarithromycin (500 mg orally twice daily) combined with amoxicillin (1 g twice daily) or metronidazole (500 mg two to three times daily) and a proton pump inhibitor for 14 days is recommended as a first-line therapy for the treatment of *H. pylori* in areas where clarithromycin resistance for this pathogen is <15% and in patients without a history of macrolide exposure.<sup>337</sup> Failure is often associated with primary resistance by *H. pylori* strains to one of these antimicrobials<sup>338</sup> or with the emergence of secondary resistance, which occurs more frequently if only one antimicrobial agent is used or if two are used in the presence of primary resistance to one of them.<sup>339,340</sup> Sequential therapy involving 5 days of use of a proton pump inhibitor together with amoxicillin, followed by another 5 days of the proton pump inhibitor plus clarithromycin and a nitroimidazole, is associated with high cure rates (approximately 90%)<sup>338</sup> and is a suggested first-line therapy.<sup>337</sup>

Azithromycin is effective in the treatment of *Campylobacter* enteritis, but in some areas macrolide-resistant strains are common.<sup>153,341</sup> Because of its lower toxicity, azithromycin should be used instead of erythromycin when therapy is indicated and cost not a major factor. Azithromycin treatment for 5 days was found to be as effective as Page 387ciprofloxacin in the treatment of shigellosis in a randomized controlled trial and can be considered an alternative therapy for that condition, especially for children and pregnant women.<sup>158</sup>

Clarithromycin and azithromycin have been used effectively as the central drug or in combination with common antituberculous drugs in a variety of mycobacterial infections, including *M.*

tuberculosis, *M. avium* complex, *M. chelonae*, *M. fortuitum*, *M. genavense*, and *M. kansasii*.<sup>342</sup> In deep-seated infections and in immunocompromised patients with these infections, combination therapy with two or more active agents is suggested to decrease the chance for the emergence of resistant strains.<sup>343,344</sup> Antimicrobial resistance testing, including testing for the presence of genes conferring macrolide resistance, is recommended to guide individual treatment regimens for nontuberculous mycobacterial infections.<sup>344</sup> Limited data suggest the effectiveness of clarithromycin in treating leprosy, for which it can serve as an alternative drug.<sup>342,345</sup>

Clarithromycin (500 mg twice daily for 14–21 days) or azithromycin (500 mg for 7–10 days) is an alternative treatment of early Lyme disease<sup>346,347</sup>; however, it is less effective than doxycycline, amoxicillin,<sup>347</sup> and cefuroxime.<sup>348</sup> In some situations, such as allergy to  $\beta$ -lactam antibiotics in pregnant women or allergy to both  $\beta$ -lactam and tetracycline antibiotics, azithromycin may be an effective alternative treatment for Lyme disease. However, present guidelines do not recommend macrolides as first-line therapy for Lyme disease.<sup>348</sup> Several studies have shown favorable outcomes and safety profiles when azithromycin has been used to treat scrub typhus compared with rifamycins, tetracyclines, and fluoroquinolones.<sup>349–352</sup>

Azithromycin (when given with atovaquone) is as effective as clindamycin and quinine for the treatment of *Babesia microti* infections,<sup>353</sup> a frequent coinfecting agent with *B. burgdorferi*.<sup>161</sup> Atovaquone plus azithromycin is now the preferred treatment of babesiosis of any severity over the previous first-line treatment, clindamycin plus quinine.<sup>354,355</sup>

Azithromycin has been shown to have antimalarial activity similar to that of doxycycline in animal models.<sup>356</sup> Two human volunteer studies suggested that azithromycin has potential in the prevention of chloroquine-resistant *Plasmodium falciparum* infection.<sup>356,357</sup> Two field trials on the use of azithromycin as a single agent for malaria prophylaxis demonstrated only moderate efficacy (70%–90%) for *P. falciparum* compared with high efficacy for *Plasmodium vivax*.<sup>356,358</sup> Azithromycin in combination with other antimalarial agents, such as chloroquine, quinine, or sulfadoxine-pyrimethamine in pregnant women, produces synergistic interactions in vitro and may be effective for the treatment of uncomplicated *P. falciparum* malaria in adults.<sup>359–361</sup>

There has been great interest in the potential role of chronic infections (especially those caused by *C. pneumoniae*), and more recently in the role of inflammation per se, in potentiating atherosclerosis and vascular thromboses, including coronary artery disease.<sup>362</sup> In this regard, azithromycin has been investigated because of its activity against *C. pneumoniae* and its ability to accumulate in high concentrations in atherosclerotic plaques.<sup>363</sup> One study in a small number of men who had survived myocardial infarction, had substantial *C. pneumoniae* antibody titers, and were given azithromycin (500 mg daily for 3 or 6 days) demonstrated a significant decrease in subsequent adverse cardiovascular events compared with a similar group of men who were not treated with azithromycin.<sup>364</sup> However, clinical trials have not shown a benefit of macrolide use in preventing recurrent events in patients with acute coronary syndromes.<sup>365,366</sup> Assuming a real effect of the macrolide on such cardiac events, it is uncertain whether the result is related to its antimicrobial activity, its antiinflammatory activity on atheromata or thrombogenesis (as discussed earlier in “Antimicrobial Activity” under “Erythromycin”), both of these activities, or other unknown effects. A large randomized, placebo-controlled trial of stable patients with previous myocardial infarction and immunoglobulin G antibodies to *C. pneumoniae* treated with azithromycin for 12 weeks and followed for a median of 14 months did not demonstrate a significant reduction of clinical sequelae of coronary heart disease.<sup>367</sup> A randomized, placebo-controlled trial of clarithromycin 500 mg once daily for 2 weeks in several thousand Danish patients with stable coronary artery disease followed over 3 years showed no benefit of treatment and a significant increase in cardiovascular mortality in those receiving clarithromycin (5.1% vs. 3.5%,  $P = .01$ ).<sup>270</sup> Potential reasons for the increase in cardiovascular mortality found in those who received clarithromycin include a nonspecific random effect, unknown inequalities in risk between the two randomized groups, and proarrhythmic and drug interaction effects of clarithromycin (though cardiovascular mortality did not differ during the first month of follow-up).

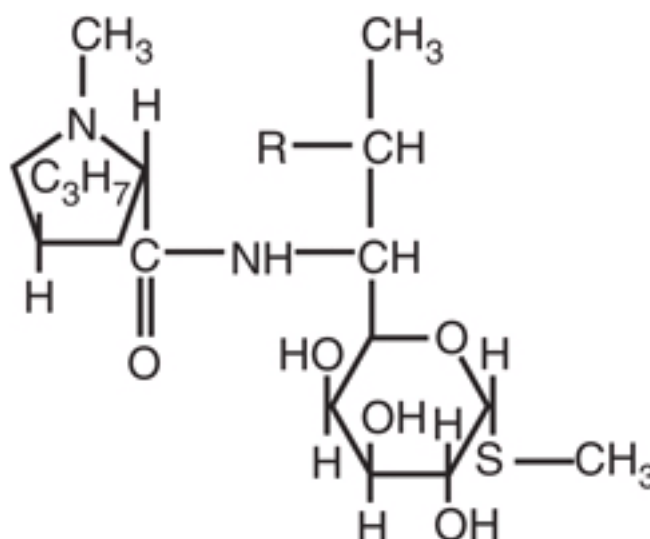
Efficacy of azithromycin has been evaluated for treatment of patients with CF. Four randomized, placebo-controlled trials comparing placebo with azithromycin administered for a period of 3 to 12 months found considerable improvement of lung function as assessed by increase of forced expiratory volume in 1 second from baseline, and reduced risk of bacterial exacerbations among

patients receiving azithromycin.<sup>368-371</sup> Azithromycin has also been shown to reduce the frequency of respiratory exacerbations of non-CF related bronchiectasis.<sup>372-374</sup> The BAT randomized controlled trial showed that daily azithromycin 250 mg by mouth for at least 6 months led to improved radiographic severity scoring of bronchiectasis, which has been shown to correlate with clinical severity of disease.<sup>375</sup> Azithromycin maintenance therapy has also been shown to reduce the incidence of respiratory exacerbations with primary ciliary dyskinesia.<sup>376</sup> The use of daily azithromycin at a dose of 250 mg once daily for 1 year was found to decrease exacerbations among patients at high risk for acute exacerbations of chronic obstructive pulmonary disease (COPD).<sup>377</sup> This effect has also been shown to be preserved over the long term, with multiple studies showing significant reductions in exacerbations of COPD and hospitalizations over several years in patients treated with maintenance azithromycin.<sup>378,379</sup> However, patients were also more likely to have hearing loss and harbor macrolide-resistant organisms.<sup>377,380</sup> Beneficial effects demonstrated in CF, COPD, and ventilator-associated pneumonia<sup>381,382</sup> are likely related not only to the bacterial growth-inhibiting effects of the macrolide but also to the general antiinflammatory activities of the macrolides,<sup>192</sup> the downregulating effects of the drug on the production of bacterial virulence factors of *P. aeruginosa* (already discussed), or both.<sup>383</sup> The use of azithromycin chronically in patients with COPD requires prior screening for QT interval prolongation and hearing loss and should not be used in those needing the administration of other drugs that are known to prolong the QT interval. Caution is also advised for such use in those with underlying heart disease.<sup>268</sup> When azithromycin is used chronically for those with COPD, it seems likely that administration three times per week rather than daily may be sufficient, given the long half-life of the drug in macrophages. Another caution comes from a recent study reporting that long-term use of azithromycin by adults with CF was associated with the development of infection with nontuberculous mycobacteria, particularly multidrug-resistant *M. abscessus*. A suggested possible mechanism was that azithromycin can block autophagosomal clearance by preventing lysosomal acidification, thereby attenuating killing of intracellular mycobacteria.<sup>192,384</sup>

The potentially useful anti-inflammatory actions of the macrolides (aside from their antimicrobial activity) were discussed earlier under “Erythromycin.”

#### Ketolides

Ketolides are a class of semisynthetic agents derived from erythromycin A that have increased acid stability and increased antibacterial potency against many bacteria resistant to macrolides. They are unable to induce the MLSB methylase type of resistance. Telithromycin (HMR 3647, or Ketek), which was approved for clinical use in the United States in April 2004 and had been in use in Europe since 2001 and in some Latin American countries, was the first member of this new class. However, its potential (although rare) for serious, even fatal, hepatotoxicity was recognized in 2006,<sup>385</sup> which severely and appropriately limited its use, and the drug was removed from the market in the United States in 2016. Solithromycin, mentioned above, is not yet FDA approved for use in the United



trans-L-4-n propylhygrinic acid

## Lincomycin and Clindamycin Derivation, Chemistry, and Preparations

Lincomycin was isolated in 1962 from an organism, *Streptomyces lincolnensis*, obtained from soil near Lincoln, Nebraska. Many of its biologic properties are similar to those of erythromycin, but it is chemically unrelated, consisting of an amino acid linked to an amino sugar (Fig. 28.4). Chemical modification provided clindamycin (7-chloro-7-deoxy-lincomycin) (see Fig. 28.4) with increased antibacterial potency and absorption after oral administration.<sup>387</sup> Because there are no therapeutic advantages for lincomycin over clindamycin, the discussion here concentrates on the latter, although both are still marketed as pharmaceuticals. Both are weak bases that are readily water soluble when provided as salts.

Although lincomycin (Lincocin) is available in the United States as the hydrochloride salt in solution (300 mg/mL) for parenteral use, it is rarely used outside of farm animals. Clindamycin is prepared as the hydrochloride salt of the base in 75-, 150-, and 300-mg capsules and of the palmitate ester as a powder for pediatric suspension (75 mg/5 mL). It is supplied as the phosphate ester for intramuscular use (150 mg/mL) and for intravenous use (150 mg/mL, 300 mg/mL, 600 mg/mL, 900 mg/mL). It is also available in topical solution, gel, lotion, foam, and pad (all at 1%) for the treatment of acne vulgaris and in a concentration of 2% in a vaginal cream and as a vaginal suppository (100 mg) for the treatment of bacterial vaginosis. Last, the lincosamide pirlimycin, is a cis-4-ethyl-L-picecolic acid amide of clindamycin that is exclusively approved for veterinary applications.<sup>388</sup>

### Mechanism of Action

The lincosamide antibiotics have, in susceptible organisms, the same or overlapping 50S ribosomal binding sites as those for the macrolides and chloramphenicol, and they may compete with these drugs for binding.<sup>10,19</sup> Protein synthesis is inhibited primarily in early chain elongation by interference with the transpeptidation reaction,<sup>388</sup> possibly by blocking of the P (peptidyl donor) site. Like the macrolides, the lincosamide antibiotics may also stimulate the dissociation of peptidyl-tRNA from ribosomes.

### Mechanisms of Resistance

There are several mechanisms of resistance to the lincosamide antibiotics. First, and most important, is the alteration in the 23S rRNA of the 50S ribosomal subunit by methylation of adenine,<sup>10,389</sup> which was discussed earlier in the "Target Site Alterations" section under "Erythromycin." It is usually plasmid mediated and provides the MLSB type of resistance, which includes that exhibited by some strains of *S. aureus*, *S. pyogenes*, and *B. fragilis* to clindamycin. This type of resistance in *S. aureus* is encoded by the *ermA* or *ermC* genes and, when of the macrolide-inducible variety, is characterized by a positive erythromycin-clindamycin "D test"—a double-disk diffusion test in which the zone of inhibition around the clindamycin disk is blunted on the side facing the erythromycin disk.<sup>390</sup> The MLSB resistance mechanism in staphylococci more frequently manifests clindamycin resistance when the strain possesses the *ermA* gene as compared with the *ermC* gene.<sup>391</sup> With that type of resistance in *S. aureus*, exposure to clindamycin (which is not an inducer) in vitro or in vivo may result in clindamycin resistance due to selection of preexisting constitutive *erm* mutants, especially when the organism is at high inoculum.<sup>391,392</sup> An additional methyltransferase enzyme, Cfr, was found to cause multidrug resistance in *S. aureus* to chloramphenicol, lincosamides, oxazolidinones, pleuromutilins, and streptogramin A through the methylation of the 23S ribosomal RNA nucleotide.<sup>393-395</sup> Second, mutations in the bacterial ribosomal RNA may confer resistance to clindamycin as demonstrated in some strains of *Mycobacterium smegmatis*.<sup>396</sup> Third, alteration in particular 50S ribosomal proteins of the receptor site confers resistance to erythromycin and often to the lincosamides<sup>25</sup>; this mechanism was previously discussed for erythromycin. Fourth, resistance is conferred through inactivation of lincomycin and clindamycin by a few isolates of staphylococci (including *S. aureus*) and *Bacteroides* spp. that possess a plasmid-mediated 3-lincomycin 4-clindamycin O-nucleotidyltransferase that catalyzes the nucleotidylation of the hydroxyl group in position 4 of clindamycin.<sup>19,397,398</sup> This adenylation of the lincosamides is associated with high-level resistance to lincomycin, but clindamycin resistance may not be detected by routine methods. The adenylation of clindamycin is associated with impaired bactericidal activity and decreased



activity at high inoculum levels. The *lnu* genes (formerly known as *linA* and *linA'*), include *lnu(A)*, *lnu(B)*, *lnu(C)*, and others, which can be chromosomal or plasmid-carried.<sup>388</sup> Plasmids carrying *lnu(A)* are found in *S. aureus* and coagulase-negative staphylococci, and *lnu(B)* has been identified in staphylococci, enterococci, and *Erysipelothrix*, whereas *lnu(C)* was first found in *Streptococcus agalactiae*.<sup>388</sup>

Finally, Enterobacterales, *Pseudomonas* spp., and *Acinetobacter* spp. are intrinsically resistant to clindamycin, apparently because of poor permeability of the cellular outer envelope to the drug.<sup>19</sup>

#### Antimicrobial Activity

In vitro susceptibilities to clindamycin are given in Table 28.5.<sup>59,399-403</sup> Clindamycin is more potent than lincomycin but similar in potency to erythromycin against staphylococci, pneumococci, *S. pyogenes*, and streptococci of the viridans group when strains are susceptible to both. However, although erythromycin demonstrates at least moderate activity against *Enterococcus*, *H. influenzae*, and *N. meningitidis*, clindamycin is generally inactive against these organisms at clinically achievable concentrations. In contrast, clindamycin has shown significantly greater activity than erythromycin against most clinically significant anaerobic bacteria, particularly *B. fragilis*.<sup>89</sup> The strong anaerobic action of clindamycin against *Bacteroides* and some clostridia is one reason why *C. difficile* can flourish in the gut of clindamycin-exposed persons. As a rule, clindamycin is not active against *Nocardia* spp., most aerobic gram-negative bacilli, or mycobacteria.

Clindamycin used to be one of the most active antibiotics available against *B. fragilis*; however, resistance is increasing to high levels (from approximately 30% to 60%).<sup>404-406</sup> With other agents available that are more reliably active (metronidazole, piperacillin-tazobactam, and carbapenems), empirical therapy with clindamycin for an anaerobic infection should be started with caution.<sup>405,407,408</sup> Estimates of resistance to clindamycin by anaerobes also include nearly 30% of clostridial species other than *C. perfringens*,<sup>406,409,410</sup> 15% to 20% of peptostreptococci,<sup>411</sup> 10% of *Fusobacterium* spp.,<sup>406,411</sup> and 30% to 50% of *Prevotella* strains.<sup>406,409-411</sup> All the Enterobacterales are resistant to clindamycin.

Clindamycin theoretically provides high activity against pneumococci and group A streptococci; however, clinical isolates showing resistance to clindamycin and erythromycin have been increasingly reported from different areas, as already discussed under "Erythromycin." The majority of strains of *S. pneumoniae* remain clindamycin-susceptible, with resistance rates approaching 10% in some areas of the United States.<sup>324</sup> Most surveys conducted on macrolide-resistant *S. pneumoniae* and *S. pyogenes* strains isolated through the 1990s found them to be of the M phenotype, which does not show cross-resistance to clindamycin.<sup>52</sup> However, a recent survey of more than 17,000 invasive *S. pyogenes* isolates in the United States noted clindamycin nonsusceptibility in 14.6%.<sup>57</sup> Other reports document resistance rates closer in the 20%–25% range.<sup>412</sup> Emergence of resistance might be expected during clindamycin treatment of macrolide-resistant *S. pyogenes*, initially showing clindamycin susceptibility, when the strain possesses the MLSB-inducible phenotype. It is therefore suggested that the double disc diffusion method (D-test) be performed on macrolide-resistant *S. pyogenes* to rule out inducible resistance to clindamycin.<sup>61</sup> A study of *S. pyogenes* isolates from the United States predicted 12.7% resistance to erythromycin and clindamycin based on whole-genome sequencing,<sup>413</sup> which matches separate data of clinical resistance found by D-testing of clinical isolates.<sup>61</sup>

The antibacterial activity of clindamycin against *S. pyogenes* displays several characteristics that have the potential to be clinically advantageous and are less prominently demonstrated or absent in the penicillin family. In contrast to  $\beta$ -lactam drugs, clindamycin acts on stationary growth-phase bacteria, suppressing the production of bacterial proteins such as *S. pyogenes* superantigens,<sup>414</sup> supported by an observational clinical study finding that clindamycin treatment improved survival of patients with streptococcal toxic shock syndrome.<sup>415</sup> However, the rise in clindamycin resistance in group A *Streptococcus* is causing concern for its ongoing effectiveness in this setting.<sup>416</sup> In vitro studies suggest that subinhibitory concentrations of clindamycin paradoxically induce virulence mechanisms in *S. pyogenes*, suggesting that therapeutic dosing is important.<sup>414</sup>

Group B *Streptococcus* (*S. agalactiae*) used to be predictably susceptible to clindamycin but that is no longer the case, with nearly 45% of strains resistant to clindamycin in some studies.<sup>417,418</sup>

Both constitutive and inducible resistance are important issues in group B *Streptococcus*, which remains on the CDC's list of antibiotic-resistant threat agents.<sup>412</sup> Clindamycin continues to have substantial activity against methicillin-susceptible *S. aureus* but against far fewer strains that are methicillin resistant.<sup>419</sup> Cross-resistance of *S. aureus* between lincomycin and clindamycin is complete. The MICs of clindamycin and erythromycin in vitro are generally similar for *S. aureus* strains that are susceptible to both agents; however, resistance can be selected in vitro by serial subculture in the presence of subinhibitory concentrations of either drug, and it occurs slowly for clindamycin and more rapidly for erythromycin.<sup>420</sup> In contrast, strains that are susceptible to clindamycin and resistant to erythromycin can be rapidly selected for clindamycin resistance by serial subculture on clindamycin. Consistent with these in vitro observations, the emergence of clindamycin-resistant *S. aureus* has been noted in clindamycin-treated patients, in particular when the organisms had demonstrated erythromycin resistance at the onset of treatment (i.e., the dissociated resistance of Garrod [inducible MLSB]).<sup>387,420</sup> The bacterial strains possessing the inducible MLSB resistance mechanism can often be detected in the laboratory with the D test, as discussed earlier. That test should be performed when clindamycin treatment is being considered for the treatment of *S. aureus* or *S. pyogenes* infections when the bacterial strain shows macrolide resistance. A positive D test suggests that emergence of resistance to clindamycin will often occur if that drug is used in treatment. A survey of nasal strains of *S. aureus* isolated in 2003 and 2004 showed that, among 237 that were methicillin susceptible, 2% were resistant to clindamycin with only a constitutive MLSB phenotype, and 22.4% were resistant when including both inducible and constitutive MLSB phenotype strains.<sup>421</sup> In the same study, among 134 strains that were methicillin resistant, 33.6% were resistant to clindamycin with only a constitutive MLSB phenotype, and 62.4% were resistant when including both inducible and constitutive MLSB phenotype strains. When methicillin-resistant *S. aureus* strains are of the community-associated variety, clindamycin resistance has been less common than among hospital-associated methicillin-resistant strains<sup>422</sup>; however, the proportion of the community-associated strains that are clindamycin resistant, often of the MLSB inducible phenotype, has been increasing.<sup>423</sup>

The antibacterial activity of lincomycin and clindamycin has been shown, in limited in vitro studies, to be bactericidal for *S. pneumoniae*, *S. pyogenes*, and *S. aureus*. Its killing activity is similar to that of erythromycin and therefore probably varies with the concentration, bacterial species, and inoculum. It is more slowly bactericidal for *S. aureus* than are the penicillins,<sup>424</sup> and it is inconsistently bactericidal for *B. fragilis*.<sup>425</sup> Studies in vitro have demonstrated that clindamycin is capable of suppressing the production of some *S. aureus* extracellular toxin proteins.<sup>426</sup>

Clindamycin has substantial in vitro activity against *T. gondii* in infected human fibroblasts.<sup>427</sup> Clindamycin and its three major metabolites are inhibitory against *P. falciparum*.<sup>428</sup>

## Clinical Pharmacology

Peak serum levels achieved after oral administration of clindamycin occur earlier and are at least twice as high as those of lincomycin. Absorption of clindamycin is about 90% and is slightly delayed, but not decreased, by ingestion of food, whereas that of lincomycin is markedly decreased.<sup>387</sup> Mean peak serum concentrations of clindamycin in adults after single oral doses of 150 and 300 mg occur at 1

TABLE 28.5

In vitro Susceptibilities to Clindamycin

| ORGANISM                           | MINIMAL INHIBITORY CONCENTRATION (μg/mL) |                   |
|------------------------------------|------------------------------------------|-------------------|
|                                    | Range                                    | MIC <sub>50</sub> |
| <i>Streptococcus pneumoniae</i>    | ≤0.25 to ≥128                            | ≤0.25             |
| <i>Streptococcus pyogenes</i>      | ≤0.06 to >64                             | ≤0.06             |
| Viridans-group streptococci        | ≤0.06 to >256                            | ≤0.06 to 0.12     |
| <i>Enterococcus</i>                | 0.12 to >8                               | >8                |
| <i>Staphylococcus aureus</i>       | ≤0.06 to >8                              | 0.25              |
| <i>Staphylococcus epidermidis</i>  | <0.12 to >8                              | 0.12              |
| <i>Clostridium perfringens</i>     | ≤0.06 to >4                              | 1                 |
| <i>Neisseria gonorrhoeae</i>       | 0.01 to 6.3                              | 3.1               |
| <i>Haemophilus influenzae</i>      | 0.4 to 50                                | 12.5              |
| <i>Bacteroides fragilis</i> group  | 0.03 to >128                             | 1–3               |
| <i>Bacteroides melaninogenicus</i> | ≤0.1 to 1                                | <0.1              |
| <i>Bacteroides thetaotaomicron</i> | 0.06 to >128                             | 1.7 to 4          |
| <i>Fusobacterium</i> spp.          | ≤0.008 to >16                            | 0.25              |
| <i>Peptococcus</i> spp.            | ≤0.1 to >100                             | ≤0.5              |
| <i>Peptostreptococcus</i> spp.     | ≤0.06 to >32                             | 0.125             |
| <i>Prevotella</i> spp.             | 0.03 to >128                             | 0.12              |
| <i>Mycoplasma pneumoniae</i>       | 1.6 to 3.1                               | 3.1               |

MIC<sub>50</sub>, Minimal inhibitory concentration for 50% of isolates (μg/mL).

hour and are 2.5 and 3.6 µg/mL, respectively; at 6 hours, they are 0.7 and 1.1 µg/mL, respectively. The esters clindamycin palmitate in suspension for oral use and clindamycin phosphate for parenteral use are absorbed as the inactive ester and are rapidly hydrolyzed in the blood to the active base. After intramuscular administration, which causes little pain, mean peak serum levels are reached in 3 hours and are about 6 µg/mL after a 300-mg dose and 9 µg/mL after a 600-mg dose; at 12 hours, they are 0.7 and 0.9 µg/mL, respectively.<sup>429</sup> In healthy adult volunteers, immediately after 20- to 45-minute intravenous infusions of 600, 900, or 1200 mg of clindamycin phosphate, serum levels of base are 10, 11, and 14 µg/mL, respectively. Higher levels after intravenous infusion have been reported in infected patients under treatment.<sup>430</sup> Dose regimens of intravenous clindamycin using 900 mg every 8 hours or 600 mg every 6 hours are considered acceptable.<sup>431</sup>

Limited studies have demonstrated good penetration of most tissues by the lincosamides, except for clinically insignificant entry of clindamycin into the CSF, even with meningitis.<sup>432</sup> Thus clindamycin should not be used to treat infections of the central nervous system. The concentration in bone compared with serum is particularly high.<sup>433</sup> Clindamycin administered to pregnant women readily passes the placental barrier and enters fetal blood and tissues.<sup>118</sup> Clindamycin is actively transported into polymorphonuclear leukocytes and macrophages<sup>434</sup> and is present in relatively high concentrations, compared with peak serum levels, in experimental abscesses.<sup>435</sup>

The normal half-life of clindamycin is 2.4 hours. Most of the absorbed drug is metabolized, probably by the liver, to products with variable antibacterial activity, including N-demethyl-clindamycin (more active than the parent compound) and clindamycin sulfoxide (less active), which have been detected in bile and urine but not in serum.<sup>429</sup> High bioactivity is found in bile, mostly as the N-demethyl metabolite; this represents a minor route of excretion and accounts for the activity assayed in feces after parenteral administration.<sup>429,436</sup> Clindamycin activity in feces persists for at least 5 days after 48 hours of parenteral administration and is associated with a major reduction in the population of susceptible bacteria in the colon that lasts for up to 14 days.<sup>437</sup> Clindamycin concentration in bile is markedly diminished or absent when the common bile duct is obstructed.<sup>438</sup> High clindamycin bioactivity, also mostly in the N-demethyl form, is found in the urine and persists for up to 4 days after a single dose, suggesting slow release from tissues.<sup>430</sup> Accurate data on the proportion of absorbed clindamycin that is excreted in the urine are not available because of the variable activity of the metabolites and their unknown proportions in urine.

The half-life of clindamycin is increased from 2.4 to about 6 hours in patients with severe renal failure, and peak blood levels after parenteral administration are about twice those in healthy people.<sup>439</sup> However, dose modifications are not recommended. Some prolongation of clindamycin activity in serum is noted in patients with severe liver disease.<sup>440</sup> Appreciable dose modification could be considered when a patient exhibits concomitant severe renal and hepatic disease. Neither hemodialysis nor peritoneal dialysis removes significant amounts of clindamycin.

#### Adverse Reactions

In a report from the CDC of emergency department visits in 2004 to 2006 for antibiotic-associated adverse events, those associated with clindamycin use were among the highest, with 18.5 emergency department visits per 10,000 outpatient prescription visits.<sup>441</sup> Cutaneous reactions are common adverse responses to clindamycin, including delayed maculopapular eruptions, urticaria, erythema multiforme, fixed drug eruptions, drug rash with eosinophilia and systemic symptoms, Stevens-Johnson syndrome, toxic epidermal necrolysis, and acute generalized exanthematous pustulosis.<sup>442</sup> Recent data from China suggest a genetic predisposition to cutaneous adverse events in persons with an HLA-B\*51:01 genotype.<sup>442</sup>

Diarrhea occurs in up to 20% of clindamycin-treated patients and is more common with oral administration. However, the major toxicity of lincomycin and clindamycin that now appreciably limits their use is the occurrence of pseudomembranous colitis caused by toxins secreted by *C. difficile* that overgrows in the presence of these antibiotics.<sup>443</sup> In the early reports of that infection, it was noted in 0.01% to 10% of clindamycin-treated patients.<sup>444</sup> The infection may occur in association with the administration of other antibiotics or occasionally without a history of recent antibiotic use, and it has become more frequent and severe.<sup>445</sup> Antibiotics other than

clindamycin—particularly cephalosporins and fluoroquinolones, which have less potency than clindamycin to change the normal balance of intestinal microbiota and thereby allow the overgrowth of *C. difficile*—have become more frequently involved as inducers of the infection because they are now used more frequently than clindamycin.<sup>126,446</sup> *C. difficile* infection has been reported after the use of clindamycin vaginal cream in a patient being treated for bacterial vaginosis, suggesting systemic absorption via this route of administration.<sup>447</sup>

Minor reversible elevation of transaminase levels, unassociated with other evidence of liver abnormality, has commonly been observed in patients receiving clindamycin, especially by the parenteral route. Some of these may have been false-positive reactions associated with colorimetric rather than specific enzymatic measurements.<sup>387</sup> However, rare cases of frank hepatotoxicity, including jaundice associated with hepatocellular damage, have been observed.<sup>448</sup>

Isolated cases of reversible neutropenia, thrombocytopenia, and agranulocytosis associated with lincomycin or clindamycin therapy have been reported; their relation to the administration of the antibiotic was uncertain.

Hypotension and electrocardiographic changes have occasionally been reported. Cardiopulmonary arrest has occurred rarely, when large intravenous doses of lincomycin were given rapidly. These effects have not been reported with clindamycin.

Local irritative reactions are rare with these drugs. Intramuscular or intravenous administration is generally well tolerated.

#### Drug Interactions

Clindamycin may block neuromuscular transmission and may enhance the action of other blocking agents.<sup>449</sup> Clindamycin may decrease the effect of cyclosporine and tacrolimus.<sup>450,451</sup> Clindamycin phosphate in solution is physically incompatible with ampicillin, diphenylhydantoin, barbiturates, aminophylline, calcium gluconate, and magnesium sulfate.

#### Uses of Clindamycin

The higher activity and absorption properties of clindamycin compared with lincomycin, along with no greater potential for toxicity, favor the former in all indications for use of these antibiotics. The lincosamides have been used in a variety of infections, often with good effect; however, appreciation of the potential for serious or even fatal toxicity with pseudomembranous colitis and the availability of safer alternative antibiotics should now generally limit the use of clindamycin to a few indications as an alternative-choice antibiotic.

Clindamycin is largely used for gram-positive or anaerobic bacterial infections of the soft tissues, or both, though increasing resistance among pathogens has limited its utility.<sup>452</sup> It is recommended to add clindamycin on to empiric broad-spectrum antimicrobial coverage for necrotizing soft tissue infections to modulate toxin production, but it should not be used as the primary agent for coverage of staphylococci, streptococci, or anaerobic organisms due to increasing resistance.<sup>453,454</sup> Clindamycin has also been shown to reduce the rate of limb amputation in necrotizing soft tissue infections.<sup>455</sup> While clindamycin has been known as an alternative choice for the treatment of infections outside of the central nervous system that are likely to involve *B. fragilis* or other penicillin-resistant anaerobic bacteria, it has lost favor as an empirical agent for polymicrobial intraabdominal infections.<sup>456,457</sup> It has also fallen out of favor for perioperative prophylaxis for many surgical sites due to increasing resistance rates.<sup>458-460</sup>

Clindamycin may offer some advantage over penicillin G in the treatment of anaerobic bronchopulmonary infections<sup>461</sup>; in addition, it may serve as an alternative in patients who are allergic to penicillin. In a prospective, randomized study of 39 patients with community-acquired putrid lung abscess, clindamycin was more effective than penicillin in the time until eradication of fever and fetid sputum and in the “overall response” to treatment.<sup>462</sup> The study involved small numbers of patients and had some flaws in the analysis<sup>463</sup>; however, the superiority of clindamycin for some patients was demonstrated and may relate to observations that 15% to 25% of anaerobic pulmonary infections involve  $\beta$ -lactamase-producing strains of *B. fragilis*, *B. melaninogenicus*, *Prevotella ruminicola*, and *B. ureolyticus*, which are resistant to penicillin.<sup>463</sup>

Another similar study also demonstrated a higher failure rate with penicillin than with clindamycin and attributed it to penicillin-resistant anaerobes.<sup>464</sup>

FIG. 28.3 Clarithromycin base.

Follow for extended description

#### Antimicrobial Activity

Although one of the most widely prescribed antibiotics in the United States, azithromycin is about two- to fourfold less active than erythromycin against *S. pneumoniae* and *S.*

*pyogenes*.<sup>205,206,214</sup> Clarithromycin is highly active against gram-positive bacteria, being two- to fourfold more active than erythromycin against most streptococci, including *S. pneumoniae* and *S. pyogenes*, and methicillin-sensitive *S. aureus*.<sup>47,205,206</sup> Streptococci and staphylococci that are resistant to erythromycin are resistant to clarithromycin and azithromycin.<sup>51,52,206,215</sup> Selection for penicillin and multidrug-resistant *S. pneumoniae* appears to be associated with use of macrolides and cephalosporins.<sup>214</sup> (Additional information regarding the emergence of macrolide resistance in clinical isolates of *S. pneumoniae* and *S. pyogenes* was discussed earlier under "Erythromycin.") Most methicillin-resistant staphylococci are resistant to the newer macrolides.<sup>63,64</sup> An active metabolite of clarithromycin, 14-hydroxyclearithromycin, has slightly greater activity than the parent compound against *S. aureus* and *S. pneumoniae* and is additive in vitro to the activity of clarithromycin.<sup>14,215</sup>

The activity of clarithromycin against many gram-negative bacteria is similar to that of erythromycin,<sup>14,206</sup> although it is slightly more active against *M. catarrhalis*. 14-Hydroxyclearithromycin also has slightly greater activity for *M. catarrhalis* and *H. influenzae* than the parent compound.<sup>215</sup> Azithromycin is more active than erythromycin or clarithromycin against gram-negative bacteria, especially against *H. influenzae* and *M. catarrhalis*, likely because of its higher penetration into the bacteria because of higher lipophilicity.<sup>205,216,217</sup> The greater activity of azithromycin against the Enterobacterales is of questionable clinical significance. However, it is of interest that prolonged incubation of *P. aeruginosa* strains with macrolides (especially azithromycin) at clinically achievable concentrations is associated with decreased viability and diminished protein synthesis.<sup>218</sup> That said, ribosomal mutations appear to confer reduced macrolide susceptibility in *P. aeruginosa* clinical isolates, which has been observed in patients with CF.<sup>219</sup> Macrolides have also been shown to inhibit the expression of quorum sensing of *P. aeruginosa*, thereby downregulating quorum sensing-regulated genes that encode virulence factors such as adhesion molecules and those that bring about cytotoxicity and inflammation.<sup>220</sup> Azithromycin is more active than erythromycin against *N. meningitidis* and *N. gonorrhoeae*; however, azithromycin resistance among *N. gonorrhoeae* isolates has increased up to 5% to 8% as of 2018 in the United States and European countries.<sup>221</sup>Page 384

Azithromycin and clarithromycin have equal or slightly better in vitro activities than erythromycin against *L. pneumophila*.<sup>42,92</sup> All three macrolides generally have good activity against *M. pneumoniae* and *C. pneumoniae*,<sup>42,90,215</sup> although macrolide-resistant *M. pneumoniae* isolates have been increasingly recognized globally, exhibiting marked geographic variability.<sup>222</sup> In the United States macrolide resistance among *M. pneumoniae* isolates hovers in the 5% to 10% range but is >50% in parts of the Western Pacific.<sup>222</sup> Azithromycin and clarithromycin have significantly greater activity than erythromycin against *C. trachomatis* and *U. urealyticum*.<sup>40,90</sup> and somewhat greater activity against *Borrelia burgdorferi*.<sup>206,207</sup> The small in vitro differences in potency among these macrolides may not have any clinical significance with regard to efficacy.<sup>223</sup> Both azithromycin and clarithromycin also have significant and approximately equal activity against *Toxoplasma gondii* in tissue culture systems.<sup>224,225</sup> Azithromycin or clarithromycin plus pyrimethamine plus leucovorin have demonstrated activity, with mixed clinical outcomes, against *T. gondii* in small cohorts of patients or case reports including few patients.<sup>226</sup> However, the dose of clarithromycin used in the case report is one that should not be used in patients with human immunodeficiency virus because of an association with increased mortality at doses higher than 500 mg/day.<sup>227</sup>

The macrolides generally show little activity against *M. tuberculosis*.<sup>103,228</sup> Clarithromycin shows substantial activity against *Mycobacterium leprae* and is superior in this respect to erythromycin and azithromycin.<sup>103,215</sup> Clarithromycin and azithromycin have appreciable activities against *M. avium* complex. Clarithromycin is about fourfold more active than azithromycin against this organism in vitro<sup>229</sup> and is somewhat more active in slowing its replication in infected human macrophages.<sup>230</sup>

Azithromycin and clarithromycin demonstrate activity against *Mycobacterium abscessus* complex species; however, this organism produces an inducible 23S rRNA methylase encoded by *erm*(41), which promotes macrolide resistance. Overall, clarithromycin induces greater *erm*(41) expression leading to higher macrolide resistance compared with azithromycin, but levels of inducible clarithromycin resistance are distinguishable among the subspecies of *M. abscessus*.<sup>231</sup> *M. abscessus* subsp. *bolletii* demonstrate inducible resistance, as does *M. abscessus* subsp. *abscessus*, if the *erm*(41) sequevar T28 is present. Conversely, *erm*(41) sequevar C28 is associated with clarithromycin susceptibility among *M. abscessus* subsp. *abscessus*. *M. abscessus* subsp. *massiliense* may also harbor *erm*(41) but remain susceptible to clarithromycin.<sup>232,233</sup>

#### Clinical Pharmacology

The oral bioavailability of azithromycin after a single 250 mg and 500 mg dose is 38% and 37%, respectively.<sup>234</sup> Food increases the maximum serum concentration by approximately 50%,<sup>235</sup> but absorption is unaffected by magnesium- or aluminum-containing antacids.<sup>236</sup> The maximum serum concentration achieved after a single 500-mg oral dose was 0.41 µg/mL within 2 to 4 hours.<sup>206</sup> The maximum concentration and 24-hour area under the curve (AUC) after a single 2-g dose of the sustained-release formulation are two and three times higher than those achieved with a total treatment dose of 1.5 g of conventional, immediate-release azithromycin.<sup>237</sup>

Protein binding of azithromycin in serum varies between 7% and 50% depending on the drug concentration.<sup>206</sup> Azithromycin is widely distributed in tissues, and for most tissues the drug concentration exceeds that in serum by 10- to 100-fold,<sup>234</sup> particularly in sputum and lung. Very high concentrations were found in alveolar macrophages and neutrophils.<sup>238</sup> The extensive tissue uptake of azithromycin has been attributed to cell uptake of this basic compound into relatively acidic lysosomes because of ionic trapping.<sup>238</sup> Very low concentrations were noted in CSF in patients without meningitis and in the aqueous humor of the uninflamed eye.<sup>239</sup> However, appreciable concentrations of azithromycin have been detected in the brains of patients undergoing resections of brain tumors after they received 500 mg orally.<sup>239</sup> The average half-life in many tissues is between 2 and 4 days,<sup>234</sup> so it is estimated that significant antibacterial activity against many pathogens persists in tissue for at least 5 days after a 5-day course of treatment.<sup>234</sup> The average terminal half-life is 68 hours, consistent with a slow release of drug from tissues, followed by elimination from the vascular compartment. About 6% of an oral dose appears as unchanged drug in the urine within 1 week of administration, and another small proportion is metabolized to inactive compounds, particularly by demethylation.<sup>234</sup> Most of the drug that is absorbed remains unmetabolized and is probably eliminated in feces by way of biliary excretion and possibly transintestinal elimination.<sup>234</sup> Biliary concentrations of azithromycin are higher than in the serum, and most of the drug in the bile is unchanged.<sup>234</sup> There are no data available on dose adjustments required with severe renal or hepatic failure.

Clarithromycin is well absorbed after oral administration, with or without food, and is approximately 55% bioavailable.<sup>206</sup> Mean peak serum concentrations at steady state with oral doses of 250 and 500 mg every 12 hours are 1 and 2 to 3 µg/mL, respectively. The elimination half-lives for those two regimens are 3 to 4 and 5 to 7 hours, respectively. Clarithromycin is appreciably metabolized in the liver by oxidation and hydrolysis to a number of compounds, accounting for a recovery of 78%. The major metabolite, 14-hydroxycarithromycin, has antibacterial activity and accounts for 20% of the metabolites.<sup>206</sup> With the 250-mg and 500-mg oral doses given every 12 hours, about 20% to 30%, respectively, of the drug is excreted into the urine unchanged and 10% to 15% as the hydroxy metabolite. At higher doses, there is some nonlinearity of half-life, apparently because of saturation of metabolic mechanisms with a higher proportion of unchanged drug eliminated in the urine.<sup>206</sup> About 65% to 70% of the drug is bound to protein in the serum. With renal insufficiency involving creatinine clearances of less than 30 mL/min, there is a marked increase in the half-life of clarithromycin.<sup>215</sup> Dose adjustment is suggested in patients with severe renal failure, including recommendations for a 500-mg loading dose, followed by 250 mg once or twice daily depending on the type of infection.<sup>215</sup> No dosage adjustment is recommended at present for severe hepatic dysfunction.<sup>215</sup>

Clarithromycin is widely distributed and penetrates well into various tissues, generally exceeding peak maximum serum levels by severalfold.<sup>215</sup> Concentrations of clarithromycin and its 14-

hydroxy metabolite in middle ear fluids of children with acute otitis media exceeded the plasma concentrations by approximately ninefold and fourfold, respectively, 12 hours after the sixth dose when the drug was given every 12 hours.<sup>240</sup> Concentrations generally exceed the MIC of most strains of middle ear pathogens, except for highly penicillin-resistant *S. pneumoniae*. Data indicate minimal penetration of clarithromycin and its 14-hydroxy metabolite into CSF in patients without meningitis.<sup>241</sup> The levels achieved were below the MIC for the usual pathogens associated with bacterial meningitis and were 1% to 2% of the corresponding plasma levels. However, CSF penetration appears to increase in the setting of infection and might achieve therapeutic levels.<sup>242</sup> Clarithromycin, like the other macrolides, penetrates well into phagocytic cells.<sup>243</sup>

#### Adverse Reactions

Adverse reactions to clarithromycin and azithromycin at the usual doses have been rare.<sup>205,206,244</sup> The most common complaints are gastrointestinal (diarrhea, nausea, abdominal pain), and discontinuance of therapy is rarely required. One retrospective study noted a small increased risk for hypertrophic pyloric stenosis in neonates exposed to azithromycin in the first 42 days of life, but this effect is more pronounced with exposure to other macrolides.<sup>245</sup> A proposed mechanism to explain this association is the capacity of macrolides to act as motilin agonists, interacting with motilin receptors and thereby directly stimulating contraction of the gastrointestinal smooth muscles, which could produce hypertrophy of the pylorus.<sup>246</sup> Azithromycin has also been shown to have a profound effect on the gastrointestinal microbiome. When compared with household contacts who did not receive antibiotics, patients who received either azithromycin or amoxicillin experienced substantial decreases in microbiota diversity that were sustained throughout the 6-month study period.<sup>247</sup> Studies in mice show long-term effects on the microbiome from macrolides, especially in young animals,<sup>248,249</sup> and studies in schoolchildren in Finland confirm the more profound and prolonged perturbation compared with  $\beta$ -lactam antibiotics.<sup>250</sup>Page 385

Acute psychosis or “mania,” akathisia, and visual hallucinations have been noted in a few patients receiving clarithromycin or azithromycin.<sup>251-255</sup> High doses of clarithromycin in animals have been associated with teratogenic effects, and therefore that drug is not recommended for use in pregnancy but definitive teratogenic effects in humans have not been substantiated.<sup>256,257</sup> Azithromycin is classified as pregnancy category B.

Abnormalities in liver function are occasionally encountered in patients treated with these drugs, and reversible cholestatic hepatitis has been reported with azithromycin.<sup>258</sup> With the high doses of these drugs used in the treatment of *M. avium* complex, tinnitus, dizziness, and reversible hearing loss have been reported.<sup>259,260</sup> Rarely, severe allergic reactions have occurred with the use of azithromycin, including erythema multiforme, Stevens-Johnson syndrome, and acute generalized exanthematous pustulosis.<sup>261-263</sup>

The risk of macrolide-associated torsades de pointes (a polymorphic ventricular tachycardia) has been associated with increasing age, female sex, chronic use, hemodialysis-dependent end stage renal failure, and concomitant drug use, especially with cisapride.<sup>264-266</sup> Chronic exposure to azithromycin in an animal model promoted intracellular sodium loading, which could lead to proarrhythmia.<sup>265</sup> A retrospective study reported that patients taking a 5-day course of azithromycin had a 2.88 increased risk of cardiovascular death compared with those taking no antibiotics, with risk being most pronounced among patients at high risk for cardiovascular disease.<sup>267,268</sup> Additionally, in an observational case series of patients who received treatment for *H. pylori* with either clarithromycin or amoxicillin, there was an increased risk of myocardial infarction, arrhythmia, and cardiac mortality during or shortly after clarithromycin therapy. However, clarithromycin use was not associated with cardiovascular risks long term.<sup>269</sup> After following patients for a year or longer in a large randomized, multicenter study, an increased number of deaths were observed among patients with underlying coronary artery disease who received 2 weeks of clarithromycin.<sup>270</sup> Conversely, a recent retrospective cohort study found that macrolide antibiotics were not associated with a higher 30-day risk of ventricular arrhythmia or all-cause mortality compared with nonmacrolide antibiotics.<sup>271</sup> Although these conflicting observations are of concern, definitive association of increased cardiovascular-related death and exposure to azithromycin or clarithromycin cannot be based on a retrospective cohort design, given residual confounding that may have influenced the outcome.<sup>272</sup> However, as a result of that

study, the US Food and Drug Administration has required changes in the labeling of azithromycin warning about the risk of QT interval prolongation and cardiac arrhythmias, as well as changes in the labeling of clarithromycin warning about the increased risk of death among patients with underlying heart disease even in the setting of short courses of therapy based on the study by Jespersen and coauthors.<sup>270,273</sup> It is advised to avoid prescribing a macrolide antibiotic together with another drug known to prolong the QT interval and to exercise caution in the use of a macrolide antibiotic, even alone, in patients with underlying heart disease.

#### Drug Interactions

Azithromycin minimally inhibits CYP3A4 and therefore appears to be the safest macrolide derivative from a drug interaction perspective.<sup>274</sup>

Clarithromycin has been reported to be associated with increased concentrations of several drugs that undergo hepatic metabolism by the CYP3A subclass of the cytochrome P-450 enzyme system (see Table 28.3).<sup>275</sup> As with erythromycin, these interactions can lead to serious toxicity. It is not yet clear whether the clarithromycin interactions can occur with all the drugs that interact with erythromycin, but a conservative clinical approach would be to consider that potential. These interactions with the hepatic metabolism of other drugs have not been documented with azithromycin, which does not appear to induce or bind and inactivate the cytochrome P-450 enzymes, probably because of its different azalide structure.<sup>149,276</sup>

As a potent P-glycoprotein inhibitor, clarithromycin coadministration with direct oral anticoagulants (DOACs) has led to twofold increases in AUCs of dabigatran and rivaroxaban.<sup>277</sup> To better understand the clinical ramifications of this interaction, a study demonstrated higher rates of hospitalization with hemorrhage associated with coprescribing clarithromycin (0.77%) versus azithromycin (0.43%) with DOACs (adjusted hazard ratio, 1.71; 95% CI, 1.20 to 2.45).<sup>278</sup> Clarithromycin may decrease the serum concentration of zidovudine by unknown mechanisms when the two drugs are taken at the same time.<sup>279</sup> However, in another study with a somewhat different design, there was no significant alteration of zidovudine bioavailability in volunteers with acquired immunodeficiency syndrome (AIDS) who took the two drugs 2 hours apart.<sup>280</sup>

Clarithromycin, like erythromycin, may occasionally lead to digoxin toxicity, possibly by diminishing the bacterial metabolism of digoxin in the gut.<sup>281</sup>

#### Uses of Clarithromycin and Azithromycin

Clarithromycin and azithromycin have several indications for use as the drug of choice and some important applications as an alternative drug (see Table 28.4).<sup>14</sup> Clarithromycin and azithromycin were as effective as other commonly used antimicrobial agents when employed in randomized multicenter trials for the treatment of pharyngitis, sinusitis, community-acquired pneumonia (including *M. pneumoniae* and *C. pneumoniae* pneumonia), and skin infections.<sup>163,205,206,282</sup> Clarithromycin and azithromycin have been considered alternatives to penicillin in treating group A  $\beta$ -hemolytic streptococcal pharyngitis in patients with penicillin allergies.<sup>283</sup> Treatment with a 10- to 14-day course of either clarithromycin (250 mg twice daily) or penicillin VK (250 mg every 6 hours) is equally effective.<sup>206,284</sup> Azithromycin administered for 5 days (500 mg on day 1, followed by 250 mg daily for 4 days) is effective in eradicating group A streptococci from the pharynx,<sup>285</sup> although shorter courses of 3-day treatment with azithromycin were associated with lower levels of bacteriologic eradication when compared with a 10-day course of penicillin V.<sup>286</sup> One study demonstrated that a 10-day course of treatment with clarithromycin (250 mg twice daily) led to more bacteriologic eradication compared with a 5-day course with azithromycin (500 mg on day 1, followed by 250 mg daily).<sup>287</sup> Clinical studies have demonstrated equal or superior tolerability and also clinical and bacterial efficacy of azithromycin (12 mg/kg, once daily for 5 days) compared with a 10-day therapy with penicillin V in the treatment of pediatric streptococcal pharyngitis.<sup>288,289</sup> However, the effectiveness of a 5-day course of azithromycin in preventing acute rheumatic fever is unknown.<sup>290</sup> Moreover, with the variable prevalence of macrolide resistance in group A streptococci, the effectiveness of these agents in treating such infections when the susceptibility of the organism is unknown should be questioned.

Azithromycin administered for 5 days or clarithromycin for 7 to 10 days for the treatment of acute otitis media in children has been effective,<sup>291,292</sup> but use is limited because of increasing



macrolide resistance among *S. pneumoniae* isolates, and cases of meningitis have been reported during oral therapy.<sup>293</sup>

Guidelines available for the treatment of adult outpatients with community-acquired pneumonia include the use of a macrolide alone as a first-line agent in patients in areas where macrolide resistance in pneumococci is documented to be <25%, and the patient has no significant medical problems or risk factors for antibiotic resistant pathogens, or as combination therapy with a  $\beta$ -lactam agent in patients with comorbidities such as chronic heart, lung, or liver disease; diabetes; or immunosuppressing conditions.<sup>163,294</sup> The aforementioned guideline is now further complicated for patients with heart disease by the concern for potentiating cardiac arrhythmias with either a macrolide or a fluoroquinolone. All three macrolides have good activity against most pathogens that commonly cause community-acquired pneumonia, including many strains of *S. pneumoniae* (though resistance is emerging) and almost all strains of *M. pneumoniae*, *C. pneumoniae*, *Legionella* spp., and *M. catarrhalis* (see Table 28.1).<sup>47</sup> However, the rising prevalence, already discussed, of macrolide resistance by strains of *S. pneumoniae*, especially those that are penicillin resistant, requires caution in the use of any of the macrolides as a sole agent when that organism may be causing pneumonia. That concern would be greatest for older adults or persons with underlying medical conditions. A study by Martinez and colleagues<sup>295</sup> suggested lower mortality benefit with combination therapy including a macrolide in community-acquired pneumonia associated with *S. pneumoniae*, including penicillin-susceptible strains, although sample size was small and the study was Page 386retrospective in design. Macrolides, with their ability to concentrate intracellularly, are effective against intracellular pathogens, including *C. pneumoniae*, *Legionella* spp., and *Coxiella burnetii* (the agent of Q fever).<sup>296</sup> Treatment with macrolides of lower respiratory tract infection with *M. pneumoniae* and *C. pneumoniae* in children shortens the clinical course of infection.<sup>297</sup> In the treatment of patients with nonpneumococcal pneumonia, erythromycin and azithromycin were equally effective (76% and 79%, respectively) in achieving clinical and radiologic resolution.<sup>223</sup> The longer half-life of azithromycin allows a shorter duration of therapy, and in one retrospective study, a 3-day course of azithromycin (daily dose, 500 mg) was as effective as a 5-day course (500 mg in a single dose the first day, followed by 250 mg daily for 4 days) in the treatment of atypical pneumonia.<sup>298</sup>

The generally recommended empirical therapy for community-acquired pneumonia requiring hospitalization includes the combination of a macrolide and a  $\beta$ -lactam,<sup>294</sup> which was associated with briefer hospital stays and a lower mortality rate than treatment with a cephalosporin alone.<sup>299,300</sup> In addition, two retrospective studies found that in cases of bacteremic pneumococcal pneumonia, dual antimicrobial therapy, including a macrolide, reduced mortality.<sup>301,302</sup>

*B. pertussis* is susceptible in vitro to azithromycin and clarithromycin. Furthermore, data from a multicenter randomized trial in North America demonstrated that azithromycin was as effective as erythromycin, better tolerated, and associated with improved compliance.<sup>303</sup> The Centers for Disease Control and Prevention (CDC) and others recommend that azithromycin should be used for all neonates younger than 1 month, and erythromycin, clarithromycin, or azithromycin are noted to be acceptable for the treatment of pertussis in persons age 1 month or older.<sup>304,305</sup> Resistance to azithromycin has been rarely reported, but limited studies are available on its clinical effectiveness.<sup>306</sup> In a study involving a small number of children with pertussis in Japan, a 5-day course of azithromycin and a 7-day course of clarithromycin were as effective in eradicating *B. pertussis* from cultures 1 week after treatment as in historical control subjects who had been treated with erythromycin for 2 weeks.<sup>152</sup>

Clarithromycin (500 mg orally twice daily) or azithromycin (500–600 mg orally once daily) in addition to ethambutol with or without rifabutin are now considered the drugs of choice in the treatment of disseminated *M. avium* complex infections in persons living with AIDS. However, recent US National Institutes of Health guidelines no longer recommend primary prophylaxis for *M. avium* complex infections in persons living with AIDS who immediately initiate antiretroviral therapy (ART).<sup>226</sup> Relatively few mutations are associated with macrolide resistance in mycobacteria, and all are in genes encoding the peptidyl transferase region of the 23S rRNA.<sup>307</sup> Therefore the use of macrolides alone is often associated with clinical relapse and the emergence of macrolide-resistant organisms.<sup>308</sup> In persons living with AIDS who are not receiving ART or who remain viremic on ART but have no current options for a fully suppressive ART regimen and

have CD4 counts <50 cells/mm<sup>3</sup>, prophylaxis of disseminated *M. avium* complex infection with clarithromycin (500 mg orally twice daily) or azithromycin (1200 mg orally once weekly) is effective.<sup>226,309,310</sup> In patients who develop disseminated infection despite prophylaxis, macrolide-resistant isolates are frequently found.<sup>309</sup>

While some studies have shown that mass administration of a single weight-based dose of azithromycin every 6 months improved childhood mortality in sub-Saharan Africa, such approaches require further validation and careful attention to potential adverse effects of mass antibacterial use.<sup>311</sup> The MORDOR I trial randomized children aged 1 to 59 months to receive biennial single doses of azithromycin or placebo, and found that the groups that received azithromycin resulted in approximately a third fewer deaths due to meningitis and dysentery, and a fifth fewer deaths due to malaria and pneumonia.<sup>311</sup> The MORDOR II trial showed that this benefit to childhood mortality persisted after 3 years of biennial azithromycin administration.<sup>312</sup> In a recent study of azithromycin mass administration, childhood mortality was incidentally found to be 13.5% lower ( $P < .001$ ) among children in communities who received azithromycin than in those who received placebo.<sup>313</sup> A single oral dose of azithromycin (1-g single dose oral dose for adults or 20 mg/kg for children) is a highly effective treatment for trachoma caused by *C. trachomatis*, and studies have been performed assessing the value of mass administration on childhood trachoma rates as well.<sup>314,315</sup> As a result, the WHO recommends mass azithromycin administration campaigns and hygiene interventions in areas with high rates of trachoma in children.<sup>316,317</sup> Additionally, biennial mass administration of azithromycin has been shown to decrease rates of yaws (*Treponema pallidum* subsp. *pertenue*) in children in yaws-endemic areas.<sup>318,319</sup> Biennial mass administration of azithromycin has been shown to have a low incidence of side effects, but concerns exist about the effect of mass antibiotic exposure on rates of antimicrobial resistance and effects on the human microbiome in developing nations.<sup>315,320-323</sup>

As mentioned in “Erythromycin,” erythromycin has been recommended in addition to ampicillin for prevention of endometritis in PPRM for pregnant patients before 34 weeks’ gestational age.<sup>177-179</sup> Azithromycin has also been studied and is an acceptable alternative and may in fact result in a lower rate of endometritis and chorioamnionitis in these patients compared with erythromycin.<sup>179</sup> Several studies suggest that the addition of azithromycin to standard perioperative antimicrobial prophylaxis may reduce the risk of infection after cesarean delivery.<sup>324-327</sup> One randomized controlled trial added azithromycin or placebo to cefuroxime for perioperative prophylaxis for cesarean deliveries and showed that patients who received azithromycin had a lower incidence of postoperative cesarean scar defect.<sup>328</sup>

Azithromycin is no longer recommended as first line therapy for *C. trachomatis* urethritis, cervicitis, proctitis, or pharyngitis due to increasing rates of macrolide resistance.<sup>329,330</sup> Doxycycline 100 mg by mouth twice per day for 7 days is now recommended.<sup>329,330</sup> One randomized controlled trial showed that doxycycline 100 mg by mouth twice per day for 7 days resulted in a 26% (95% CI, 16% to 36%;  $P < .001$ ) greater rate of microbiologic cure than a single oral 1-g dose of azithromycin.<sup>331</sup> Another study in adolescents with urogenital *C. trachomatis* infection showed that azithromycin single oral 1-g dose was associated with a 97% rate of treatment success compared with a 100% treatment success rate with doxycycline 100 mg by mouth twice per day for 7 days.<sup>332</sup> Doxycycline 100 mg by mouth twice a day for 7 days was also showed to be superior to azithromycin 1 g as a single oral dose for rectal *C. trachomatis*.<sup>333</sup> Azithromycin is also no longer considered first-line treatment for lymphogranuloma venereum cause by *C. trachomatis* serovars L1, L2, or L3, but it is still considered an alternate treatment.<sup>334</sup> Azithromycin is no longer recommended for *M. genitalium* infection because of rising rates of resistance in North America, Europe, and Australia.<sup>335</sup> Azithromycin (1-g single oral dose) is as effective as ceftriaxone (250-mg IM dose) in treating chancroid (*Haemophilus ducreyi*) and is considered first-line treatment.<sup>336</sup> A single 2-g dose of azithromycin is not recommended as treatment for *N. gonorrhoeae* infection because of gastrointestinal side effects, expense, and emerging antibiotic resistance.<sup>336</sup>

Clarithromycin (500 mg orally twice daily) combined with amoxicillin (1 g twice daily) or metronidazole (500 mg two to three times daily) and a proton pump inhibitor for 14 days is recommended as a first-line therapy for the treatment of *H. pylori* in areas where clarithromycin resistance for this pathogen is <15% and in patients without a history of macrolide exposure.<sup>337</sup>

Failure is often associated with primary resistance by *H. pylori* strains to one of these antimicrobials<sup>338</sup> or with the emergence of secondary resistance, which occurs more frequently if only one antimicrobial agent is used or if two are used in the presence of primary resistance to one of them.<sup>339,340</sup> Sequential therapy involving 5 days of use of a proton pump inhibitor together with amoxicillin, followed by another 5 days of the proton pump inhibitor plus clarithromycin and a nitroimidazole, is associated with high cure rates (approximately 90%)<sup>338</sup> and is a suggested first-line therapy.<sup>337</sup>

Azithromycin is effective in the treatment of *Campylobacter* enteritis, but in some areas macrolide-resistant strains are common.<sup>153,341</sup> Because of its lower toxicity, azithromycin should be used instead of erythromycin when therapy is indicated and cost not a major factor. Azithromycin treatment for 5 days was found to be as effective as Page 387ciprofloxacin in the treatment of shigellosis in a randomized controlled trial and can be considered an alternative therapy for that condition, especially for children and pregnant women.<sup>158</sup>

Clarithromycin and azithromycin have been used effectively as the central drug or in combination with common antituberculous drugs in a variety of mycobacterial infections, including *M. tuberculosis*, *M. avium* complex, *M. chelonae*, *M. fortuitum*, *M. genavense*, and *M. kansasii*.<sup>342</sup> In deep-seated infections and in immunocompromised patients with these infections, combination therapy with two or more active agents is suggested to decrease the chance for the emergence of resistant strains.<sup>343,344</sup> Antimicrobial resistance testing, including testing for the presence of genes conferring macrolide resistance, is recommended to guide individual treatment regimens for nontuberculous mycobacterial infections.<sup>344</sup> Limited data suggest the effectiveness of clarithromycin in treating leprosy, for which it can serve as an alternative drug.<sup>342,345</sup>

Clarithromycin (500 mg twice daily for 14–21 days) or azithromycin (500 mg for 7–10 days) is an alternative treatment of early Lyme disease<sup>346,347</sup>; however, it is less effective than doxycycline, amoxicillin,<sup>347</sup> and cefuroxime.<sup>348</sup> In some situations, such as allergy to  $\beta$ -lactam antibiotics in pregnant women or allergy to both  $\beta$ -lactam and tetracycline antibiotics, azithromycin may be an effective alternative treatment for Lyme disease. However, present guidelines do not recommend macrolides as first-line therapy for Lyme disease.<sup>348</sup> Several studies have shown favorable outcomes and safety profiles when azithromycin has been used to treat scrub typhus compared with rifamycins, tetracyclines, and fluoroquinolones.<sup>349–352</sup>

Azithromycin (when given with atovaquone) is as effective as clindamycin and quinine for the treatment of *Babesia microti* infections,<sup>353</sup> a frequent coinfecting agent with *B. burgdorferi*.<sup>161</sup> Atovaquone plus azithromycin is now the preferred treatment of babesiosis of any severity over the previous first-line treatment, clindamycin plus quinine.<sup>354,355</sup>

Azithromycin has been shown to have antimalarial activity similar to that of doxycycline in animal models.<sup>356</sup> Two human volunteer studies suggested that azithromycin has potential in the prevention of chloroquine-resistant *Plasmodium falciparum* infection.<sup>356,357</sup> Two field trials on the use of azithromycin as a single agent for malaria prophylaxis demonstrated only moderate efficacy (70%–90%) for *P. falciparum* compared with high efficacy for *Plasmodium vivax*.<sup>356,358</sup> Azithromycin in combination with other antimalarial agents, such as chloroquine, quinine, or sulfadoxine-pyrimethamine in pregnant women, produces synergistic interactions in vitro and may be effective for the treatment of uncomplicated *P. falciparum* malaria in adults.<sup>359–361</sup>

There has been great interest in the potential role of chronic infections (especially those caused by *C. pneumoniae*), and more recently in the role of inflammation per se, in potentiating atherosclerosis and vascular thromboses, including coronary artery disease.<sup>362</sup> In this regard, azithromycin has been investigated because of its activity against *C. pneumoniae* and its ability to accumulate in high concentrations in atherosclerotic plaques.<sup>363</sup> One study in a small number of men who had survived myocardial infarction, had substantial *C. pneumoniae* antibody titers, and were given azithromycin (500 mg daily for 3 or 6 days) demonstrated a significant decrease in subsequent adverse cardiovascular events compared with a similar group of men who were not treated with azithromycin.<sup>364</sup> However, clinical trials have not shown a benefit of macrolide use in preventing recurrent events in patients with acute coronary syndromes.<sup>365,366</sup> Assuming a real effect of the macrolide on such cardiac events, it is uncertain whether the result is related to its antimicrobial activity, its antiinflammatory activity on atheromata or thrombogenesis (as discussed

earlier in “Antimicrobial Activity” under “Erythromycin”), both of these activities, or other unknown effects. A large randomized, placebo-controlled trial of stable patients with previous myocardial infarction and immunoglobulin G antibodies to *C. pneumoniae* treated with azithromycin for 12 weeks and followed for a median of 14 months did not demonstrate a significant reduction of clinical sequelae of coronary heart disease.<sup>367</sup> A randomized, placebo-controlled trial of clarithromycin 500 mg once daily for 2 weeks in several thousand Danish patients with stable coronary artery disease followed over 3 years showed no benefit of treatment and a significant increase in cardiovascular mortality in those receiving clarithromycin (5.1% vs. 3.5%,  $P = .01$ ).<sup>270</sup> Potential reasons for the increase in cardiovascular mortality found in those who received clarithromycin include a nonspecific random effect, unknown inequalities in risk between the two randomized groups, and proarrhythmic and drug interaction effects of clarithromycin (though cardiovascular mortality did not differ during the first month of follow-up).

Efficacy of azithromycin has been evaluated for treatment of patients with CF. Four randomized, placebo-controlled trials comparing placebo with azithromycin administered for a period of 3 to 12 months found considerable improvement of lung function as assessed by increase of forced expiratory volume in 1 second from baseline, and reduced risk of bacterial exacerbations among patients receiving azithromycin.<sup>368-371</sup> Azithromycin has also been shown to reduce the frequency of respiratory exacerbations of non-CF related bronchiectasis.<sup>372-374</sup> The BAT randomized controlled trial showed that daily azithromycin 250 mg by mouth for at least 6 months led to improved radiographic severity scoring of bronchiectasis, which has been shown to correlate with clinical severity of disease.<sup>375</sup> Azithromycin maintenance therapy has also been shown to reduce the incidence of respiratory exacerbations with primary ciliary dyskinesia.<sup>376</sup> The use of daily azithromycin at a dose of 250 mg once daily for 1 year was found to decrease exacerbations among patients at high risk for acute exacerbations of chronic obstructive pulmonary disease (COPD).<sup>377</sup> This effect has also been shown to be preserved over the long term, with multiple studies showing significant reductions in exacerbations of COPD and hospitalizations over several years in patients treated with maintenance azithromycin.<sup>378,379</sup> However, patients were also more likely to have hearing loss and harbor macrolide-resistant organisms.<sup>377,380</sup> Beneficial effects demonstrated in CF, COPD, and ventilator-associated pneumonia<sup>381,382</sup> are likely related not only to the bacterial growth-inhibiting effects of the macrolide but also to the general antiinflammatory activities of the macrolides,<sup>192</sup> the downregulating effects of the drug on the production of bacterial virulence factors of *P. aeruginosa* (already discussed), or both.<sup>383</sup> The use of azithromycin chronically in patients with COPD requires prior screening for QT interval prolongation and hearing loss and should not be used in those needing the administration of other drugs that are known to prolong the QT interval. Caution is also advised for such use in those with underlying heart disease.<sup>268</sup> When azithromycin is used chronically for those with COPD, it seems likely that administration three times per week rather than daily may be sufficient, given the long half-life of the drug in macrophages. Another caution comes from a recent study reporting that long-term use of azithromycin by adults with CF was associated with the development of infection with nontuberculous mycobacteria, particularly multidrug-resistant *M. abscessus*. A suggested possible mechanism was that azithromycin can block autophagosomal clearance by preventing lysosomal acidification, thereby attenuating killing of intracellular mycobacteria.<sup>192,384</sup>

The potentially useful anti-inflammatory actions of the macrolides (aside from their antimicrobial activity) were discussed earlier under “Erythromycin.”

### Ketolides

Ketolides are a class of semisynthetic agents derived from erythromycin A that have increased acid stability and increased antibacterial potency against many bacteria resistant to macrolides. They are unable to induce the MLSB methylase type of resistance. Telithromycin (HMR 3647, or Ketek), which was approved for clinical use in the United States in April 2004 and had been in use in Europe since 2001 and in some Latin American countries, was the first member of this new class. However, its potential (although rare) for serious, even fatal, hepatotoxicity was recognized in 2006,<sup>385</sup> which severely and appropriately limited its use, and the drug was removed from the market in the United States in 2016. Solithromycin, mentioned above, is not yet FDA approved for use in the United States.<sup>386</sup>Page 388

FIG. 28.4 The lincosamide antibiotics. In lincomycin, R = OH; in clindamycin, R = Cl.

Follow for extended description

Lincomycin and Clindamycin

Derivation, Chemistry, and Preparations

Lincomycin was isolated in 1962 from an organism, *Streptomyces lincolnensis*, obtained from soil near Lincoln, Nebraska. Many of its biologic properties are similar to those of erythromycin, but it is chemically unrelated, consisting of an amino acid linked to an amino sugar (Fig. 28.4). Chemical modification provided clindamycin (7-chloro-7-deoxy-lincomycin) (see Fig. 28.4) with increased antibacterial potency and absorption after oral administration.<sup>387</sup> Because there are no therapeutic advantages for lincomycin over clindamycin, the discussion here concentrates on the latter, although both are still marketed as pharmaceuticals. Both are weak bases that are readily water soluble when provided as salts.

Although lincomycin (Lincocin) is available in the United States as the hydrochloride salt in solution (300 mg/mL) for parenteral use, it is rarely used outside of farm animals. Clindamycin is prepared as the hydrochloride salt of the base in 75-, 150-, and 300-mg capsules and of the palmitate ester as a powder for pediatric suspension (75 mg/5 mL). It is supplied as the phosphate ester for intramuscular use (150 mg/mL) and for intravenous use (150 mg/mL, 300 mg/mL, 600 mg/mL, 900 mg/mL). It is also available in topical solution, gel, lotion, foam, and pad (all at 1%) for the treatment of acne vulgaris and in a concentration of 2% in a vaginal cream and as a vaginal suppository (100 mg) for the treatment of bacterial vaginosis. Last, the lincosamide pirlimycin, is a cis-4-ethyl-L-picecolic acid amide of clindamycin that is exclusively approved for veterinary applications.<sup>388</sup>

#### Mechanism of Action

The lincosamide antibiotics have, in susceptible organisms, the same or overlapping 50S ribosomal binding sites as those for the macrolides and chloramphenicol, and they may compete with these drugs for binding.<sup>10,19</sup> Protein synthesis is inhibited primarily in early chain elongation by interference with the transpeptidation reaction,<sup>388</sup> possibly by blocking of the P (peptidyl donor) site. Like the macrolides, the lincosamide antibiotics may also stimulate the dissociation of peptidyl-tRNA from ribosomes.

#### Mechanisms of Resistance

There are several mechanisms of resistance to the lincosamide antibiotics. First, and most important, is the alteration in the 23S rRNA of the 50S ribosomal subunit by methylation of adenine,<sup>10,389</sup> which was discussed earlier in the “Target Site Alterations” section under “Erythromycin.” It is usually plasmid mediated and provides the MLSB type of resistance, which includes that exhibited by some strains of *S. aureus*, *S. pyogenes*, and *B. fragilis* to clindamycin. This type of resistance in *S. aureus* is encoded by the *ermA* or *ermC* genes and, when of the macrolide-inducible variety, is characterized by a positive erythromycin-clindamycin “D test”—a double-disk diffusion test in which the zone of inhibition around the clindamycin disk is blunted on the side facing the erythromycin disk.<sup>390</sup> The MLSB resistance mechanism in staphylococci more frequently manifests clindamycin resistance when the strain possesses the *ermA* gene as compared with the *ermC* gene.<sup>391</sup> With that type of resistance in *S. aureus*, exposure to clindamycin (which is not an inducer) in vitro or in vivo may result in clindamycin resistance due to selection of preexisting constitutive *erm* mutants, especially when the organism is at high inoculum.<sup>391,392</sup> An additional methyltransferase enzyme, Cfr, was found to cause multidrug resistance in *S. aureus* to chloramphenicol, lincosamides, oxazolidinones, pleuromutilins, and streptogramin A through the methylation of the 23S ribosomal RNA nucleotide.<sup>393-395</sup> Second, mutations in the bacterial ribosomal RNA may confer resistance to clindamycin as demonstrated in some strains of *Mycobacterium smegmatis*.<sup>396</sup> Third, alteration in particular 50S ribosomal proteins of the receptor site confers resistance to erythromycin and often to the lincosamides<sup>25</sup>; this mechanism was previously discussed for erythromycin. Fourth, resistance is conferred through inactivation of lincomycin and clindamycin by a few isolates of staphylococci (including *S. aureus*) and *Bacteroides* spp. that possess a plasmid-mediated 3-lincomycin 4-clindamycin O-nucleotidyltransferase that catalyzes the nucleotidylation of the hydroxyl group in position 4 of clindamycin.<sup>19,397,398</sup> This adenylation of the lincosamides is associated with high-level resistance to lincomycin, but clindamycin resistance may not be detected by routine methods. The adenylation of clindamycin is associated with impaired bactericidal activity and decreased

activity at high inoculum levels. The *lnu* genes (formerly known as *linA* and *linA'*), include *lnu(A)*, *lnu(B)*, *lnu(C)*, and others, which can be chromosomal or plasmid-carried.<sup>388</sup> Plasmids carrying *lnu(A)* are found in *S. aureus* and coagulase-negative staphylococci, and *lnu(B)* has been identified in staphylococci, enterococci, and *Erysipelothrix*, whereas *lnu(C)* was first found in *Streptococcus agalactiae*.<sup>388</sup>

Finally, Enterobacterales, *Pseudomonas* spp., and *Acinetobacter* spp. are intrinsically resistant to clindamycin, apparently because of poor permeability of the cellular outer envelope to the drug.<sup>19</sup>

#### Antimicrobial Activity

In vitro susceptibilities to clindamycin are given in Table 28.5.<sup>59,399-403</sup> Clindamycin is more potent than lincomycin but similar in potency to erythromycin against staphylococci, pneumococci, *S. pyogenes*, and streptococci of the viridans group when strains are susceptible to both. However, although erythromycin demonstrates at least moderate activity against *Enterococcus*, *H. influenzae*, and *N. meningitidis*, clindamycin is generally inactive against these organisms at clinically achievable concentrations. In contrast, clindamycin has shown significantly greater activity than erythromycin against most clinically significant anaerobic bacteria, particularly *B. fragilis*.<sup>89</sup> The strong anaerobic action of clindamycin against *Bacteroides* and some clostridia is one reason why *C. difficile* can flourish in the gut of clindamycin-exposed persons. As a rule, clindamycin is not active against *Nocardia* spp., most aerobic gram-negative bacilli, or mycobacteria.

Clindamycin used to be one of the most active antibiotics available against *B. fragilis*; however, resistance is increasing to high levels (from approximately 30% to 60%).<sup>404-406</sup> With other agents available that are more reliably active (metronidazole, piperacillin-tazobactam, and carbapenems), empirical therapy with clindamycin for an anaerobic infection should be started with caution.<sup>405,407,408</sup> Estimates of resistance to clindamycin by anaerobes also include nearly 30% of clostridial species other than *C. perfringens*,<sup>406,409,410</sup> 15% to 20% of peptostreptococci,<sup>411</sup> 10% of *Fusobacterium* spp.,<sup>406,411</sup> and 30% to 50% of *Prevotella* strains.<sup>406,409-411</sup> All the Enterobacterales are resistant to clindamycin.

Clindamycin theoretically provides high activity against pneumococci and group A streptococci; however, clinical isolates showing resistance to clindamycin and erythromycin have been increasingly reported from different areas, as already discussed under "Erythromycin." The majority of strains of *S. pneumoniae* remain clindamycin-susceptible, with resistance rates approaching 10% in some areas of the United States.<sup>324</sup> Most surveys conducted on macrolide-resistant *S. pneumoniae* and *S. pyogenes* strains isolated through the 1990s found them to be of the M phenotype, which does not show cross-resistance to clindamycin.<sup>52</sup> However, a recent survey of more than 17,000 invasive *S. pyogenes* isolates in the United States noted clindamycin nonsusceptibility in 14.6%.<sup>57</sup> Other reports document resistance rates closer in the 20%–25% range.<sup>412</sup> Emergence of resistance might be expected during clindamycin treatment of macrolide-resistant *S. pyogenes*, initially showing clindamycin susceptibility, when the strain possesses the MLSB-inducible phenotype. It is therefore suggested that the double disc diffusion method (D-test) be performed on macrolide-resistant *S. pyogenes* to rule out inducible resistance to clindamycin.<sup>61</sup> A study of *S. pyogenes* isolates from the United States predicted 12.7% resistance to erythromycin and clindamycin based on whole-genome sequencing,<sup>413</sup> which matches separate data of clinical resistance found by D-testing of clinical isolates.<sup>61</sup>

TABLE 28.5

| In vitro Susceptibilities to Clindamycin |               |                   |
|------------------------------------------|---------------|-------------------|
| MINIMAL INHIBITORY CONCENTRATION (μg/mL) |               |                   |
| ORGANISM                                 | Range         | MIC <sub>50</sub> |
| <i>Streptococcus pneumoniae</i>          | ≤0.25 to ≥128 | ≤0.25             |
| <i>Streptococcus pyogenes</i>            | ≤0.06 to >64  | ≤0.06             |
| Viridans-group streptococci              | ≤0.06 to >256 | ≤0.06 to 0.12     |
| <i>Enterococcus</i>                      | 0.12 to >8    | >8                |
| <i>Staphylococcus aureus</i>             | ≤0.06 to >8   | 0.25              |
| <i>Staphylococcus epidermidis</i>        | <0.12 to >8   | 0.12              |
| <i>Clostridium perfringens</i>           | ≤0.06 to >4   | 1                 |

|                                     |               |          |
|-------------------------------------|---------------|----------|
| <i>Neisseria gonorrhoeae</i>        | 0.01 to 6.3   | 3.1      |
| <i>Haemophilus influenzae</i>       | 0.4 to 50     | 12.5     |
| <i>Bacteroides fragilis</i> group   | 0.03 to >128  | 1–3      |
| <i>Bacteroides melaninogenicus</i>  | ≤0.1 to 1     | <0.1     |
| <i>Bacteroides thetaiotaomicron</i> | 0.06 to >128  | 1.7 to 4 |
| <i>Fusobacterium</i> spp.           | ≤0.008 to >16 | 0.25     |
| <i>Peptococcus</i> spp.             | ≤0.1 to >100  | ≤0.5     |
| <i>Peptostreptococcus</i> spp.      | ≤0.06 to >32  | 0.125    |
| <i>Prevotella</i> spp.              | 0.03 to >128  | 0.12     |
| <i>Mycoplasma pneumoniae</i>        | 1.6 to 3.1    | 3.1      |

MIC50, Minimal inhibitory concentration for 50% of isolates (µg/mL).

The antibacterial activity of clindamycin against *S. pyogenes* displays several characteristics that have the potential to be clinically advantageous and are less prominently demonstrated or absent in the penicillin family. In contrast to  $\beta$ -lactam drugs, clindamycin acts on stationary growth-phase bacteria, suppressing the production of bacterial proteins such as *S. pyogenes* superantigens,<sup>414</sup> supported by an observational clinical study finding that clindamycin treatment improved survival of patients with streptococcal toxic shock syndrome.<sup>415</sup> However, the rise in clindamycin resistance in group A *Streptococcus* is causing concern for its ongoing effectiveness in this setting.<sup>416</sup> In vitro studies suggest that subinhibitory concentrations of clindamycin paradoxically induce virulence mechanisms in *S. pyogenes*, suggesting that therapeutic dosing is important.<sup>414</sup>

Group B *Streptococcus* (*S. agalactiae*) used to be predictably susceptible to clindamycin but that is no longer the case, with nearly 45% of strains resistant to clindamycin in some studies.<sup>417,418</sup> Both constitutive and inducible resistance are important issues in group B *Streptococcus*, which remains on the CDC's list of antibiotic-resistant threat agents.<sup>412</sup> Clindamycin continues to have substantial activity against methicillin-susceptible *S. aureus* but against far fewer strains that are methicillin resistant.<sup>419</sup> Cross-resistance of *S. aureus* between lincomycin and clindamycin is complete. The MICs of clindamycin and erythromycin in vitro are generally similar for *S. aureus* strains that are susceptible to both agents; however, resistance can be selected in vitro by serial subculture in the presence of subinhibitory concentrations of either drug, and it occurs slowly for clindamycin and more rapidly for erythromycin.<sup>420</sup> In contrast, strains that are susceptible to clindamycin and resistant to erythromycin can be rapidly selected for clindamycin resistance by serial subculture on clindamycin. Consistent with these in vitro observations, the emergence of clindamycin-resistant *S. aureus* has been noted in clindamycin-treated patients, in particular when the organisms had demonstrated erythromycin resistance at the onset of treatment (i.e., the dissociated resistance of Garrod [inducible MLSB]).<sup>387,420</sup> The bacterial strains possessing the inducible MLSB resistance mechanism can often be detected in the laboratory with the D test, as discussed earlier. That test should be performed when clindamycin treatment is being considered for the treatment of *S. aureus* or *S. pyogenes* infections when the bacterial strain shows macrolide resistance. A positive D test suggests that emergence of resistance to clindamycin will often occur if that drug is used in treatment. A survey of nasal strains of *S. aureus* isolated in 2003 and 2004 showed that, among 237 that were methicillin susceptible, 2% were resistant to clindamycin with only a constitutive MLSB phenotype, and 22.4% were resistant when including both inducible and constitutive MLSB phenotype strains.<sup>421</sup> In the same study, among 134 strains that were methicillin resistant, 33.6% were resistant to clindamycin with only a constitutive MLSB phenotype, and 62.4% were resistant when including both inducible and constitutive MLSB phenotype strains. When methicillin-resistant *S. aureus* strains are of the community-associated variety, clindamycin resistance has been less common than among hospital-associated methicillin-resistant strains<sup>422</sup>; however, the proportion of the community-associated strains that are clindamycin resistant, often of the MLSB inducible phenotype, has been increasing.<sup>423</sup>

The antibacterial activity of lincomycin and clindamycin has been shown, in limited in vitro studies, to be bactericidal for *S. pneumoniae*, *S. pyogenes*, and *S. aureus*. Its killing activity is similar to that of erythromycin and therefore probably varies with the concentration, bacterial species, and inoculum. It is more slowly bactericidal for *S. aureus* than are the penicillins,<sup>424</sup> and it is inconsistently bactericidal for *B. fragilis*.<sup>425</sup> Studies in vitro have demonstrated that clindamycin is capable of suppressing the production of some *S. aureus* extracellular toxin proteins.<sup>426</sup>

Clindamycin has substantial in vitro activity against *T. gondii* in infected human fibroblasts.<sup>427</sup> Clindamycin and its three major metabolites are inhibitory against *P. falciparum*.<sup>428</sup>

#### Clinical Pharmacology

Peak serum levels achieved after oral administration of clindamycin occur earlier and are at least twice as high as those of lincomycin. Absorption of clindamycin is about 90% and is slightly delayed, but not decreased, by ingestion of food, whereas that of lincomycin is markedly decreased.<sup>387</sup> Mean peak serum concentrations of clindamycin in adults after single oral doses of 150 and 300 mg occur at 1 hour and are 2.5 and 3.6 µg/mL, respectively; at 6 hours, they are 0.7 and 1.1 µg/mL, respectively. The esters clindamycin palmitate in suspension for oral use and clindamycin phosphate for parenteral use are absorbed as the inactive ester and are rapidly hydrolyzed in the blood to the active base. After intramuscular administration, which causes little pain, mean peak serum levels are reached in 3 hours and are about 6 µg/mL after a 300-mg dose and 9 µg/mL after a 600-mg dose; at 12 hours, they are 0.7 and 0.9 µg/mL, respectively.<sup>429</sup> In healthy adult volunteers, immediately after 20- to 45-minute intravenous infusions of 600, 900, or 1200 mg of clindamycin phosphate, serum levels of base are 10, 11, and 14 µg/mL, respectively. Higher levels after intravenous infusion have been reported in infected patients under treatment.<sup>430</sup> Dose regimens of intravenous clindamycin using 900 mg every 8 hours or 600 mg every 6 hours are considered acceptable.<sup>431</sup>

Limited studies have demonstrated good penetration of most tissues by the lincosamides, except for clinically insignificant entry of clindamycin into the CSF, even with meningitis.<sup>432</sup> Thus clindamycin should not be used to treat infections of the central nervous system. The concentration in bone compared with serum is particularly high.<sup>433</sup> Clindamycin administered to pregnant women readily passes the placental barrier and enters fetal blood and tissues.<sup>118</sup> Clindamycin is actively transported into polymorphonuclear leukocytes and macrophages<sup>434</sup> and is present in relatively high concentrations, compared with peak serum levels, in experimental abscesses.<sup>435</sup>

The normal half-life of clindamycin is 2.4 hours. Most of the absorbed drug is metabolized, probably by the liver, to products with variable antibacterial activity, including N-demethyl-clindamycin (more active than the parent compound) and clindamycin sulfoxide (less active), which have been detected in bile and urine but not in serum.<sup>429</sup> High bioactivity is found in bile, mostly as the N-demethyl metabolite; this represents a minor route of excretion and accounts for the activity assayed in feces after parenteral administration.<sup>429,436</sup> Clindamycin activity in feces persists for at least 5 days after 48 hours of parenteral administration and is associated with a major reduction in the population of susceptible bacteria in the colon that lasts for up to 14 days.<sup>437</sup> Clindamycin concentration in bile is markedly diminished or absent when the common bile duct is obstructed.<sup>438</sup> High clindamycin bioactivity, also mostly in the N-demethyl form, is found in the urine and persists for up to 4 days after a single dose, suggesting slow release from tissues.<sup>430</sup> Accurate data on the proportion of absorbed clindamycin that is excreted in the urine are not available because of the variable activity of the metabolites and their unknown proportions in urine.

The half-life of clindamycin is increased from 2.4 to about 6 hours in patients with severe renal failure, and peak blood levels after parenteral administration are about twice those in healthy people.<sup>439</sup> However, dose modifications are not recommended. Some prolongation of clindamycin activity in serum is noted in patients with severe liver disease.<sup>440</sup> Appreciable dose modification could be considered when a patient exhibits concomitant severe renal and hepatic disease. Neither hemodialysis nor peritoneal dialysis removes significant amounts of clindamycin.

#### Adverse Reactions

In a report from the CDC of emergency department visits in 2004 to 2006 for antibiotic-associated adverse events, those associated with clindamycin use were among the highest, with 18.5 emergency department visits per 10,000 outpatient prescription visits.<sup>441</sup> Cutaneous reactions are common adverse responses to clindamycin, including delayed maculopapular eruptions, urticaria, erythema multiforme, fixed drug eruptions, drug rash with eosinophilia and systemic symptoms, Stevens-Johnson syndrome, toxic epidermal necrolysis, and acute generalized exanthematous pustulosis.<sup>442</sup> Recent data from China suggest a genetic predisposition to cutaneous adverse events in persons with an HLA-B\*51:01 genotype.<sup>442</sup>



Diarrhea occurs in up to 20% of clindamycin-treated patients and is more common with oral administration. However, the major toxicity of lincomycin and clindamycin that now appreciably limits their use is the occurrence of pseudomembranous colitis caused by toxins secreted by *C. difficile* that overgrows in the presence of these antibiotics.<sup>443</sup> In the early reports of that infection, it was noted in 0.01% to 10% of clindamycin-treated patients.<sup>444</sup> The infection may occur in association with the administration of other antibiotics or occasionally without a history of recent antibiotic use, and it has become more frequent and severe.<sup>445</sup> Antibiotics other than clindamycin—particularly cephalosporins and fluoroquinolones, which have less potency than clindamycin to change the normal balance of intestinal microbiota and thereby allow the overgrowth of *C. difficile*—have become more frequently involved as inducers of the infection because they are now used more frequently than clindamycin.<sup>126,446</sup> *C. difficile* infection has been reported after the use of clindamycin vaginal cream in a patient being treated for bacterial vaginosis, suggesting systemic absorption via this route of administration.<sup>447</sup>

Minor reversible elevation of transaminase levels, unassociated with other evidence of liver abnormality, has commonly been observed in patients receiving clindamycin, especially by the parenteral route. Some of these may have been false-positive reactions associated with colorimetric rather than specific enzymatic measurements.<sup>387</sup> However, rare cases of frank hepatotoxicity, including jaundice associated with hepatocellular damage, have been observed.<sup>448</sup>

Isolated cases of reversible neutropenia, thrombocytopenia, and agranulocytosis associated with lincomycin or clindamycin therapy have been reported; their relation to the administration of the antibiotic was uncertain.

Hypotension and electrocardiographic changes have occasionally been reported. Cardiopulmonary arrest has occurred rarely, when large intravenous doses of lincomycin were given rapidly. These effects have not been reported with clindamycin.

Local irritative reactions are rare with these drugs. Intramuscular or intravenous administration is generally well tolerated.

#### Drug Interactions

Clindamycin may block neuromuscular transmission and may enhance the action of other blocking agents.<sup>449</sup> Clindamycin may decrease the effect of cyclosporine and tacrolimus.<sup>450,451</sup> Clindamycin phosphate in solution is physically incompatible with ampicillin, diphenylhydantoin, barbiturates, aminophylline, calcium gluconate, and magnesium sulfate.

#### Uses of Clindamycin

The higher activity and absorption properties of clindamycin compared with lincomycin, along with no greater potential for toxicity, favor the former in all indications for use of these antibiotics. The lincosamides have been used in a variety of infections, often with good effect; however, appreciation of the potential for serious or even fatal toxicity with pseudomembranous colitis and the availability of safer alternative antibiotics should now generally limit the use of clindamycin to a few indications as an alternative-choice antibiotic.

Clindamycin is largely used for gram-positive or anaerobic bacterial infections of the soft tissues, or both, though increasing resistance among pathogens has limited its utility.<sup>452</sup> It is recommended to add clindamycin on to empiric broad-spectrum antimicrobial coverage for necrotizing soft tissue infections to modulate toxin production, but it should not be used as the primary agent for coverage of staphylococci, streptococci, or anaerobic organisms due to increasing resistance.<sup>453,454</sup> Clindamycin has also been shown to reduce the rate of limb amputation in necrotizing soft tissue infections.<sup>455</sup> While clindamycin has been known as an alternative choice for the treatment of infections outside of the central nervous system that are likely to involve *B. fragilis* or other penicillin-resistant anaerobic bacteria, it has lost favor as an empirical agent for polymicrobial intraabdominal infections.<sup>456,457</sup> It has also fallen out of favor for perioperative prophylaxis for many surgical sites due to increasing resistance rates.<sup>458-460</sup>

Clindamycin may offer some advantage over penicillin G in the treatment of anaerobic bronchopulmonary infections<sup>461</sup>; in addition, it may serve as an alternative in patients who are allergic to penicillin. In a prospective, randomized study of 39 patients with community-acquired putrid lung abscess, clindamycin was more effective than penicillin in the time until eradication of fever and fetid sputum and in the “overall response” to treatment.<sup>462</sup> The study involved small numbers of patients and had some flaws in the analysis<sup>463</sup>; however, the superiority of clindamycin for some patients was demonstrated and may relate to observations that 15% to 25% of anaerobic pulmonary infections involve  $\beta$ -lactamase-producing strains of *B. fragilis*, *B. melaninogenicus*, *Prevotella ruminicola*, and *B. ureolyticus*, which are resistant to penicillin.<sup>463</sup> Another similar study also demonstrated a higher failure rate with penicillin than with clindamycin and attributed it to penicillin-resistant anaerobes.<sup>464</sup> That study was problematic in that penicillin oral therapy was used to complete the course of treatment of some patients in the penicillin group. Nevertheless, clindamycin may be preferable for treatment of this condition,<sup>465</sup> particularly in seriously ill patients and in those who have responded poorly to penicillin.

Clindamycin is useful as an alternative to penicillin in treatment of infections with *C. perfringens*, which is generally susceptible to this drug.<sup>406,409</sup> Clindamycin was more effective than penicillin in reducing mortality in a mouse model of *C. perfringens* gas gangrene and in vitro in suppressing the  $\alpha$ -toxin activity produced by that organism.<sup>466,467</sup>

Depending upon susceptibilities, clindamycin can be useful in the treatment of staphylococcal infections, especially when therapy is appropriate by the oral route for a methicillin-resistant strain and when there is a history of  $\beta$ -lactam allergy for a methicillin-susceptible strain. However, the more limited bactericidal rate with clindamycin for staphylococci compared with that of the  $\beta$ -lactams, the real potential for the emergence of clindamycin-resistant strains Page 391 in treated patients, and the substantial potency of clindamycin for sometimes inducing *C. difficile* colitis are disadvantages. The problem of emergence of clindamycin resistance, as discussed earlier and noted especially but not only with erythromycin-resistant strains, appreciably limits its effectiveness as therapy for deep-seated, high-bacterial density staphylococcal infections, particularly endocarditis.<sup>468</sup> The use of the D test (discussed earlier) is recommended to detect the likelihood of emergence of resistance to clindamycin when a strain is resistant to erythromycin. In general, vancomycin, daptomycin (not for pneumonia), or linezolid for methicillin-resistant strains or  $\beta$ -lactams for methicillin-susceptible strains are better choices for treatment of staphylococcal infections. Several studies have shown, however, that combination therapy with clindamycin and rifampin is a generally safe, effective, and well-tolerated regimen for clindamycin-susceptible *Staphylococcus* bone and joint infections.<sup>469-471</sup> Although high concentrations of clindamycin are achieved in bone, an advantage of clindamycin for the treatment of osteomyelitis over non-clindamycin-based regimens has not been established.<sup>433</sup> A combination of clindamycin and trimethoprim-sulfamethoxazole with or without rifampin was also found to be a potential option for early oral switch therapy for *S. aureus* endocarditis.<sup>472</sup>

The topical clindamycin/benzoyl peroxide gel is more effective in the treatment of acne vulgaris than topical clindamycin alone.<sup>473</sup> However, pseudomembranous colitis associated with the use of topical clindamycin has been reported.<sup>474</sup> In the treatment of bacterial vaginosis, clindamycin vaginal cream (2% for 3–7 days) appears to be similar in efficacy and in the incidence of side effects to oral metronidazole (both used for 7 days).<sup>475</sup> Alternative dosing regimens, including oral clindamycin (300 mg twice daily for 7 days) or intravaginal clindamycin ovules (100 mg once daily for 3 days), can also be used for this condition. A case report of *C. difficile* colitis associated with clindamycin vaginal cream has already been noted, but this appears to be exceedingly rare.<sup>447</sup> Bacterial vaginosis is a risk factor for preterm delivery, and studies of clindamycin treatment for bacterial vaginosis have shown mixed results with respect to preventing preterm labor. Thus opinions differ as to the capacity of clindamycin to prevent preterm birth in women being treated for bacterial vaginosis.<sup>476,477</sup>

Clindamycin is an effective alternative regimen in treating animals experimentally infected with toxoplasma and when combined with pyrimethamine (and leucovorin) in treating patients with toxoplasmosis of the central nervous system.<sup>478</sup> This is a preferred alternative regimen for patients with cerebral toxoplasmosis who cannot tolerate sulfadiazine or do not respond to first-line therapy.<sup>479</sup> This combination, however, does not prevent *Pneumocystis jirovecii* infection; therefore additional prophylaxis must be administered when it is used.

Clindamycin in combination with primaquine is an effective and well-tolerated regimen for the treatment of mild and moderately severe *P. jirovecii* pneumonia in patients with AIDS. In a comparative trial, this combination showed an efficacy similar to that of trimethoprim with sulfamethoxazole or trimethoprim with dapsons.<sup>480</sup> An analysis of 82 AIDS cases with *P. jirovecii* pneumonia from Copenhagen, London, and Milan, which had previously been reported in observational studies and had initial treatment regimens that were switched to other regimens after at least 5 days because of poor clinical response, showed that in 22 cases in which the switch was made to clindamycin with primaquine, the response rate was 73% and was comparable with that for trimethoprim-sulfamethoxazole (68%) and superior to that for intravenous pentamidine (44%).<sup>481</sup>

Clindamycin in combination with quinine is effective in the treatment of falciparum malaria.<sup>482</sup> That regimen used for 4 days was found to be superior to quinine used alone for 7 days in a randomized trial in Gabonese children with severe disease.<sup>483</sup> The same combination has also been reported to be useful in the treatment of babesiosis.<sup>353</sup> A randomized controlled trial involving 100 Gabonese children with uncomplicated *P. falciparum* malaria showed that a 3-day oral regimen of artesunate with clindamycin was highly successful and at least comparable in efficacy and safety with a 3-day oral regimen of quinine with clindamycin.<sup>484</sup>

The coexistence of  $\beta$ -lactamase-producing *S. aureus* or *Bacteroides* spp. and group A streptococci may be associated with the failure of penicillin to eradicate the latter, resulting in recurrent tonsillitis. Limited evidence suggests that recurrence rates may be lowered when clindamycin is used.<sup>485</sup> In another study, patients with group A streptococcal pharyngitis that recurred after 10 days of treatment with phenoxymethyl penicillin were randomly allocated to re-treatment for 10 days with phenoxymethyl penicillin or clindamycin.<sup>486</sup> Group A streptococci were not recovered from any of the patients receiving clindamycin, but the same (or similar) strain was cultured from 64% of those re-treated with the penicillin. However, widespread use of clindamycin for this common problem can increase the risk of *C. difficile* colitis, as well as selection for clindamycin-resistant strains of group A streptococci.

Although penicillin has been the traditional drug of choice for the treatment of group A streptococcal infections, on the basis of data, already discussed, clindamycin has been advocated as a combination agent with penicillin for the empirical treatment of life-threatening group A streptococcal infections, until susceptibility data are available.<sup>487</sup> This adjunctive use of clindamycin is aimed at reducing the production of virulence factors such as superantigens and other toxins. Increasing antimicrobial resistance to clindamycin is tempering this enthusiasm and leading some to recommend alternative agents such as oxazolidinones in this setting.<sup>488</sup>

The dosage of clindamycin for adults depends on the site and severity of infection and the condition of the patient. Oral doses are usually 150 to 450 mg every 6 hours, and parenteral doses, given every 6 to 12 hours, usually total 600 to 2700 mg/day, occasionally higher.

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